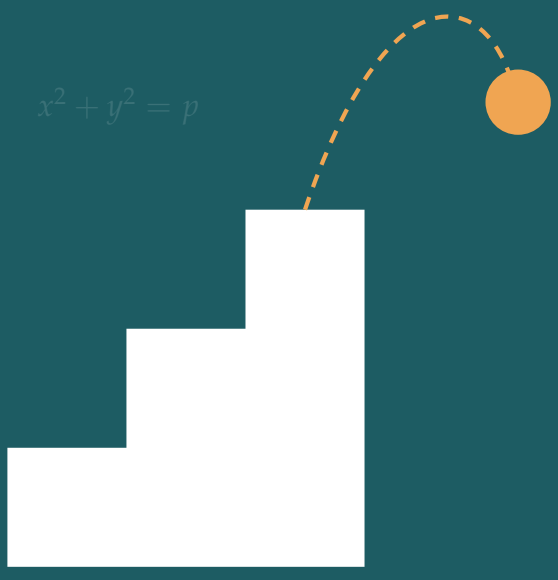
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2015 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2015 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2015 Worked Solutions

Worked solutions to all 39 questions of the 2015 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2015, Question 1

Pure • Curve Sketching

STEP I 2015, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q1

The key idea

Factorise everything you can. The quadratic factorises, so the zeros are immediate, and the derivative is again e^x times a quadratic that factorises, so the turning points are immediate too. Once the sketch is right, counting solutions of "curve $= k$" is just sliding a horizontal line up and down and watching how many times it meets the curve. For part (ii), notice that the new function is the old one evaluated at x^2, so we can reuse almost everything.

Solution

Part (i)

Let $f(x) = e^x(2x^2 - 5x + 2) = e^x(2x - 1)(x - 2)$.

Zeros and sign. Since $e^x > 0$, $f(x) = 0$ exactly when $x = \frac{1}{2}$ or $x = 2$. Also f is positive for $x < \frac{1}{2}$ and $x > 2$, and negative on $(\frac{1}{2}, 2)$. The curve crosses the y-axis at $f(0) = 2$.

Turning points. By the product rule,

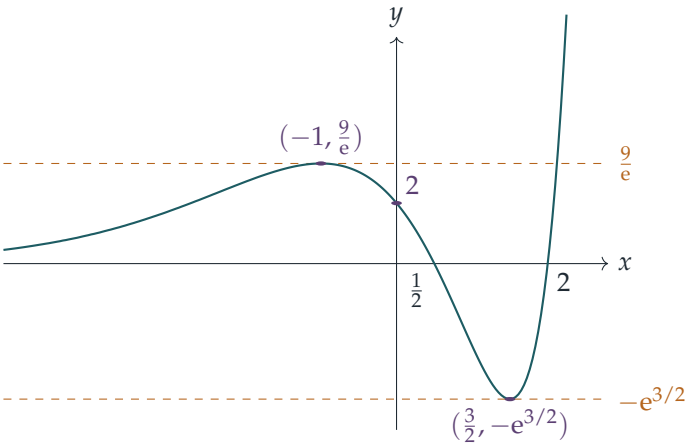
$$f'(x) = e^x(2x^2 - 5x + 2) + e^x(4x - 5) = e^x(2x^2 - x - 3) = e^x(2x - 3)(x + 1).$$

So $f'(x) = 0$ at $x = -1$ and $x = \frac{3}{2}$. The sign of f' is the sign of $(2x - 3)(x + 1)$: positive for $x < -1$, negative on $(-1, \frac{3}{2})$, positive for $x > \frac{3}{2}$. Hence f increases, then decreases, then increases, and

$$f(-1) = e^{-1}(2 + 5 + 2) = \frac{9}{e} \approx 3.31 \qquad \text{(local maximum)},$$

$$f(\tfrac{3}{2}) = e^{3/2}(\tfrac{9}{2} - \tfrac{15}{2} + 2) = -e^{3/2} \approx -4.48 \qquad \text{(local minimum)}.$$

Behaviour at the ends. As $x \to \infty$ both factors grow, so $f(x) \to \infty$. As $x \to -\infty$, the given fact (with $n = 2, 1, 0$) shows that each of $2x^2e^x$, $-5xe^x$ and $2e^x$ tends to 0, so $f(x) \to 0$. Since $f > 0$ for $x < \frac{1}{2}$, the curve approaches the negative x-axis from above: the x-axis is an asymptote.



Counting solutions of $f(x) = k$. The number of solutions is the number of times the horizontal line $y = k$ meets the curve. Going through the three monotonic pieces of the curve:

- on $(-\infty, -1]$, f increases strictly and takes every value in $(0, \frac{9}{e}]$ exactly once (it never reaches 0);
- on $[-1, \frac{3}{2}]$, f decreases strictly from $\frac{9}{e}$ to $-e^{3/2}$, taking each value in between exactly once;
- on $[\frac{3}{2}, \infty)$, f increases strictly from $-e^{3/2}$ to ∞, taking each value exactly once.

Adding up the contributions (and taking care not to count $x = -1$ or $x = \frac{3}{2}$ twice):

$k < -e^{3/2}$	no solutions
$k = -e^{3/2}$	one solution ($x = \frac{3}{2}$)
$-e^{3/2} < k \leq 0$	two solutions
$0 < k < \frac{9}{e}$	three solutions
$k = \frac{9}{e}$	two solutions ($x = -1$ and one with $x > 2$)
$k > \frac{9}{e}$	one solution

Ask yourself Why does $k = 0$ give two solutions, not three?

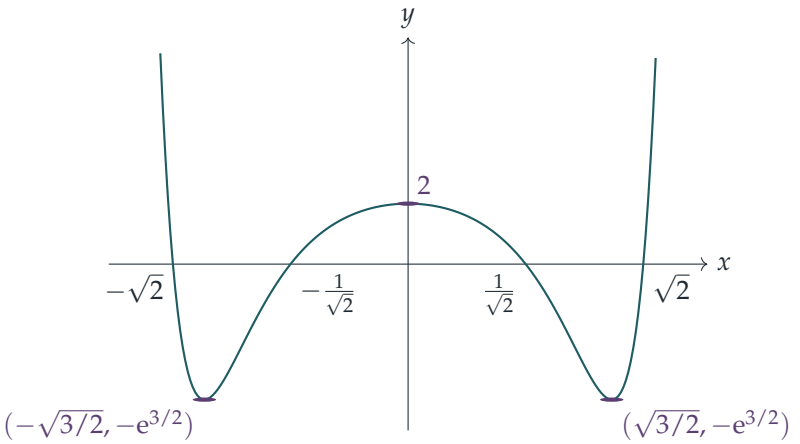
The x-axis is an asymptote on the left, and $f(x) > 0$ for every $x < \frac{1}{2}$, so the far-left branch never actually reaches 0. The only solutions of $f(x) = 0$ are $x = \frac{1}{2}$ and $x = 2$. For any small $k > 0$, though, the left branch does reach height k, which is why the count jumps to three.

Part (ii)

Let $g(x) = e^{x^2}(2x^4 - 5x^2 + 2)$. Then $g(x) = f(x^2)$, so g is even and its graph is symmetric in the y-axis. As x runs over $[0, \infty)$, $u = x^2$ runs over $[0, \infty)$ increasingly, so for $x \geq 0$ the graph of g just retraces the part of the graph of f with $u \geq 0$:

- $g(0) = f(0) = 2$;
- zeros where $x^2 = \frac{1}{2}$ or $x^2 = 2$, that is $x = \pm\frac{1}{\sqrt{2}}$ and $x = \pm\sqrt{2}$;
- by the chain rule, $g'(x) = 2x\,f'(x^2) = 2x\,\mathrm{e}^{x^2}(2x^2 - 3)(x^2 + 1)$, which vanishes only at $x = 0$ and $x = \pm\sqrt{3/2}$. (The turning point of f at $u = -1$ has disappeared, since $x^2 = -1$ has no real solution.)

For $0 < x < \sqrt{3/2}$ we have $g'(x) < 0$, and for $x > \sqrt{3/2}$ we have $g'(x) > 0$. So $(0, 2)$ is a local maximum (by symmetry $g' > 0$ just to the left of 0), and $x = \pm\sqrt{3/2}$ are minima with value $f(\frac{3}{2}) = -\mathrm{e}^{3/2}$. Finally $g(x) \to \infty$ as $x \to \pm\infty$, very rapidly because of the e^{x^2} factor. The curve is a "W" shape:



Checks

- $f(-1) = 9/\mathrm{e} \approx 3.311$ and $f(1.5) = -\mathrm{e}^{1.5} \approx -4.482$, and evaluating f at a few nearby points confirms that these are a maximum and a minimum.
- Near $x = 0$, $g(x) = f(x^2) \approx f(0) + f'(0)x^2 = 2 - 3x^2$, a downward parabola, which confirms the local maximum at $(0, 2)$.
- The root counts were confirmed by computer for sample values of k in each range (for example $k = 2$ gives three roots and $k = -1$ gives two).
- $f(-40) \approx 1.4 \times 10^{-14}$: positive, and tiny, as the asymptote predicts.

Common slips

- Counting three solutions when $k = 0$, by forgetting that the curve only approaches the x-axis on the left.
- Missing the boundary cases $k = \frac{9}{\mathrm{e}}$ and $k = -\mathrm{e}^{3/2}$, where the line is tangent to the curve and two solutions merge into one.

- In part (ii), keeping a turning point corresponding to $x^2 = -1$.

Beyond STEP

The counting argument rests on the intermediate value theorem together with strict monotonicity: a continuous strictly increasing function on an interval hits each value between its endpoint values exactly once. University analysis courses prove these facts carefully, along with the limit $x^n e^x \to 0$ that you were allowed to assume. Abbott's *Understanding Analysis* is a very readable place to see this done.

WORKED SOLUTION

STEP I 2015, Question 2

Pure • Trigonometric Solution of Cubics

> **STEP I 2015, Question 2**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/15-s1-q2

The key idea

The cubic $4x^3 - 3x$ is exactly what the triple-angle formula produces: $\cos 3\alpha = 4\cos^3\alpha - 3\cos\alpha$. So a cubic of this shape can be solved by guessing a cosine. Once we know one root, we can factorise and solve the leftover quadratic. In the final part a rescaling $y = 2x$ turns the given cubic into this shape, and part (i) supplies the exact values we need.

Solution

Part (i)

Write $15^\circ = 45^\circ - 30^\circ$ and use the compound-angle formulae:

$$\cos 15^\circ = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{3}+1}{2\sqrt{2}},$$

as required. In the same way,

$$\sin 15^\circ = \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}}.$$

Rationalising the denominators gives the equivalent forms

$$\cos 15^\circ = \frac{\sqrt{6}+\sqrt{2}}{4}, \qquad \sin 15^\circ = \frac{\sqrt{6}-\sqrt{2}}{4},$$

which are the handier ones in part (iii).

Part (ii)

Write $c = \cos\alpha$ and $s = \sin\alpha$. Expanding $\cos(2\alpha + \alpha)$,

$$\begin{aligned}\cos 3\alpha &= \cos 2\alpha \cos\alpha - \sin 2\alpha \sin\alpha \\ &= (2c^2 - 1)c - 2s^2 c \\ &= 2c^3 - c - 2(1 - c^2)c \\ &= 4c^3 - 3c.\end{aligned}$$

So $4c^3 - 3c - \cos 3\alpha = 0$: $x = \cos\alpha$ is a root.

For the other roots, replace $\cos 3\alpha$ by $4c^3 - 3c$ and factorise:

$$\begin{aligned}4x^3 - 3x - (4c^3 - 3c) &= 4(x^3 - c^3) - 3(x - c) \\ &= (x - c)\big(4x^2 + 4cx + 4c^2 - 3\big).\end{aligned}$$

Ask yourself How do we simplify the discriminant of the quadratic factor?

The discriminant is $16c^2 - 16(4c^2 - 3) = 48(1 - c^2) = 48s^2$, using $c^2 + s^2 = 1$. Its square root is $4\sqrt{3}\,|s|$, but since we take both signs $\pm$ we can drop the modulus. The quadratic formula gives

$$x = \frac{-4c \pm 4\sqrt{3}\,s}{8},$$

so the other two roots are

$$x = -\tfrac{1}{2}\cos\alpha + \tfrac{\sqrt{3}}{2}\sin\alpha \quad \text{and} \quad x = -\tfrac{1}{2}\cos\alpha - \tfrac{\sqrt{3}}{2}\sin\alpha.$$

These are $\cos(\alpha - 120^\circ)$ and $\cos(\alpha + 120^\circ)$. That makes sense: $3(\alpha \pm 120^\circ) = 3\alpha \pm 360^\circ$, so these angles have the same $\cos 3\theta$ as α.

Part (iii)

Ask yourself The given cubic is $y^3 - 3y - \sqrt{2}$, not $4x^3 - 3x - \cdots$. How do we match the two?

We want the cubic and linear coefficients to be in the ratio 4 to 3, as in part (ii). Put $y = 2x$:

$$8x^3 - 6x - \sqrt{2} = 0 \iff 4x^3 - 3x - \frac{\sqrt{2}}{2} = 0.$$

This is the equation of part (ii) with $\cos 3\alpha = \frac{\sqrt{2}}{2} = \cos 45^\circ$, so we may take $\alpha = 15^\circ$. Part (ii) then gives three roots x. They are distinct (see below), so they are all the roots of the cubic, and $y = 2x$ gives

$$y = 2\cos\alpha, \qquad y = -\cos\alpha \pm \sqrt{3}\sin\alpha, \qquad \alpha = 15^\circ.$$

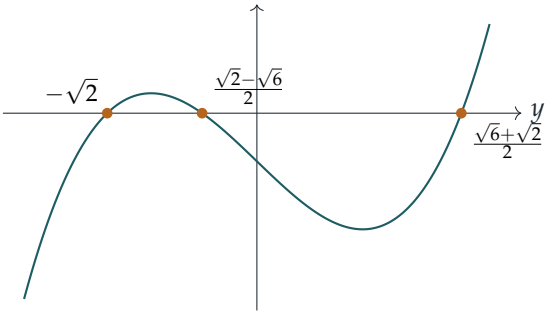
Now substitute the values from part (i).

- $y = 2\cos 15^\circ = \dfrac{\sqrt{3}+1}{\sqrt{2}} = \dfrac{\sqrt{6}+\sqrt{2}}{2}$.
- $y = -\cos 15^\circ + \sqrt{3}\sin 15^\circ = \dfrac{-(\sqrt{6}+\sqrt{2})+(\sqrt{18}-\sqrt{6})}{4}$, which simplifies to $\dfrac{2\sqrt{2}-2\sqrt{6}}{4} = \dfrac{\sqrt{2}-\sqrt{6}}{2}$.
- $y = -\cos 15^\circ - \sqrt{3}\sin 15^\circ = \dfrac{-(\sqrt{6}+\sqrt{2})-(3\sqrt{2}-\sqrt{6})}{4} = -\sqrt{2}$.

Here we used $\sqrt{3}\sqrt{6} = \sqrt{18} = 3\sqrt{2}$. These three values are clearly different. So the solutions are

$$y = \frac{\sqrt{6}+\sqrt{2}}{2}, \qquad y = \frac{\sqrt{2}-\sqrt{6}}{2}, \qquad y = -\sqrt{2}.$$

In terms of angles they are $2\cos 15^\circ$, $2\cos 255^\circ$ and $2\cos 135^\circ$.



Checks

- The y^2 coefficient of $y^3 - 3y - \sqrt{2}$ is zero, so the roots must sum to zero. Indeed $\frac{\sqrt{6}+\sqrt{2}}{2} + \frac{\sqrt{2}-\sqrt{6}}{2} - \sqrt{2} = 0$.
- The product of the roots should be $\sqrt{2}$. Since $(\sqrt{2}+\sqrt{6})(\sqrt{2}-\sqrt{6}) = 2-6 = -4$, the product is $(-1)\cdot(-\sqrt{2}) = \sqrt{2}$.
- The root $y = -\sqrt{2}$ is easy to check directly: $-2\sqrt{2} + 3\sqrt{2} - \sqrt{2} = 0$. Numerically, $\frac{\sqrt{6}+\sqrt{2}}{2} \approx 1.93185$, and $(1.93185)^3 - 3(1.93185) \approx 1.41421 = \sqrt{2}$.
- In part (ii), $\alpha = 0$ gives the equation $4x^3 - 3x - 1 = 0$, i.e. $(x-1)(2x+1)^2 = 0$. The formulae give roots $1, -\frac{1}{2}, -\frac{1}{2}$, as they should.

Common slips

- Forgetting to rescale in part (iii): solving $4x^3 - 3x - \sqrt{2} = 0$ instead. That equation would need $\cos 3\alpha = \sqrt{2} > 1$, which is impossible.
- Giving the answers for x rather than for $y = 2x$.

- Writing $\sqrt{3(1-c^2)} = \sqrt{3}\sin\alpha$ without noting that the $\pm$ sign takes care of the case $\sin\alpha < 0$.

Beyond STEP

This is the trigonometric method for solving a cubic with three real roots, due to Viète. Any cubic can be shifted to remove its square term, giving $t^3 + pt + q = 0$. When $p < 0$, the scaling $t = 2\sqrt{-p/3}\,x$ turns it into $4x^3 - 3x = k$. Cardano's formula fails to help here, because it routes through square roots of negative numbers even though all three roots are real. Seeing why that happens, and why the cosine trick sidesteps it, leads naturally to complex numbers and the roots of unity. Needham's *Visual Complex Analysis* gives a beautiful geometric picture of this.

WORKED SOLUTION

STEP I 2015, Question 3

Pure • Coordinate Geometry Visibility

STEP I 2015, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

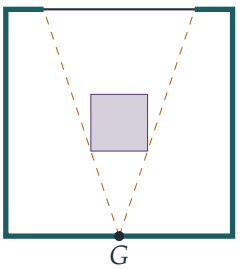
step.maths.org/questions/15-s1-q3

The key idea

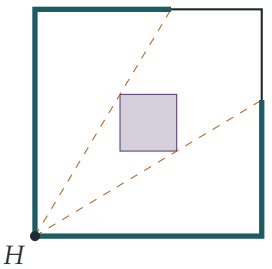
A guard's line of sight is cut off by the building, and since the building is a convex square, the lines that just graze one of its corners mark where the visible part of a wall ends. So for each position we draw these grazing lines, find where they meet the perimeter, and add up the visible lengths. Similar triangles (or the equation of a straight line) do all the work. The number $3a$ turns up as the value of b at which the grazing line from the middle of a wall reaches a corner of the courtyard.

Solution

Put the origin at the centre, so the perimeter wall is the square with sides on $x = \pm\frac{b}{2}$, $y = \pm\frac{b}{2}$, and the building is the square $|x| \leq \frac{a}{2}$, $|y| \leq \frac{a}{2}$, with $0 < a < b$. The total length of the perimeter is $4b$. A guard sees a point Q of the wall exactly when the segment from the guard to Q does not pass through the building.



middle of a wall ($b > 3a$)



corner

The thick (teal) parts of the wall are the ones the guard can see; the dashed lines graze corners of the building.

A guard at the middle of a wall

Take the guard at $G = (0, -\frac{b}{2})$, the middle of the bottom wall. The whole bottom wall is visible, since the building lies strictly above it: that is length b. By symmetry about the y-axis we only need to look to the right.

A line of sight going up and to the right misses the building exactly when it passes to the right of the corner $K = (\frac{a}{2}, -\frac{a}{2})$ nearest to it. (The line meets $y = -\frac{a}{2}$ first and its x-coordinate increases as it rises, so if it is to the right of the building at height $-\frac{a}{2}$, it stays to the right; if it is to the left of K there, it enters the building, because at height $-\frac{a}{2}$ its x-coordinate is between 0 and $\frac{a}{2}$.) So the boundary of what G sees is the line GK, which rises by $\frac{b-a}{2}$ over a run of $\frac{a}{2}$, so its gradient is $\frac{b-a}{a}$.

This line meets the right-hand wall $x = \frac{b}{2}$ at height

$$y_1 = -\frac{b}{2} + \frac{b}{2} \cdot \frac{b-a}{a} = -\frac{b}{2} + \frac{b(b-a)}{2a}.$$

It reaches the top of that wall ($y_1 \geq \frac{b}{2}$) exactly when $\frac{b-a}{a} \geq 2$, that is, when $b \geq 3a$.

? **Ask yourself** Why does the comparison of b with $3a$ come in here?

It decides which wall the grazing line GK hits: the right-hand wall if $b < 3a$, or the far (top) wall if $b > 3a$. The two cases give different formulas for the visible length.

Case $b < 3a$. The guard sees the right-hand wall from its bottom corner up to height y_1, a length

$$y_1 + \frac{b}{2} = \frac{b(b-a)}{2a},$$

and nothing beyond it (every line to a higher point of that wall, or to the top wall, passes to the left of K). With the matching piece on the left, the total visible length is

$$M = b + 2 \cdot \frac{b(b-a)}{2a} = b + \frac{b(b-a)}{a} = \frac{b^2}{a}.$$

Case $b > 3a$. Now the whole right-hand wall (length b) is visible, and the line GK meets the top wall $y = \frac{b}{2}$ after rising by b, so at

$$x = \frac{a}{b-a} \cdot b = \frac{ab}{b-a},$$

which is less than $\frac{b}{2}$ because $2a < b - a$. The guard sees the top wall from there to the corner, a length $\frac{b}{2} - \frac{ab}{b-a} = \frac{b(b-3a)}{2(b-a)}$. Adding the bottom wall, both side walls and both pieces of the top wall:

$$M = 3b + \frac{b(b-3a)}{b-a} = \frac{3b(b-a) + b(b-3a)}{b-a} = \frac{b(4b-6a)}{b-a}.$$

A guard in a corner

Take the guard at $H = (-\frac{b}{2}, -\frac{b}{2})$. The two walls through H are fully visible: length $2b$. The picture is symmetric about the diagonal $y = x$, so look at the right-hand wall. By the same argument as before, the visible part is cut off by the line through H and the corner $K = (\frac{a}{2}, -\frac{a}{2})$, which has gradient

$$\frac{\frac{b}{2} - \frac{a}{2}}{\frac{b}{2} + \frac{a}{2}} = \frac{b-a}{b+a}.$$

Over the run of b to the wall $x = \frac{b}{2}$ it rises by $\frac{b(b-a)}{a+b}$, which is less than b, so this line always hits the right-hand wall, whatever the values of a and b. The guard sees a length $\frac{b(b-a)}{a+b}$ of the right-hand wall, the same of the top wall, and none of the rest of either. Hence

$$C = 2b + \frac{2b(b-a)}{a+b} = \frac{2b(a+b) + 2b(b-a)}{a+b} = \frac{4b^2}{a+b}.$$

Comparing the two positions

Case $b < 3a$.

$$C - M = \frac{4b^2}{a+b} - \frac{b^2}{a} = \frac{b^2\big(4a - (a+b)\big)}{a(a+b)} = \frac{b^2(3a-b)}{a(a+b)} > 0,$$

so the **corner** is preferable.

Case $b > 3a$.

$$M - C = \frac{b(4b-6a)}{b-a} - \frac{4b^2}{a+b} = \frac{b\big[(4b-6a)(a+b) - 4b(b-a)\big]}{(b-a)(a+b)} = \frac{2ab(b-3a)}{(b-a)(a+b)} > 0,$$

since $(4b-6a)(a+b) - 4b(b-a) = 4b^2 - 2ab - 6a^2 - 4b^2 + 4ab = 2ab - 6a^2$. So the **middle of a wall** is preferable.

When $b = 3a$ both formulas for M give $9a$, and $C = \frac{36a^2}{4a} = 9a$ too, so the two positions are equally good.

Checks

- Take $a = 1$, $b = 2$: $M = 4$ and $C = \frac{16}{3} \approx 5.33$, so the corner wins. With $a = 1$, $b = 5$: $M = \frac{5 \cdot 14}{4} = 17.5$ and $C = \frac{100}{6} \approx 16.67$, so the middle wins. These values were confirmed by a computer check that tested the line of sight to thousands of points around the wall.
- If $b \to a$ the building fills the courtyard: $M \to a$ (only the guard's own wall) and $C \to 2a$ (the two walls meeting at the corner), as they should.
- As $b \to \infty$ the building becomes negligible: $M = \frac{b(4b-6a)}{b-a}$ and $C = \frac{4b^2}{a+b}$ both approach the full perimeter $4b$, and both stay below it.

Common slips

- Forgetting that for $b > 3a$ the guard at the middle of a wall can see part of the opposite wall, not only the two side walls.
- Grazing the wrong corner of the building: from below, the corners that matter are the two bottom ones, $(\pm\frac{a}{2}, -\frac{a}{2})$.
- Leaving out the guard's own wall (or walls) from the total.

Beyond STEP

Questions about which parts of a region can be seen from a point belong to computational geometry: the classic "art gallery theorem" says that $\lfloor n/3 \rfloor$ guards always suffice to watch every point of a polygonal gallery with n walls, and sometimes that many are needed. Working out a visibility region, as we did here, is the basic step in such problems, and in computer graphics, where hidden-surface removal works the same way. For more problems of this style, Siklos, *Advanced Problems in Mathematics*, is a good next step.

WORKED SOLUTION

STEP I 2015, Question 4

Pure • Parametric Curves and Area

STEP I 2015, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q4

The key idea

Use the gradient of the curve as the parameter. If the tangent at the midpoint makes angle θ with the x-axis, the gradient there is $\tan\theta$, which fixes the midpoint. Each end is then a distance b from the midpoint along the unit vector $(\cos\theta, \sin\theta)$. For the area, the half-rod AP lies along the tangent at P, so the region is just the gap between a parabola and one of its tangents, and that integrand is a perfect square.

Solution

The locus of an end

Let the midpoint be P and let the rod make angle θ with the positive x-axis. Since $y = \frac{1}{4}x^2$ has $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2}x$, and the rod is tangent to the curve at P,

$$\tfrac{1}{2}x_P = \tan\theta, \qquad \text{so} \qquad P = \left(2\tan\theta,\ \tan^2\theta\right).$$

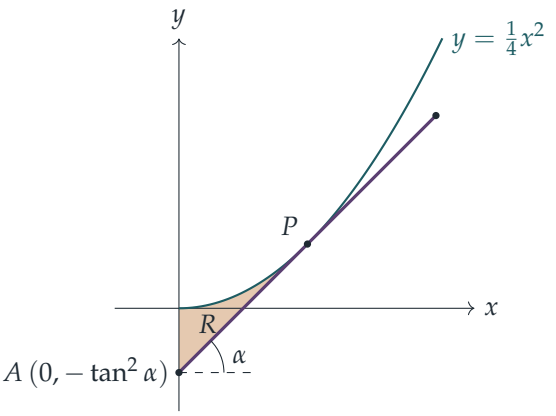
The condition $x \geq 0$ gives $\tan\theta \geq 0$, so we can take $0 \leq \theta < \frac{1}{2}\pi$. As θ runs over this range, $2\tan\theta$ takes every value in $[0, \infty)$, so every position of the midpoint arises from exactly one θ.

A unit vector along the rod is $(\cos\theta, \sin\theta)$, and each end is a distance b from P. So the ends are at $P \pm b(\cos\theta, \sin\theta)$. Taking the minus sign (the lower, left-hand end),

$$\begin{aligned} x &= 2\tan\theta - b\cos\theta, \\ y &= \tan^2\theta - b\sin\theta, \end{aligned}$$

as required. (The other end traces out the same equations with $+b$ in place of $-b$.)

The area of R



The end that reaches the y-axis is the one with $x = 2\tan\theta - b\cos\theta$: the other end has $x = 2\tan\theta + b\cos\theta > 0$ always. So at $\theta = \alpha$,

$$2\tan\alpha = b\cos\alpha.$$

The y-coordinate of A is $\tan^2\alpha - b\sin\alpha$, and we can eliminate b using the last equation:

$$b\sin\alpha = (b\cos\alpha)\tan\alpha = 2\tan^2\alpha, \qquad \text{so} \qquad A = \big(0, -\tan^2\alpha\big).$$

Ask yourself Where is A relative to the origin, and what is the equation of the line AP?

A is below the origin, and the line AP is the tangent to the parabola at P. Write $t = \tan\alpha$. The tangent at $P = (2t, t^2)$ has gradient t, so its equation is

$$y = t^2 + t(x - 2t) = tx - t^2,$$

which indeed passes through $A = (0, -t^2)$. The parabola lies above this tangent, since

$$\tfrac{1}{4}x^2 - (tx - t^2) = \big(\tfrac{1}{2}x - t\big)^2 \geq 0.$$

So R is the set of points with $0 \leq x \leq 2t$ lying between the line AP (below) and the curve (above). Its left edge is the segment of the y-axis from A up to O. Therefore

$$\begin{aligned}\text{Area of } R &= \int_0^{2t} \Big(\tfrac{1}{4}x^2 - (tx - t^2)\Big)\,\mathrm{d}x = \int_0^{2t} \big(\tfrac{1}{2}x - t\big)^2\,\mathrm{d}x \\ &= \Big[\tfrac{2}{3}\big(\tfrac{1}{2}x - t\big)^3\Big]_0^{2t} = 0 - \tfrac{2}{3}(-t)^3 = \tfrac{2}{3}t^3.\end{aligned}$$

So the area of R is $\frac{2}{3}\tan^3\alpha$, as required.

Notice that b has dropped out completely: the area depends only on where the tangent at P meets the y-axis. Of course b and α are linked by $2\sin\alpha = b\cos^2\alpha$, so b fixes α.

Checks

- Take $\alpha = \frac{1}{4}\pi$ (as in the diagram). Then $b\cos\alpha = 2$ gives $b = 2\sqrt{2}$, with $P = (2,1)$ and $A = (0,-1)$. The area is $\int_0^2 (\frac{1}{4}x^2 - x + 1)\,\mathrm{d}x = \frac{2}{3} - 2 + 2 = \frac{2}{3} = \frac{2}{3}\tan^3\frac{1}{4}\pi$.
- For $b = 2$, solving $2\tan\alpha = 2\cos\alpha$ numerically gives $\alpha \approx 0.6662$, and the area found by numerical integration is 0.32391, matching $\frac{2}{3}\tan^3\alpha$.
- As $b \to 0$, $\alpha \to 0$ and the region shrinks to nothing, as the formula predicts.

Common slips

- Splitting the region at the x-axis and forgetting the triangle below it. The single integral between the tangent and the curve handles both pieces at once.
- Using the wrong end of the rod: the end with $+b\cos\theta$ can never reach the y-axis.
- Leaving b in the answer instead of using $b\cos\alpha = 2\tan\alpha$ to eliminate it.

Beyond STEP

The midpoint traces the parabola, and each end traces a curve at constant distance b along the tangent. Building new curves from the tangents of an old one is a standard theme of the differential geometry of plane curves (involutes, envelopes and the tractrix are classic examples), met in first-year courses alongside arc length and curvature. The neat answer for the area recalls Archimedes' quadrature of the parabola, which shows that a parabolic segment has $\frac{4}{3}$ the area of a certain inscribed triangle. For more questions in this spirit, try Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 2015, Question 5

Pure • Integrals with a Parameter

STEP I 2015, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q5

The key idea

Both integrals have elementary antiderivatives in t, with x held fixed as a parameter, so the instruction "by evaluating the integral" is the whole hint: find a closed form in x, then sketch that. The care lies in the details. In (i) the integrand can be unbounded at $t = 1$, and in (ii) the square roots evaluate to *absolute values*, which makes g a piecewise function with a surprisingly simple shape.

Solution

Part (i)

For fixed $x > 0$, an antiderivative of $(t-1)^{x-1}$ is $\frac{1}{x}(t-1)^x$. So

$$f(x) = \left[\frac{(t-1)^x}{x}\right]_1^3 = \frac{2^x - 0}{x} = \frac{2^x}{x}.$$

When $0 < x < 1$ the integrand $(t-1)^{x-1}$ blows up as $t \to 1^+$, so the integral is improper; but $(t-1)^x \to 0$ as $t \to 1^+$ because $x > 0$, so it converges and the value above stands.

Now sketch $y = 2^x/x$ for $x > 0$.

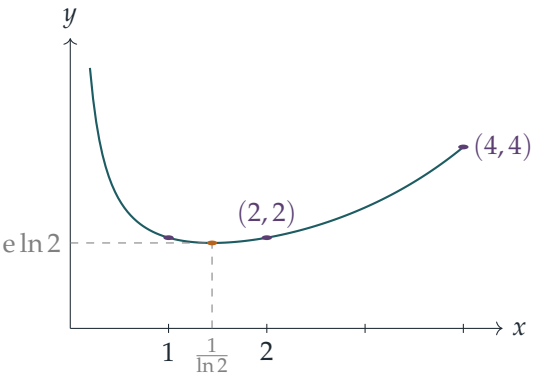
- As $x \to 0^+$, $2^x \to 1$, so $f(x) \to +\infty$: the y-axis is an asymptote.
- As $x \to \infty$, exponentials beat powers, so $f(x) \to \infty$ (and faster than any straight line).
- Differentiating, using $\frac{\mathrm{d}}{\mathrm{d}x}2^x = 2^x \ln 2$,

$$f'(x) = \frac{2^x \ln 2 \cdot x - 2^x}{x^2} = \frac{2^x(x \ln 2 - 1)}{x^2},$$

which is negative for $x < 1/\ln 2$ and positive for $x > 1/\ln 2$. So there is a single stationary point, a minimum, at $x = 1/\ln 2 \approx 1.44$.

- There, $2^{1/\ln 2} = e^{(\ln 2)/\ln 2} = e$, so the minimum value is
$$f\left(\frac{1}{\ln 2}\right) = e\ln 2 \approx 1.88.$$
- Useful points: $f(1) = f(2) = 2$, $f(4) = 4$.

The curve is also convex: $\ln f(x) = x\ln 2 - \ln x$ is convex (its second derivative is $1/x^2 > 0$), and the exponential of a convex function is convex. So the sketch is a single U-shaped curve.



Part (ii)

First, the integrand makes sense: for $-1 \le t \le 1$,
$$1 - 2xt + x^2 \ge 1 - 2|x| + x^2 = (1 - |x|)^2 \ge 0,$$
with equality only when $|x| = 1$ and $t = \pm 1$ (an endpoint). We return to that case below.

For $x = 0$ the integrand is 1, so $g(0) = 2$. For $x \ne 0$, treat x as a constant: since $\frac{\mathrm{d}}{\mathrm{d}t}(1 - 2xt + x^2) = -2x$, an antiderivative is $-\frac{1}{x}\sqrt{1 - 2xt + x^2}$. Hence
$$g(x) = -\frac{1}{x}\left[\sqrt{1 - 2xt + x^2}\right]_{-1}^{1} = \frac{\sqrt{(1+x)^2} - \sqrt{(1-x)^2}}{x} = \frac{|1+x| - |1-x|}{x}.$$

? **Ask yourself** Why not simply write $\sqrt{(1+x)^2} = 1 + x$ and $\sqrt{(1-x)^2} = 1 - x$?

Because $\sqrt{u^2} = |u|$, not u. Doing it carelessly gives $g(x) = 2$ for every x, which is certainly wrong for large $|x|$, where the integrand is roughly $1/|x|$ and so $g(x) \approx 2/|x|$. We must split into cases.

- $-1 \le x \le 1$, $x \ne 0$: both $1 + x$ and $1 - x$ are ≥ 0, so $g(x) = \dfrac{(1+x) - (1-x)}{x} = 2$. This also agrees with $g(0) = 2$.

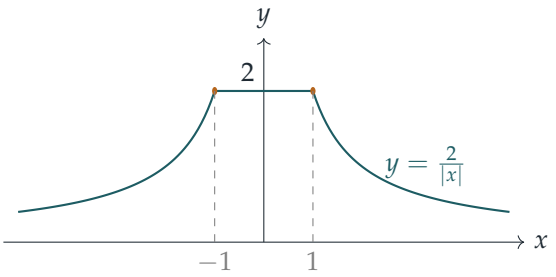
- $x > 1$: $|1+x| = 1+x$ and $|1-x| = x-1$, so $g(x) = \dfrac{2}{x}$.
- $x < -1$: $|1+x| = -1-x$ and $|1-x| = 1-x$, so $g(x) = \dfrac{-2}{x} = \dfrac{2}{|x|}$.

At $x = \pm 1$ the integrand is unbounded at one endpoint (for $x = 1$ it is $1/\sqrt{2-2t}$, which blows up as $t \to 1$), so the integral is improper. The antiderivative $-\sqrt{2-2t}$ is continuous up to $t = 1$, so it converges, and the formula still gives $g(\pm 1) = 2$.

So

$$g(x) = \begin{cases} 2, & |x| \le 1, \\ \dfrac{2}{|x|}, & |x| \ge 1. \end{cases}$$

The function is even (as it must be: replacing t by $-t$ shows $g(-x) = g(x)$). It is continuous everywhere, constant on $[-1, 1]$, and decays like $2/|x|$ towards the asymptote $y = 0$. At $x = \pm 1$ there is a corner: the one-sided gradients are 0 (inside) and ∓ 2 (outside), so g is not differentiable there.



Checks

- Part (i) at $x = 1$: the integrand is 1, so $f(1) = 2$, matching $2^1/1$. At $x = 2$ the integral is $\int_1^3 (t-1)\,\mathrm{d}t = 2$, matching $2^2/2$.
- Part (ii) at $x = 0$: the integrand is 1 and $g(0) = 2$. For large $|x|$ the integrand is close to $1/|x|$ over an interval of length 2, matching $2/|x|$.
- Numerically, $\int_1^3 (t-1)^{0.5}\,\mathrm{d}t = 1.8856\ldots = 2^{1.5}/1.5$; the integral in (ii) with $x = 1.2$ evaluates to $1.6667 = 2/1.2$, and with $x = 0.8$ to 2.0000.

Common slips

- Writing $\sqrt{(1-x)^2} = 1 - x$ for all x, and concluding that g is constant.
- Forgetting that the integral in (i) is improper for $0 < x < 1$, or that $(t-1)^x$ vanishes at $t = 1$ only because $x > 0$.
- Drawing g smooth at $x = \pm 1$: the two pieces meet with different gradients.

Beyond STEP

Part (ii) is Newton's shell theorem in disguise. With $t = \cos\theta$, the quantity $1/\sqrt{1-2xt+x^2}$ is the reciprocal distance from a point at distance x from the centre of a unit sphere to a point on the sphere, so g is (up to a constant) the gravitational potential of a uniform spherical shell: constant inside, and that of a point mass outside. The same expression is the generating function $\sum_n P_n(t)x^n$ of the Legendre polynomials, and $g(x) = 2$ for $|x| < 1$ reflects the fact that every P_n with $n \geq 1$ integrates to zero over $[-1, 1]$. Part (i), integrating t^{x-1} with x as a parameter, is the first step towards the Gamma and Beta functions. Mary Boas's *Mathematical Methods in the Physical Sciences* covers Legendre polynomials and the Gamma function at just the right level.

WORKED SOLUTION

STEP I 2015, Question 6

Pure • Vectors in Geometry

STEP I 2015, Question 6

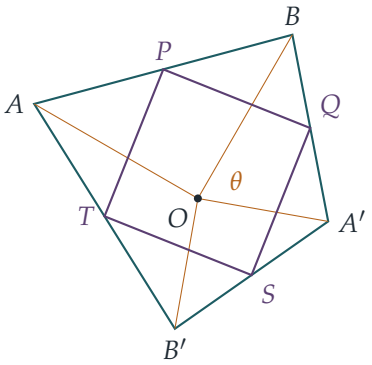
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q6

The key idea

With O as origin, the conditions $OA = OB$ and $\angle AOB = 90°$ say that B is A turned through a right angle about O, and likewise B' is A' turned the same way. Write R for that quarter-turn. Then each side of the midpoint quadrilateral turns out to be the previous side with R applied, and that is exactly what makes a square. Once the side is known in terms of $\mathbf{a}$ and $\mathbf{a}'$, the area is a cosine-rule calculation.

Solution



The midpoints form a square

Take O as origin and write $\mathbf{a}, \mathbf{b}, \mathbf{a}', \mathbf{b}'$ for the position vectors of A, B, A', B'. Let R be rotation through $90°$ clockwise about O. It is linear, it preserves lengths, and it turns every non-zero vector through a right angle.

Because the vertices go round clockwise and O lies inside the quadrilateral, going from OA to OB is a clockwise turn of $90°$. Since $OA = OB$, this gives $\mathbf{b} = R\mathbf{a}$. In the same way $\mathbf{b}' = R\mathbf{a}'$.

Let P, Q, S, T be the midpoints of AB, BA', $A'B'$, $B'A$:

$$\mathbf{p} = \tfrac{1}{2}(\mathbf{a}+\mathbf{b}), \quad \mathbf{q} = \tfrac{1}{2}(\mathbf{b}+\mathbf{a}'), \quad \mathbf{s} = \tfrac{1}{2}(\mathbf{a}'+\mathbf{b}'), \quad \mathbf{t} = \tfrac{1}{2}(\mathbf{b}'+\mathbf{a}).$$

Now take the four sides in turn. Let $\mathbf{u} = \overrightarrow{PQ}$:

$$\begin{aligned}
\overrightarrow{PQ} &= \tfrac{1}{2}(\mathbf{a}'-\mathbf{a}) = \mathbf{u},\\
\overrightarrow{QS} &= \tfrac{1}{2}(\mathbf{b}'-\mathbf{b}) = \tfrac{1}{2}R(\mathbf{a}'-\mathbf{a}) = R\mathbf{u},\\
\overrightarrow{ST} &= \tfrac{1}{2}(\mathbf{a}-\mathbf{a}') = -\mathbf{u},\\
\overrightarrow{TP} &= \tfrac{1}{2}(\mathbf{b}-\mathbf{b}') = -R\mathbf{u}.
\end{aligned}$$

Here $\mathbf{u} \neq \mathbf{0}$, because $A \neq A'$. So PQ and ST are equal and opposite, which makes $PQST$ a parallelogram. Its adjacent side QS is $\mathbf{u}$ turned through a right angle with no change in length. A parallelogram with equal, perpendicular adjacent sides is a square, so $PQST$ is a square, as required.

Ask yourself Why must the quarter-turn taking A to B be the same one that takes A' to B'?

Both pairs come one after the other in the clockwise labelling, and O is inside, so in each case the turn from the first ray to the second is clockwise. If one turn were clockwise and the other anticlockwise, the step from $\overrightarrow{QS}$ to $R\mathbf{u}$ would break down, and in general the figure would not be a square.

Maximising the area

Let $OA = OB = p$ and $OA' = OB' = q$, both fixed, and let $\theta = \angle BOA'$. The four angles at O add up to $360°$, and two of them are right angles, so $\angle B'OA = 180° - \theta$. As O is inside, both θ and $180° - \theta$ lie strictly between $0°$ and $180°$, so $0 < \theta < 180°$.

The square has side $|\mathbf{u}| = \frac{1}{2}AA'$, so its area is $\frac{1}{4}AA'^2$. Turning from OA through OB to OA', the angle between $\mathbf{a}$ and $\mathbf{a}'$ is $90° + \theta$, and $\cos(90° + \theta) = -\sin\theta$. By the cosine rule (or by expanding $|\mathbf{a}' - \mathbf{a}|^2$),

$$AA'^2 = p^2 + q^2 - 2pq\cos(90° + \theta) = p^2 + q^2 + 2pq\sin\theta,$$

so

$$\text{area} = \tfrac{1}{4}\left(p^2 + q^2 + 2pq\sin\theta\right).$$

With p and q fixed, this is largest when $\sin\theta$ is largest. On $0 < \theta < 180°$ that happens exactly at

$$\angle BOA' = \theta = 90°,$$

and the maximum area is $\frac{1}{4}(p+q)^2$. In this case all four angles at O are right angles, so A' lies on AO extended beyond O, and the side of the square is $\frac{1}{2}(p+q)$.

Checks

- Take $p = 2$, $q = 3$. The area formula gives $\frac{1}{4}(13 + 12\sin\theta)$, which is 3.25 near $\theta = 0$, rises to $6.25 = \frac{1}{4} \cdot 5^2$ at $\theta = 90°$, and falls back towards 3.25. A fine scan over θ by computer gives its peak at $90.0°$.
- Symmetric case $p = q$, $\theta = 90°$: $ABA'B'$ is itself a square centred at O with side $p\sqrt{2}$, and its midpoint square has side p, area $p^2 = \frac{1}{4}(2p)^2$, as the formula says.
- With random choices of p, q and θ, the midpoints were confirmed by computer to form a square with area $\frac{1}{4}(p^2 + q^2 + 2pq\sin\theta)$.

Common slips

- Taking the angle between OA and OA' to be θ rather than $90° + \theta$. That leads to the area $\frac{1}{4}(p^2 + q^2 - 2pq\cos\theta)$ and the wrong conclusion that θ should approach $180°$.
- Showing only that the four sides are equal. That gives a rhombus. You also need a right angle, and here it comes free from $\overrightarrow{QS} = R\,\overrightarrow{PQ}$.

Beyond STEP

A rotation through a right angle is a linear map, $R(x, y) = (y, -x)$, and in the complex plane it is multiplication by $-\mathrm{i}$. The whole argument is a computation with linear maps: $\mathbf{q} - \mathbf{p}$ and $\mathbf{s} - \mathbf{q}$ differ by R because R commutes with subtraction. The same idea gives quick proofs of results such as Napoleon's theorem and van Aubel's theorem. Needham's *Visual Complex Analysis* develops geometry through complex multiplication in exactly this way.

WORKED SOLUTION

STEP I 2015, Question 7

Pure • Maxima on an Interval

STEP I 2015, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q7

The key idea

The greatest value of a smooth function on a closed interval is attained either at a stationary point inside the interval or at an endpoint. Here the stationary points are at $x = 0$ and $x = \frac{a}{3}$, and the second one moves to the right as a grows, eventually leaving the interval. So we list the candidates, then compare them as a varies: the three ranges come from where the comparisons switch and from where $\frac{a}{3}$ passes $x = 1$.

Solution

The candidates

Differentiating,

$$f'(x) = 6ax - 18x^2 = 6x(a - 3x),$$

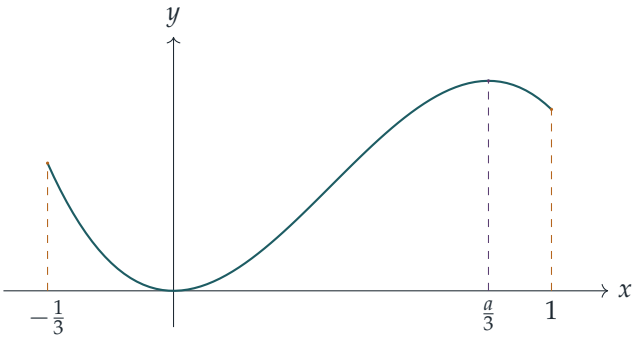
so the stationary points are $x = 0$ and $x = \frac{a}{3}$. For $a \geq 0$ the sign of f' tells us the shape:

- for $x < 0$, both $6x < 0$ and $a - 3x > 0$, so $f'(x) < 0$ and f is decreasing;
- for $0 < x < \frac{a}{3}$, $f'(x) > 0$ and f is increasing;
- for $x > \frac{a}{3}$, $f'(x) < 0$ and f is decreasing.

So $x = 0$ is a local minimum and $x = \frac{a}{3}$ a local maximum (when $a = 0$ the two coincide and $f = -6x^3$ is decreasing throughout). The values we need are

$$f\left(-\tfrac{1}{3}\right) = \frac{a}{3} + \frac{2}{9}, \qquad f\left(\tfrac{a}{3}\right) = \frac{a^3}{3} - \frac{2a^3}{9} = \frac{a^3}{9}, \qquad f(1) = 3a - 6.$$

Because f decreases on $[-\frac{1}{3}, 0]$, the largest value on that piece is $f(-\frac{1}{3})$; the value $f(0) = 0$ is never the greatest. So $\mathrm{M}(a)$ is the largest of $f(-\frac{1}{3})$ and the greatest value of f on $[0, 1]$.



The sketch shows $a = \frac{5}{2}$: the three candidates are the two endpoints (amber) and the local maximum (plum), which here is the winner.

Case $0 \le a \le 3$

Now $\frac{a}{3} \in [0,1]$, so on $[0,1]$ the function rises to $x = \frac{a}{3}$ and then falls. Its greatest value there is $f(\frac{a}{3}) = \frac{a^3}{9}$, and we must compare this with $f(-\frac{1}{3})$:

$$\frac{a^3}{9} - \left(\frac{a}{3} + \frac{2}{9}\right) = \frac{a^3 - 3a - 2}{9} = \frac{(a+1)^2(a-2)}{9}.$$

Ask yourself How do we spot the factorisation of $a^3 - 3a - 2$?

Try small integers: $a = -1$ gives $-1 + 3 - 2 = 0$ and $a = 2$ gives $8 - 6 - 2 = 0$. The sum of the three roots is 0 (there is no a^2 term), so the third root is -1 again, giving $(a+1)^2(a-2)$.

Since $(a+1)^2 > 0$ for $a \ge 0$, the difference has the sign of $a - 2$. Hence

$$\mathrm{M}(a) = \frac{a}{3} + \frac{2}{9} \quad \text{for } 0 \le a \le 2, \qquad \mathrm{M}(a) = \frac{a^3}{9} \quad \text{for } 2 \le a \le 3,$$

with both formulas giving $\frac{8}{9}$ at $a = 2$.

Case $a > 3$

Now $\frac{a}{3} > 1$, so f is increasing on all of $[0,1]$ and its greatest value there is $f(1) = 3a - 6$. Comparing with the left endpoint,

$$(3a - 6) - \left(\frac{a}{3} + \frac{2}{9}\right) = \frac{8a}{3} - \frac{56}{9} = \frac{8}{9}(3a - 7),$$

which is positive since $a > 3 > \frac{7}{3}$. So $\mathrm{M}(a) = 3a - 6$ for $a \ge 3$; at $a = 3$ this is $3 = \frac{27}{9}$, matching the middle formula.

Conclusion

$$\mathrm{M}(a) = \begin{cases} \dfrac{a}{3} + \dfrac{2}{9}, & 0 \le a \le 2, \\ \dfrac{a^3}{9}, & 2 \le a \le 3, \\ 3a - 6, & a \ge 3. \end{cases}$$

In words: for small a the left endpoint wins, for moderate a the local maximum wins, and once the local maximum has left the interval the right endpoint wins.

Checks

- M is continuous: $\frac{8}{9}$ at $a = 2$ from both sides and 3 at $a = 3$ from both sides, as it must be, since the greatest value changes continuously with a.
- $a = 0$: $f(x) = -6x^3$ is decreasing, so its greatest value is $f(-\frac{1}{3}) = \frac{2}{9}$, agreeing with the first formula.
- $a = \frac{5}{2}$: the formula gives $\frac{125}{72} \approx 1.736$, while the endpoints give $f(-\frac{1}{3}) \approx 1.056$ and $f(1) = 1.5$, both smaller. A computer search over a fine grid of x, for several hundred values of a between 0 and 12, agreed with the formula to within 10^{-8}.

Common slips

- Forgetting the left endpoint $x = -\frac{1}{3}$. Because f decreases for $x < 0$, this endpoint can beat the local maximum, and it does for $a < 2$.
- Using $f(\frac{a}{3})$ for $a > 3$, when $\frac{a}{3}$ is outside the interval and so not available.
- Assuming that a local maximum is automatically the greatest value on the interval.

Beyond STEP

Here $\mathrm{M}(a) = \max_{x \in [-1/3, 1]} f(x)$ is a function defined by optimising over a family, and such functions are typically continuous but only piecewise smooth: you might check that the left and right derivatives of M disagree at $a = 2$. That a continuous function on a closed bounded interval attains its maximum at all is the Extreme Value Theorem, proved in a first analysis course; see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 2015, Question 8

Pure • Divisibility of Power Sums

STEP I 2015, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q8

The key idea

Write the sum twice, once forwards and once backwards, and add: this is Gauss's trick for part (i), and it is exactly what part (ii) is built for when the terms are kth powers with k odd. Pairing r^k with $(n-r)^k$ shows that $2S$ is a multiple of n; pairing r^k with $(n+1-r)^k$ shows that $2S$ is also a multiple of $n+1$. Since n and $n+1$ share no common factor, the two facts combine to give divisibility by $\frac{1}{2}n(n+1)$.

Solution

Part (i)

Let $T = 1+2+\cdots+n$. Writing the sum in reverse order,

$$T = 1+2+\cdots+(n-1)+n,$$
$$T = n+(n-1)+\cdots+2+1.$$

Adding the columns, each of the n columns sums to $n+1$, so $2T = n(n+1)$ and

$$1+2+\cdots+n = \tfrac{1}{2}n(n+1).$$

Part (ii)

Working modulo N, we have $N-m \equiv -m$, so

$$(N-m)^k + m^k \equiv (-m)^k + m^k = -m^k + m^k = 0 \pmod{N},$$

where $(-m)^k = -m^k$ because k is odd. Hence N divides $(N-m)^k + m^k$.

If you prefer to avoid congruences: for odd k we have the factorisation

$$a^k + b^k = (a+b)\left(a^{k-1} - a^{k-2}b + a^{k-3}b^2 - \cdots + b^{k-1}\right),$$

(the signs alternate and there are k terms in the bracket, so the last sign is $+$). Taking $a = N-m$, $b = m$ gives $a+b = N$ times an integer. Nothing here needs $m \le N$: if $m > N$ then $N-m$ is negative, but it is still an integer.

Divisibility by n or $\frac{1}{2}n$

Let $S = 1^k + 2^k + \cdots + n^k$ with k odd. We copy the trick of part (i). Since $0^k = 0$, we may write

$$S = \sum_{r=0}^{n} r^k = \sum_{r=0}^{n} (n-r)^k,$$

the second form being the same sum in reverse order. Adding,

$$2S = \sum_{r=0}^{n} \Big[(n-r)^k + r^k \Big].$$

By part (ii) with $N = n$ and $m = r$, every bracket is divisible by n, so n divides $2S$. Say $2S = nM$ with M an integer.

- If n is odd, then n divides $2S$ and shares no factor with 2, so n divides S.
- If n is even, then $S = \frac{1}{2}n \cdot M$ with $\frac{1}{2}n$ an integer, so $\frac{1}{2}n$ divides S.

? **Ask yourself** Why can't we simply say "n divides S" when n is even?

Because the step from "$n \mid 2S$" to "$n \mid S$" needs n to be coprime to 2. Indeed it can fail: with $n = 2$, $k = 1$ we get $S = 3$, which is not a multiple of 2. All we can say for even n is that half of n divides S.

Divisibility by $1 + 2 + \cdots + n$

By part (i) we need $\frac{1}{2}n(n+1) \mid S$. Pair the terms differently: reversing $1, 2, \ldots, n$ sends r to $n+1-r$, so

$$2S = \sum_{r=1}^{n} \Big[(n+1-r)^k + r^k \Big],$$

and by part (ii) with $N = n+1$ and $m = r$, every bracket is divisible by $n+1$. So $n+1$ divides $2S$.

Now n and $n+1$ are consecutive integers, so they have no common factor other than 1; the same is then true of any factor of n and any factor of $n+1$. We use the fact that if two coprime integers each divide S, then so does their product.

- *n odd.* We showed $n \mid S$. Also $n+1$ is even and $n+1 \mid 2S$, so writing $2S = (n+1)L$ gives $S = \frac{1}{2}(n+1)L$, i.e. $\frac{1}{2}(n+1) \mid S$. Since n and $\frac{1}{2}(n+1)$ are coprime, their product $\frac{1}{2}n(n+1)$ divides S.
- *n even.* We showed $\frac{1}{2}n \mid S$. Also $n+1$ is odd and divides $2S$, so (being coprime to 2) it divides S. Since $\frac{1}{2}n$ and $n+1$ are coprime, their product $\frac{1}{2}n(n+1)$ divides S.

In both cases S is divisible by $\frac{1}{2}n(n+1) = 1 + 2 + \cdots + n$, as required.

Ask yourself Why is it legitimate to combine two divisibility facts into one by multiplying the divisors?

If $a \mid S$ and $b \mid S$ with $\gcd(a, b) = 1$, write $S = aq$. Then $b \mid aq$ and b shares no prime factor with a, so every prime power dividing b must divide q; hence $b \mid q$ and $ab \mid S$. Without coprimality this fails: $4 \mid 12$ and $6 \mid 12$, but $24 \nmid 12$. That is why we were careful to pair n with $\frac{1}{2}(n+1)$, or $\frac{1}{2}n$ with $n+1$, rather than, say, n with $n+1$ when n is even.

Checks

Try $k = 3$, $n = 4$: $S = 1 + 8 + 27 + 64 = 100$, and $1 + 2 + 3 + 4 = 10$ divides 100. In fact $1^3 + \cdots + n^3 = \left(\frac{1}{2}n(n+1)\right)^2$, so for $k = 3$ the result is immediate. For $k = 5$, $n = 5$: $S = 1 + 32 + 243 + 1024 + 3125 = 4425 = 15 \times 295$. The final result was also confirmed by computer for every $n \leq 119$ and every odd $k \leq 15$. For even k the claim fails: $1^2 + 2^2 = 5$ is not divisible by 3, which shows that oddness of k is really used.

Common slips

- Concluding $n \mid S$ from $n \mid 2S$ when n is even.
- Multiplying divisors that are not coprime, for example claiming $n(n+1) \mid 2S$ and then halving without care.
- Forgetting the term $r = 0$ (or $r = n$) when pairing r^k with $(n-r)^k$: including $0^k = 0$ makes every term pair up neatly.

Beyond STEP

Power sums $1^k + \cdots + n^k$ are polynomials in n of degree $k+1$, given by Faulhaber's formula in terms of the Bernoulli numbers; for odd k they can even be written as polynomials in $\frac{1}{2}n(n+1)$, which explains the result from another angle. The argument here is a first taste of working with congruences and coprimality, the bread and butter of elementary number theory. Davenport's *The Higher Arithmetic* is a lovely introduction.

WORKED SOLUTION

STEP I 2015, Question 9

Mechanics • Projectile Motion

STEP I 2015, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q9

The key idea

"Last to hit the ground" is about the *landing time*, not the firing time: a bullet fired later at a lower angle spends less time in the air, so it may land before an earlier one. Write the landing time as firing time plus time of flight, and maximise it over the firing period. Using the angle α as the variable instead of t makes the calculus clean, and the condition on k decides whether the maximum is at a stationary point or at an end.

Solution

Setting up

Model each bullet as a particle moving freely under gravity, with no air resistance, fired from ground level. A bullet fired with elevation α has vertical position $y = u\sin\alpha\, s - \frac{1}{2}gs^2$ at time s after firing, so it returns to the ground after

$$\frac{2u\sin\alpha}{g},$$

and in that time it travels horizontally

$$R(\alpha) = u\cos\alpha\cdot\frac{2u\sin\alpha}{g} = \frac{2u^2\sin\alpha\cos\alpha}{g}.$$

The bullet with elevation α is fired at time $t = (\frac{1}{3}\pi - \alpha)/\lambda$, and as t runs over $[0, \frac{\pi}{6\lambda}]$, α runs over $[\frac{1}{6}\pi, \frac{1}{3}\pi]$, each value once. So this bullet lands at time

$$L(\alpha) = \frac{\frac{1}{3}\pi - \alpha}{\lambda} + \frac{2u\sin\alpha}{g}, \qquad \tfrac{1}{6}\pi \le \alpha \le \tfrac{1}{3}\pi.$$

The last bullet to land is the one that maximises L. Since $k = \dfrac{g}{2\lambda u}$, we have $\dfrac{2u}{g} = \dfrac{1}{\lambda k}$, and

$$L(\alpha) = \frac{1}{\lambda}\left(\tfrac{1}{3}\pi - \alpha + \frac{\sin\alpha}{k}\right), \qquad L'(\alpha) = \frac{1}{\lambda}\left(\frac{\cos\alpha}{k} - 1\right) = \frac{\cos\alpha - k}{\lambda k}.$$

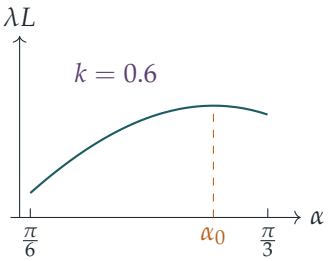
So L increases where $\cos\alpha > k$ and decreases where $\cos\alpha < k$. On our interval, $\cos\alpha$ decreases from $\frac{1}{2}\sqrt{3}$ (at $\alpha = \frac{1}{6}\pi$) to $\frac{1}{2}$ (at $\alpha = \frac{1}{3}\pi$).

Ask yourself Why does a larger α mean an *earlier* firing time, and why does that matter for the sign of L'?

The barrel is lowering, so the steep bullets go first. Raising α by a little moves the firing time earlier by $1/\lambda$ per radian, but adds $\frac{2u}{g}\cos\alpha$ per radian to the flight time. $L'(\alpha)$ is the balance of these two effects, and the sign tells us which wins.

The case $\frac{1}{2} \le k \le \frac{1}{2}\sqrt{3}$

Now k lies in the range of $\cos\alpha$, so there is exactly one $\alpha_0 \in [\frac{1}{6}\pi, \frac{1}{3}\pi]$ with $\cos\alpha_0 = k$. Because cos is decreasing on the interval, $\cos\alpha > k$ for $\alpha < \alpha_0$ and $\cos\alpha < k$ for $\alpha > \alpha_0$. So L increases up to α_0 and decreases after it, and the bullet fired at elevation α_0 is the last to land. (If $k = \frac{1}{2}$ or $k = \frac{1}{2}\sqrt{3}$, then α_0 is an endpoint and L is monotonic, but the conclusion is the same.)



(The vertical axis is shifted so that the variation in landing time is visible.)

Its range is, using $\sin\alpha_0 = \sqrt{1-k^2}$ (positive since α_0 is acute),

$$R(\alpha_0) = \frac{2u^2 \sin\alpha_0 \cos\alpha_0}{g} = \frac{2ku^2\sqrt{1-k^2}}{g},$$

which is the required distance.

The case $k < \frac{1}{2}$

Now $\cos\alpha \ge \frac{1}{2} > k$ for every α in the interval, so $L'(\alpha) > 0$ throughout and L is strictly increasing. The maximum is at $\alpha = \frac{1}{3}\pi$: the *first* bullet fired is the last to land. Its range is

$$R\left(\tfrac{1}{3}\pi\right) = \frac{2u^2}{g} \cdot \frac{\sqrt{3}}{2} \cdot \frac{1}{2} = \frac{\sqrt{3}\,u^2}{2g}.$$

So for $k < \frac{1}{2}$ the last bullet lands a distance $\dfrac{\sqrt{3}\,u^2}{2g}$ from the gun.

(For completeness: if $k > \frac{1}{2}\sqrt{3}$, then L is decreasing, the final bullet fired is the last to land, and since $\sin\frac{\pi}{3} = \sin\frac{2\pi}{3}$ the distance is again $\frac{\sqrt{3}\,u^2}{2g}$.)

Checks

- Continuity: at $k = \frac{1}{2}$ the first formula gives $\dfrac{2 \cdot \frac{1}{2}u^2 \cdot \frac{\sqrt{3}}{2}}{g} = \dfrac{\sqrt{3}\,u^2}{2g}$, agreeing with the $k < \frac{1}{2}$ answer; at $k = \frac{1}{2}\sqrt{3}$ it gives the same value again.
- Taking $u^2/g = 1$ and $k = 0.6$, the last bullet has $\cos\alpha_0 = 0.6$ and lands at distance $2(0.6)(0.8) = 0.96$. A computer check sampling twenty thousand firing times, and stepping the motion of that bullet numerically, agreed.
- Units: k is dimensionless (g over λu, which is speed per time), and u^2/g is a length.

Common slips

- Maximising the range, or the time of flight alone, instead of the landing time $t + \frac{2u\sin\alpha}{g}$.
- Assuming the last bullet fired is the last to land.
- Getting the sign of L' wrong by forgetting that t decreases as α increases.

Beyond STEP

Maximising a function over a closed interval means comparing interior stationary points with the endpoints, and here a parameter decides which one wins. This kind of "which constraint is active" switch is the starting point of constrained optimisation, which you will meet in first-year methods courses. For more mechanics in this spirit, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 2015, Question 10

Mechanics • Relative Velocity

STEP I 2015, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q10

The key idea

Rain hitting a surface is a flux: the amount per unit time is (number of drops per unit volume) $\times$ (area) $\times$ (component of the rain's velocity perpendicular to the surface). When the bus moves, work in the bus's frame, where only the horizontal component of the rain's velocity changes. Then, for the whole journey, multiply the rate by the journey time D/u and minimise over u. The modulus splits this into two cases, according to whether the bus is slower or faster than the horizontal motion of the rain.

Solution

The rate at which rain hits the bus

Model the rain as drops spread uniformly through the air, with n drops per unit volume, all moving with the same velocity. Let the bus have width b. Take north as the positive horizontal direction. Rain coming *from* the south moves northwards, so its velocity has

$$\text{horizontal component } v\cos\theta \text{ (north)}, \qquad \text{vertical component } v\sin\theta \text{ (down)}.$$

The rain has no east–west component, so the two long sides of the bus stay dry.

The case $u = 0$. In time δt, the drops that hit the roof are those in a slanting prism standing on the roof, of vertical height $v\sin\theta\,\delta t$. Its volume is $ab\,v\sin\theta\,\delta t$. The drops hitting the back (south face, area hb) fill a prism of horizontal depth $v\cos\theta\,\delta t$ behind it. The front, facing north, is sheltered. So the rate is

$$nb\left(h\,v\cos\theta + a\,v\sin\theta\right).$$

The case $u > 0$. In the frame of the bus, which is moving at a constant velocity, the rain has velocity

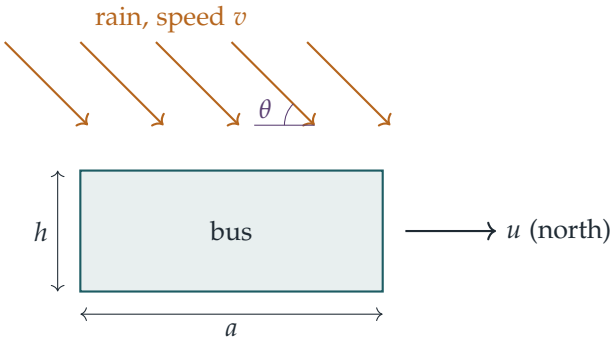
$$(v\cos\theta - u) \text{ north}, \qquad v\sin\theta \text{ down}.$$

The bus is at rest in this frame, so the argument above applies with $v\cos\theta$ replaced by $v\cos\theta - u$. The vertical component is unchanged, so the roof still collects at rate

$nab\,v\sin\theta$. If $v\cos\theta > u$, the rain still overtakes the bus and hits the back at rate $nhb\,(v\cos\theta - u)$. If $v\cos\theta < u$, the bus runs into the rain, which now hits the front at rate $nhb\,(u - v\cos\theta)$. In every case the total rate is

$$nb\left(h\,|v\cos\theta - u| + a\,v\sin\theta\right),$$

which is proportional to $h|v\cos\theta - u| + av\sin\theta$, as required.



Choosing the speed

Let the journey have length D. It takes time D/u, so the total rain collected is proportional to

$$R(u) = \frac{D}{u}\left(h|v\cos\theta - u| + av\sin\theta\right), \qquad 0 < u \leq w.$$

Write $c = v\cos\theta$ and remove the modulus.

If $u \leq c$:

$$R(u) = D\left(\frac{hv\cos\theta + av\sin\theta}{u} - h\right).$$

The numerator is positive, so R is strictly decreasing on $0 < u \leq c$.

If $u \geq c$:

$$R(u) = D\left(h + \frac{v\,(a\sin\theta - h\cos\theta)}{u}\right).$$

Whether this increases or decreases depends only on the sign of $a\sin\theta - h\cos\theta$.

? **Ask yourself** Why is going faster not always better?

Going faster shortens the journey, which always helps the roof: the roof's share is $Dav\sin\theta/u$. Once $u > c$, though, the bus runs into the rain, and the front's share $Dh(u-c)/u$ grows with u. The sign of $a\sin\theta - h\cos\theta$ tells us which effect wins.

The two formulae agree at $u = c$, so R is continuous. Now:

- If $w < v\cos\theta$, every allowed speed has $u \leq c$, where R is decreasing. So the best choice is $u = w$.

- If $a\sin\theta > h\cos\theta$, then R is decreasing on $u \le c$ and also on $u \ge c$. So R is decreasing on the whole of $(0, w]$, and again the best choice is $u = w$.

This proves the first claim.

If $w > v\cos\theta$ and $a\sin\theta < h\cos\theta$: R decreases on $(0, c]$ and strictly increases on $[c, w]$. So the minimum is at

$$u = v\cos\theta,$$

which means driving at exactly the horizontal speed of the rain. The rain then falls vertically relative to the bus, so only the roof gets wet, and the total is $R(c) = Da\tan\theta$ (in our units).

If $a\sin\theta = h\cos\theta$: for $u \ge c$, $R(u) = Dh$, which does not depend on u. If $w \ge v\cos\theta$, then any speed between $v\cos\theta$ and w is equally good. Every speed below $v\cos\theta$ is worse. (If $w < v\cos\theta$, the first case applies and $u = w$.) Physically, beyond $u = c$, going faster saves exactly as much rain on the roof as it adds on the front. The condition says $\tan\theta = h/a$: seen from the road, the rain falls parallel to a diagonal of the bus's side.

The return journey

Now the bus travels south, into the rain. Its velocity is $-u$ in our northward convention, so the relative horizontal velocity of the rain is $v\cos\theta + u > 0$, and it always hits the front. The total is proportional to

$$R(u) = \frac{D}{u}\Big(h(v\cos\theta + u) + av\sin\theta\Big) = D\left(h + \frac{v\,(h\cos\theta + a\sin\theta)}{u}\right),$$

which is strictly decreasing in u. So on the way back the driver should go as fast as possible, $u = w$, whatever the values of a, h and θ.

Checks

- With $a = 1.3$, $h = 0.8$, $v = 2$, $\theta = 0.6$ and $u = 0.7$, the formula gives a rate of 2.229 (per unit drop density and width). A computer simulation of randomly placed drops crossing the bus's cross-section gave 2.232. The agreement was equally good for $u = 0$, for $u > v\cos\theta$ and for negative u (the return journey).
- Take $\theta \to \frac{1}{2}\pi$ (rain falling vertically). Then $a\sin\theta > h\cos\theta$, so we should drive at full speed. That makes sense: only the roof gets wet, and less time in the rain means less rain.
- In the equality case with $\tan\theta = h/a$, the total Dh is the same at $u = v\cos\theta$, $1.5v\cos\theta$ and $3v\cos\theta$, as predicted.

Common slips

- Minimising the *rate* instead of the *total*: the journey time D/u must multiply the rate.

- Forgetting that for $u > v\cos\theta$ the rain hits the front, not the back. This is the reason for the modulus.
- On the return journey, using $|v\cos\theta - u|$ again; the relative speed is now $v\cos\theta + u$.

Beyond STEP

The calculation "density $\times$ area $\times$ normal velocity" is the flux of a vector field through a surface. In vector calculus this becomes the surface integral $\iint \rho\,\mathbf{v}\cdot\mathbf{n}\,\mathrm{d}S$, which is the starting point for conservation laws in fluid dynamics and electromagnetism. Changing to the bus's frame is a Galilean transformation, the classical forerunner of relativity. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both relative motion and flux arguments of this kind.

WORKED SOLUTION

STEP I 2015, Question 11

Mechanics • Friction and Moments

STEP I 2015, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q11

The key idea

Every normal reaction on a circular cylinder acts along a radius, so it passes through the centre. Taking moments about each centre therefore kills the weight and all the normal reactions, leaving only the frictional forces, which all act at the same distance from the centre. That gives part (i) almost for free. For part (ii), resolve along the plane for each cylinder, then apply the friction law at the two contacts that the inequalities concern.

Solution

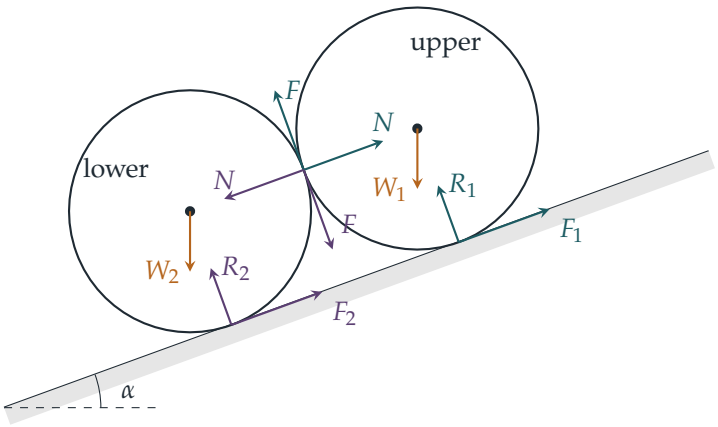
We model the cylinders as uniform rigid cylinders of radius r, both touching the plane, with the upper one higher up the slope. Since the radii are equal, the line of centres is parallel to the plane, so the point where the cylinders touch is at height r above the plane, level with both centres. The normal force between the cylinders acts along the line of centres (parallel to the plane), and the friction between them is perpendicular to the plane.

Take the x-direction up the slope and the y-direction perpendicular to the plane, away from it. Let

- R_1, R_2 be the normal reactions of the plane on the upper and lower cylinders;
- F_1, F_2 be the frictional forces of the plane on the upper and lower cylinders, measured *up* the slope (we let these be signed, so a negative value would mean down the slope);
- N be the normal force of the lower cylinder on the upper one, which acts up the slope;
- F be the friction of the lower cylinder on the upper one, measured *away from* the plane (again signed).

By Newton's third law the upper cylinder exerts N down the slope and F towards the

plane on the lower cylinder.



Forces on the upper cylinder are in teal and those on the lower cylinder in plum.

Part (i)

Take moments about the centre of the upper cylinder. Its weight acts at the centre, and R_1 and N both act along radii, so none of them has a moment. Only the two frictional forces remain, each acting along a tangent at distance r from the centre:

- F_1 acts up the slope at the lowest point of the cylinder, giving moment rF_1 anticlockwise (in the picture);
- F acts away from the plane at the point of contact, which is on the down-slope side of the centre, giving moment rF clockwise.

For equilibrium these balance, so $rF_1 = rF$, that is, $F_1 = F$.

Ask yourself Why can we ignore the normal reactions when we take moments about the centre?

A normal reaction on a circle is perpendicular to the tangent at the point of contact, so it lies along the radius and its line of action passes through the centre. Its moment about the centre is zero. This is exactly why moments about the centres are the right choice here.

For the lower cylinder, in the same way, only F_2 (up the slope at its lowest point, moment rF_2 anticlockwise) and the friction F towards the plane at the point of contact, which is on the up-slope side of its centre (moment rF clockwise) have moments. So $F_2 = F$.

Hence $F = F_1 = F_2$ as signed quantities, and in particular their magnitudes are equal, as required.

Part (ii)

Resolve parallel to the plane for each cylinder:

$$\begin{aligned} \text{upper:} \quad & F_1 + N = W_1 \sin\alpha, \\ \text{lower:} \quad & F_2 = N + W_2 \sin\alpha. \end{aligned}$$

Adding, and using $F_1 = F_2 = F$,

$$2F = (W_1 + W_2)\sin\alpha, \qquad \text{so} \qquad F = \tfrac{1}{2}(W_1 + W_2)\sin\alpha > 0,$$

which also tells us that every friction force acts in the direction we drew it. Substituting back,

$$N = W_1 \sin\alpha - F = \tfrac{1}{2}(W_1 - W_2)\sin\alpha,$$

which is positive because $W_1 > W_2$ and $\alpha > 0$, so the cylinders really do press on each other. Resolving perpendicular to the plane for the upper cylinder,

$$R_1 + F = W_1 \cos\alpha, \qquad \text{so} \qquad R_1 = W_1 \cos\alpha - F.$$

Ask yourself Which friction inequalities do we actually need?

In equilibrium, friction at each contact can be at most the coefficient of friction times the normal reaction there. The first result comes from the contact between the cylinders, and the second from the contact between the upper cylinder and the plane. (The lower contact gives a third condition, involving μ_2, which isn't asked for.)

Between the cylinders. We need $F \le \mu N$:

$$\tfrac{1}{2}(W_1 + W_2)\sin\alpha \le \mu \cdot \tfrac{1}{2}(W_1 - W_2)\sin\alpha.$$

Dividing by $\tfrac{1}{2}(W_1 - W_2)\sin\alpha$, which is positive, gives

$$\mu \ge \frac{W_1 + W_2}{W_1 - W_2},$$

as required.

Upper cylinder and plane. We need $F \le \mu_1 R_1 = \mu_1(W_1 \cos\alpha - F)$, that is,

$$(1+\mu_1)F \le \mu_1 W_1 \cos\alpha \quad \Longleftrightarrow \quad \tfrac{1}{2}(1+\mu_1)(W_1+W_2)\sin\alpha \le \mu_1 W_1 \cos\alpha.$$

The angle of an inclined plane satisfies $0 < \alpha < \tfrac{1}{2}\pi$, so $\cos\alpha > 0$, and $(1+\mu_1)(W_1 + W_2) > 0$. Dividing by both,

$$\tan\alpha \le \frac{2\mu_1 W_1}{(1+\mu_1)(W_1+W_2)},$$

as required. (Note that this also forces $R_1 > 0$, since $\mu_1 R_1 \ge F > 0$, so the upper cylinder stays on the plane.)

Checks

- With $W_1 = 3$, $W_2 = 1$ we need $\mu \geq 2$, and with $\mu_1 = 1$ as well we need $\tan\alpha \leq \frac{3}{4}$. Substituting such values into the six equilibrium equations (two resolutions and one moment equation per cylinder) and solving them directly gives the same F, N and R_1 as above.
- As $W_1 \to W_2$ the bound on μ goes to infinity. That makes sense: then $N \to 0$, so the cylinders barely press on each other, yet the friction between them must still be $\frac{1}{2}(W_1 + W_2)\sin\alpha$.
- As $\mu_1 \to \infty$ the bound becomes $\tan\alpha \leq 2W_1/(W_1 + W_2)$, which is exactly the condition $R_1 \geq 0$: even a perfectly rough plane can't help once the upper cylinder starts to lift off.

Common slips

- Taking the friction between the cylinders to act along the slope. It acts along the common tangent at the contact, which is perpendicular to the plane here.
- Forgetting the moment equations. Resolving alone gives four equations in six unknowns; the two moment equations are what fix the frictional forces.
- Leaving F out of the normal reaction R_1. The lower cylinder's friction pushes the upper one away from the plane, so R_1 is less than $W_1\cos\alpha$.

Beyond STEP

A rigid body is in equilibrium only if both the resultant force and the resultant moment vanish, and choosing a clever point to take moments about is the main skill of rigid-body statics. Here we had exactly as many equations as unknowns; when there are more unknowns than equations, the problem is *statically indeterminate*, and you need to model how the bodies deform to finish. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and torque carefully.

WORKED SOLUTION

STEP I 2015, Question 12

Probability and Statistics • Thinning a Poisson Distribution

STEP I 2015, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q12

The key idea

Conditioning on the number of arrivals turns surgery into a binomial count. Summing over all possible arrival numbers (the law of total probability) then gives the distribution of the number needing surgery, and the sum collapses into an exponential series. For the final part, the key fact is that if two independent Poisson counts with the same mean are conditioned on their total, the split between them is binomial.

Solution

Let X be the number of casualties arriving on a given day, and let Y be the number of them who need surgery.

Part (i)

Given $X = n$, each of the n casualties needs surgery with probability $\frac{1}{4}$, independently of the others. So, conditionally, $Y \sim \mathrm{B}(n, \frac{1}{4})$, and for $0 \le r \le n$

$$\mathrm{P}(Y = r \mid X = n) = \binom{n}{r}\left(\frac{1}{4}\right)^r\left(\frac{3}{4}\right)^{n-r} = \binom{n}{r}\frac{3^{n-r}}{4^n}.$$

(The probability is 0 if $r > n$.)

Part (ii)

Fix $r \ge 0$. Since $Y = r$ can only happen when $X \ge r$, the law of total probability gives

$$\begin{aligned}\mathrm{P}(Y = r) &= \sum_{n=r}^{\infty} \mathrm{P}(X = n)\,\mathrm{P}(Y = r \mid X = n)\\ &= \sum_{n=r}^{\infty} \frac{\mathrm{e}^{-8}8^n}{n!} \cdot \frac{n!}{r!\,(n-r)!}\left(\frac{1}{4}\right)^r\left(\frac{3}{4}\right)^{n-r}.\end{aligned}$$

The $n!$ cancels. Now split $8^n = 8^r \cdot 8^{n-r}$, so that $8^r(\frac{1}{4})^r = 2^r$ and $8^{n-r}(\frac{3}{4})^{n-r} = 6^{n-r}$:

$$\begin{aligned} \mathrm{P}(Y = r) &= \frac{\mathrm{e}^{-8}\,2^r}{r!} \sum_{n=r}^{\infty} \frac{6^{n-r}}{(n-r)!} \\ &= \frac{\mathrm{e}^{-8}\,2^r}{r!} \sum_{k=0}^{\infty} \frac{6^k}{k!} \qquad (k = n-r) \\ &= \frac{\mathrm{e}^{-8}\,2^r}{r!}\,\mathrm{e}^6 = \frac{\mathrm{e}^{-2}\,2^r}{r!}. \end{aligned}$$

This holds for every $r = 0, 1, 2, \ldots$, so the number needing surgery each day follows a Poisson distribution with mean $8 \times \frac{1}{4} = 2$.

Ask yourself Where did the e^6 come from?

Once everything that depends only on r is taken outside the sum, what is left is $\sum 6^k/k!$, the exponential series for e^6. Splitting 8^n as $8^r \cdot 8^{n-r}$ is exactly what makes the remaining sum depend on $n - r$ alone.

Part (iii)

Let M and T be the numbers needing surgery on Monday and Tuesday. By part (ii) each is $\mathrm{Po}(2)$. They are independent, since the arrivals on different days are independent and casualties need surgery independently of one another. Then

$$\mathrm{P}(M = 8 \mid M + T = 12) = \frac{\mathrm{P}(M = 8,\ T = 4)}{\mathrm{P}(M + T = 12)}.$$

By independence the numerator is

$$\mathrm{P}(M = 8)\,\mathrm{P}(T = 4) = \frac{\mathrm{e}^{-2}2^8}{8!} \cdot \frac{\mathrm{e}^{-2}2^4}{4!} = \frac{\mathrm{e}^{-4}\,2^{12}}{8!\,4!}.$$

For the denominator, the sum of independent Poisson variables is Poisson, with the means added, so $M + T \sim \mathrm{Po}(4)$. If you would rather not quote this, it follows directly:

$$\mathrm{P}(M + T = 12) = \sum_{m=0}^{12} \frac{\mathrm{e}^{-2}2^m}{m!} \cdot \frac{\mathrm{e}^{-2}2^{12-m}}{(12-m)!} = \frac{\mathrm{e}^{-4}2^{12}}{12!} \sum_{m=0}^{12} \binom{12}{m} = \frac{\mathrm{e}^{-4}\,4^{12}}{12!},$$

using $\sum_m \binom{12}{m} = 2^{12}$. Dividing,

$$\begin{aligned} \mathrm{P}(M = 8 \mid M + T = 12) &= \frac{\mathrm{e}^{-4}\,2^{12}}{8!\,4!} \cdot \frac{12!}{\mathrm{e}^{-4}\,4^{12}} \\ &= \binom{12}{8}\left(\frac{1}{2}\right)^{12} = \frac{495}{4096}. \end{aligned}$$

Since $495 = 3^2 \cdot 5 \cdot 11$ is odd and $4096 = 2^{12}$, the fraction is in lowest terms. So the required probability is $\dfrac{495}{4096}$.

Ask yourself Why does the answer look like a binomial probability?

Given the total of 12, each of those 12 surgical cases is equally likely to have come from Monday or Tuesday, because the two days have the same mean. So the Monday count is $\mathrm{B}(12, \frac{1}{2})$. Notice that the original mean of 8 has disappeared completely.

Checks

- The mean in part (ii) agrees with $\mathrm{E}(Y) = \mathrm{E}\big(\mathrm{E}(Y \mid X)\big) = \mathrm{E}(X/4) = 2$.
- Summing the part (ii) series numerically for $r = 0, \ldots, 14$ reproduces $\mathrm{e}^{-2}2^r/r!$ to within rounding error. For example, $\mathrm{P}(Y = 0) = \sum_n \mathrm{e}^{-8}8^n(\frac{3}{4})^n/n! = \mathrm{e}^{-8}\mathrm{e}^{6}$.
- The conditional probabilities $\binom{12}{m}/2^{12}$ for $m = 0, \ldots, 12$ sum to 1, and $495/4096 \approx 0.121$. A computer simulation of the full model (Poisson arrivals, then surgery with probability $\frac{1}{4}$ each) gave a value consistent with this.

Common slips

- Starting the sum in part (ii) at $n = 0$ instead of $n = r$: there is no way for r casualties to need surgery if fewer than r arrive.
- Using $\mathrm{Po}(8)$ for each day in part (iii). The conditioning is on surgical cases, so the relevant mean is 2. (Here the means cancel anyway, but the working should be right.)
- Leaving the answer as $\binom{12}{8}/2^{12}$ or $1980/16384$ when lowest terms were asked for.

Beyond STEP

Part (ii) is called *thinning*: if each point of a Poisson process is kept independently with probability p, the points kept form a Poisson process with rate multiplied by p, and, perhaps surprisingly, it is independent of the points discarded. Part (iii) is the reverse of this: conditioned on their total, independent Poisson counts split multinomially. These two facts underpin the theory of Poisson processes, which you meet in a first probability course. Grimmett and Welsh, *Probability: An Introduction*, covers both.

WORKED SOLUTION

STEP I 2015, Question 13

Probability and Statistics • Conditional Probability

STEP I 2015, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q13

The key idea

Condition on when the first 6 appears. If we know the first 6 is on throw n, then each of the earlier $n-1$ throws is equally likely to be any of $1,\dots,5$, independently, so questions about 4s and 5s before the first 6 become ordinary binomial questions. Summing over n then gives geometric series and their derivatives, which is exactly where the binomial expansion in the hint comes in.

Solution

Throws are independent, each face having probability $\frac{1}{6}$. Write $p_n = \mathrm{P}(A)$ for the event that the first 6 is on throw n.

Part (i)

The first 6 is on throw n exactly when throws $1,\dots,n-1$ are all non-6 and throw n is a 6. By independence,

$$\mathrm{P}(A) = p_n = \left(\tfrac{5}{6}\right)^{n-1}\tfrac{1}{6}, \qquad n = 1,2,3,\dots$$

The conditioning fact. Given A, each of throws $1,\dots,n-1$ is known only to be "not a 6". Since the die is fair and throws are independent, these $n-1$ throws are independent and each is uniform on $\{1,2,3,4,5\}$. Every later part uses this, together with the law of total probability $\mathrm{P}(X) = \sum_n \mathrm{P}(X \mid A)\, p_n$.

Part (ii)

It is easier to find $\mathrm{P}(B')$, that no 5 comes before the first 6. Given A, each of the $n-1$ earlier throws avoids 5 with probability $\frac{4}{5}$, so

$$\mathrm{P}(B') = \sum_{n=1}^{\infty} \left(\tfrac{4}{5}\right)^{n-1} \left(\tfrac{5}{6}\right)^{n-1} \tfrac{1}{6} = \tfrac{1}{6} \sum_{n=1}^{\infty} \left(\tfrac{2}{3}\right)^{n-1} = \tfrac{1}{6} \cdot \frac{1}{1-\frac{2}{3}} = \tfrac{1}{2}.$$

Hence $\mathrm{P}(B) = \frac{1}{2}$.

Part (iii)

By the inclusion–exclusion principle (in complement form),

$$\mathrm{P}(B \cap C) = 1 - \mathrm{P}(B') - \mathrm{P}(C') + \mathrm{P}(B' \cap C').$$

By symmetry between the faces 4 and 5, $\mathrm{P}(C') = \mathrm{P}(B') = \frac{1}{2}$. For $B' \cap C'$, given A each earlier throw must avoid both 4 and 5, with probability $\frac{3}{5}$:

$$\mathrm{P}(B' \cap C') = \sum_{n=1}^{\infty} \left(\tfrac{3}{5}\right)^{n-1} \left(\tfrac{5}{6}\right)^{n-1} \tfrac{1}{6} = \tfrac{1}{6} \cdot \frac{1}{1-\frac{1}{2}} = \tfrac{1}{3}.$$

So $\mathrm{P}(B \cap C) = 1 - \frac{1}{2} - \frac{1}{2} + \frac{1}{3} = \frac{1}{3}$.

Part (iv)

Given A, the number of 5s among the $n-1$ earlier throws is $\mathrm{B}(n-1, \frac{1}{5})$, so

$$\mathrm{P}(D \mid A) = (n-1)\tfrac{1}{5}\left(\tfrac{4}{5}\right)^{n-2} \qquad (n \geq 2),$$

and it is 0 for $n = 1$. Hence

$$\mathrm{P}(D) = \sum_{n=2}^{\infty} (n-1)\tfrac{1}{5}\left(\tfrac{4}{5}\right)^{n-2} \left(\tfrac{5}{6}\right)^{n-1} \tfrac{1}{6}.$$

Pull out one factor of $\frac{5}{6}$ and pair the rest: $\frac{1}{5} \cdot \frac{5}{6} \cdot \frac{1}{6} = \frac{1}{36}$ and $\frac{4}{5} \cdot \frac{5}{6} = \frac{2}{3}$. With $m = n-1$,

$$\mathrm{P}(D) = \tfrac{1}{36} \sum_{m=1}^{\infty} m \left(\tfrac{2}{3}\right)^{m-1} = \tfrac{1}{36} \left(1 - \tfrac{2}{3}\right)^{-2} = \tfrac{9}{36} = \tfrac{1}{4},$$

using the given expansion with exponent -2: $(1-x)^{-2} = 1 + 2x + 3x^2 + \cdots$.

Ask yourself How do we get $\mathrm{P}(D \cup E)$ without listing all the ways that one or both can happen?

Part (v)

Use $\mathrm{P}(D \cup E) = \mathrm{P}(D) + \mathrm{P}(E) - \mathrm{P}(D \cap E)$, with $\mathrm{P}(E) = \mathrm{P}(D) = \frac{1}{4}$ by the 4–5 symmetry. So we only need $\mathrm{P}(D \cap E)$.

Given A, exactly one 4 and exactly one 5 among the $n-1$ earlier throws means choosing which throw is the 5 and which is the 4 ($(n-1)(n-2)$ ordered choices), with the remaining $n-3$ throws all in $\{1,2,3\}$:

$$\mathrm{P}(D \cap E \mid A) = (n-1)(n-2)\left(\tfrac{1}{5}\right)^2\left(\tfrac{3}{5}\right)^{n-3} \qquad (n \geq 3).$$

Then

$$\mathrm{P}(D \cap E) = \sum_{n=3}^{\infty}(n-1)(n-2)\tfrac{1}{25}\left(\tfrac{3}{5}\right)^{n-3}\left(\tfrac{5}{6}\right)^{n-1}\tfrac{1}{6}.$$

Taking out $\left(\frac{5}{6}\right)^2$ gives the constant $\frac{1}{25}\cdot\frac{25}{36}\cdot\frac{1}{6} = \frac{1}{216}$, and $\frac{3}{5}\cdot\frac{5}{6} = \frac{1}{2}$. With $k = n-3$,

$$\begin{aligned}\mathrm{P}(D \cap E) &= \tfrac{1}{216}\sum_{k=0}^{\infty}(k+2)(k+1)\left(\tfrac{1}{2}\right)^k = \tfrac{2}{216}\sum_{k=0}^{\infty}\frac{(k+2)!}{k!\,2!}\left(\tfrac{1}{2}\right)^k \\ &= \tfrac{1}{108}\left(1-\tfrac{1}{2}\right)^{-3} = \tfrac{8}{108} = \tfrac{2}{27},\end{aligned}$$

using the given expansion with exponent -3, whose coefficient of x^k is $\frac{(k+2)!}{k!\,2!}$. Therefore

$$\mathrm{P}(D \cup E) = \tfrac{1}{4} + \tfrac{1}{4} - \tfrac{2}{27} = \tfrac{27-4}{54} = \tfrac{23}{54}.$$

Checks

There is a slicker way to see every answer, which makes an excellent check. Ignore all throws of 1, 2, 3 (and, for (ii) and (iv), also 4): they never affect the events. The remaining relevant throws form a sequence in which each relevant face is equally likely.

- (ii): the first throw that is a 5 or a 6 is equally likely to be either, so $\mathrm{P}(B) = \frac{1}{2}$.
- (iii): among 4, 5, 6, each of the three faces is equally likely to be the last of them to make its first appearance, and $B \cap C$ says 6 is last, so $\frac{1}{3}$.
- (iv): D means the 5-or-6 sequence starts 5, 6: probability $\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$.
- (v): $D \cap E$ means the 4-5-6 sequence starts $4,5,6$ or $5,4,6$: probability $2\left(\frac{1}{3}\right)^3 = \frac{2}{27}$.

All five answers were also confirmed by a computer simulation of 3×10^5 sequences of throws, which gave $\mathrm{P}(D \cup E) \approx 0.426$, against $\frac{23}{54} \approx 0.4259$.

Common slips

- Treating the earlier throws as uniform on all six faces: given A, they cannot be 6, so each 5 has probability $\frac{1}{5}$, not $\frac{1}{6}$.
- Forgetting the factor $(n-1)(n-2)$ in (v), or using $\binom{n-1}{2}$, which does not distinguish which of the two throws is the 4.
- Adding $\mathrm{P}(D) + \mathrm{P}(E)$ for (v) without subtracting the overlap.

Beyond STEP

Conditioning on a random stopping time and summing, as here, is the law of total probability in its most useful form, and the series you met are generating functions for the negative binomial distribution. The "ignore the irrelevant throws" check is an instance of a powerful idea: a process watched only at the moments something relevant happens is often far simpler than the original. Grimmett and Welsh, *Probability: An Introduction*, develops both ideas.

WORKED SOLUTION

STEP II 2015, Question 1

Pure • Inequalities and Series

STEP II 2015, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q1

The key idea

Each inequality between a function and a logarithm becomes, once we substitute a well-chosen value of x, an inequality between a single term of the series and a *difference* of logarithms. Summing those differences makes almost everything cancel (the sum telescopes), leaving one or two logarithms. The art is in choosing the substitution so that the logarithm splits into a difference.

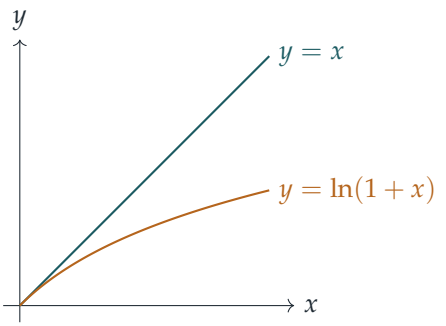
Solution

Part (i)

Let $f(x) = x - \ln(1+x)$ for $x \geq 0$. Then $f(0) = 0 - \ln 1 = 0$ and

$$f'(x) = 1 - \frac{1}{1+x} = \frac{x}{1+x} > 0 \qquad \text{for } x > 0.$$

So f is strictly increasing on $[0, \infty)$, and for every $x > 0$ we have $f(x) > f(0) = 0$. That is, $x > \ln(1+x)$ for all positive x, as required.



Now take $x = \frac{1}{k}$ for a positive integer k:

$$\frac{1}{k} > \ln\left(1 + \frac{1}{k}\right) = \ln\left(\frac{k+1}{k}\right) = \ln(k+1) - \ln k.$$

Adding these for $k = 1, 2, \ldots, n$, the right-hand sides telescope:

$$\begin{aligned}\sum_{k=1}^{n} \frac{1}{k} &> \big(\ln 2 - \ln 1\big) + \big(\ln 3 - \ln 2\big) + \cdots + \big(\ln(n+1) - \ln n\big) \\ &= \ln(n+1) - \ln 1 = \ln(n+1).\end{aligned}$$

This is the required result. (As a by-product, the harmonic series diverges, since $\ln(n+1) \to \infty$.)

Part (ii)

Let $g(x) = x + \ln(1-x)$ for $0 \le x < 1$. Then $g(0) = 0$ and

$$g'(x) = 1 - \frac{1}{1-x} = \frac{-x}{1-x} < 0 \qquad \text{for } 0 < x < 1,$$

since $1 - x > 0$ there. So g is strictly decreasing and $g(x) < 0$ for $0 < x < 1$; that is,

$$x < -\ln(1-x) \qquad (0 < x < 1).$$

Ask yourself What value of x should we substitute, and why can't we use $k = 1$?

We want $\frac{1}{k^2}$ on the left, so take $x = \frac{1}{k^2}$. This needs $x < 1$, so $k \ge 2$; for $k = 1$ the logarithm $\ln(1-1)$ is undefined. That is why the term $\frac{1}{1} = 1$ is set aside, and it is exactly the 1 in the bound $1 + \ln 2$. For $k \ge 2$, the key is that $1 - \frac{1}{k^2}$ factorises:

$$1 - \frac{1}{k^2} = \frac{(k-1)(k+1)}{k^2},$$

so

$$\frac{1}{k^2} < -\ln\frac{(k-1)(k+1)}{k^2} = \big(\ln k - \ln(k-1)\big) - \big(\ln(k+1) - \ln k\big).$$

Write $a_k = \ln k - \ln(k-1)$, so the right-hand side is $a_k - a_{k+1}$. Summing from $k = 2$ to N telescopes:

$$\sum_{k=2}^{N} \frac{1}{k^2} < a_2 - a_{N+1} = \ln 2 - \ln\left(\frac{N+1}{N}\right) < \ln 2,$$

because $\ln\frac{N+1}{N} > 0$.

So the partial sums $\sum_{k=1}^{N} \frac{1}{k^2}$ increase with N and are all less than $1 + \ln 2$. An increasing sequence that is bounded above converges, so the infinite series converges, and its sum is at most $1 + \ln 2$.

To make the inequality strict, note that passing to the limit only preserves $\leq$, so we keep a little room. The $k = 2$ inequality, $\frac{1}{4} < \ln\frac{4}{3}$, is strict; let $\delta = \ln\frac{4}{3} - \frac{1}{4} > 0$. Then for every $N \geq 2$ the same telescoping gives

$$\sum_{k=2}^{N} \frac{1}{k^2} < \ln 2 - \ln\left(\frac{N+1}{N}\right) - \delta < \ln 2 - \delta,$$

so in the limit $\sum_{k=2}^{\infty} \frac{1}{k^2} \leq \ln 2 - \delta < \ln 2$. Adding the $k = 1$ term,

$$\sum_{k=1}^{\infty} \frac{1}{k^2} < 1 + \ln 2,$$

which is the required result.

Checks

- Part (i) with $n = 1$: $1 > \ln 2 \approx 0.693$. With $n = 10$: $\sum_{k=1}^{10} \frac{1}{k} \approx 2.929$, while $\ln 11 \approx 2.398$.
- Part (ii): the true sum is $\frac{\pi^2}{6} \approx 1.6449$, comfortably below $1 + \ln 2 \approx 1.6931$. In fact our working gives the slightly sharper bound $1 + \ln 2 - \delta \approx 1.6555$, with $\delta \approx 0.0377$.
- The telescoped bound $\ln 2 - \ln\frac{N+1}{N}$ for $\sum_{k=2}^{N} \frac{1}{k^2}$ was confirmed by computer for every N up to 10^5.

Common slips

- Substituting $x = \frac{1}{k^2}$ for $k = 1$ as well: $\ln(1 - x)$ is then undefined, and forgetting to set the first term aside loses the 1 in $1 + \ln 2$.
- Showing only that $f'(x) > 0$, without $f(0) = 0$: a function can be increasing and still negative.
- Taking limits of a strict inequality and claiming the limit is strict without comment; limits only give $\leq$.

Beyond STEP

Comparing a sum with a logarithm (or an integral) is the start of the integral test, and refining part (i) shows that $\sum_{k=1}^{n} \frac{1}{k} - \ln n$ converges to Euler's constant $\gamma \approx 0.5772$. The exact value $\sum \frac{1}{k^2} = \frac{\pi^2}{6}$ is Euler's solution of the Basel problem, which first-year courses often derive from Fourier series. The monotone convergence argument we used, that bounded increasing sequences converge, is a cornerstone of real analysis; Abbott's *Understanding Analysis* develops it carefully.

WORKED SOLUTION

STEP II 2015, Question 2

Pure • Triangle Geometry

STEP II 2015, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

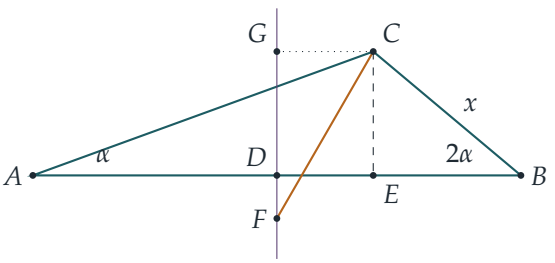
step.maths.org/questions/15-s2-q2

The key idea

The first result is the sine rule plus the triple-angle formula for $\sin 3\alpha$. For DE, measure everything along AB from one end: the right-angled triangles on either side of CE give AE and BE directly. The answer for DE turns out to be remarkably simple, and that simplicity is exactly what makes the trisection work: it tells us how far C is from the perpendicular bisector, and comparing that with $CF = x$ pins down an angle.

Solution

Throughout, the angle at C is $\angle ACB = \pi - 3\alpha$, and since 2α is acute we have $0 < \alpha < \frac{\pi}{4}$.



The length AB

By the sine rule, using $\sin(\pi - 3\alpha) = \sin 3\alpha$,

$$\frac{AB}{\sin 3\alpha} = \frac{x}{\sin \alpha}.$$

The triple-angle formula $\sin 3\alpha = 3\sin\alpha - 4\sin^3\alpha$ then gives, dividing by $\sin\alpha \neq 0$,

$$AB = x\,\frac{3\sin\alpha - 4\sin^3\alpha}{\sin\alpha} = (3 - 4\sin^2\alpha)\,x,$$

as required.

The length DE

Both angles at A and B are acute, so the foot E of the perpendicular from C lies between A and B. In the right-angled triangle CEB,

$$BE = x\cos 2\alpha, \qquad CE = x\sin 2\alpha.$$

Hence

$$AE = AB - BE = x\big(3 - 4\sin^2\alpha - (1 - 2\sin^2\alpha)\big) = x(2 - 2\sin^2\alpha) = 2x\cos^2\alpha,$$
$$AD = \tfrac{1}{2}AB = x\big(\tfrac{3}{2} - 2\sin^2\alpha\big).$$

Subtracting,

$$AE - AD = x\big(2 - 2\sin^2\alpha - \tfrac{3}{2} + 2\sin^2\alpha\big) = \tfrac{1}{2}x.$$

This is positive, so E lies between D and B, and

$$DE = \tfrac{1}{2}x,$$

independent of α.

The trisection

Let G be the foot of the perpendicular from C to the perpendicular bisector DF of AB. Since CE and GD are both perpendicular to AB, $CGDE$ is a rectangle, so $CG = DE = \frac{1}{2}x$ and CG is parallel to AB, i.e. perpendicular to CE.

Ask yourself How do we turn $CF = x$ into an angle?

The triangle CGF has a right angle at G, hypotenuse $CF = x$ and side $CG = \frac{1}{2}x$. So

$$\cos\angle GCF = \frac{CG}{CF} = \frac{1}{2}, \qquad \text{so} \qquad \angle GCF = \frac{\pi}{3}.$$

There are two points of the bisector at distance x from C, one either side of G. Measured from the ray CE (pointing down towards AB) round towards A, the ray CG is at angle $\frac{\pi}{2}$, so the two candidate rays CF are at angles $\frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}$ (the point below G) and $\frac{\pi}{2} + \frac{\pi}{3} = \frac{5\pi}{6}$ (the point above G).

Now decide which one is on the same side of AC as B. In the right-angled triangle ACE, $\angle ACE = \frac{\pi}{2} - \alpha$, so the ray CA is at angle $\frac{\pi}{2} - \alpha$ from CE on this same side. The rays from C on B's side of the line AC are those that, measured this way, are at less than $\frac{\pi}{2} - \alpha$ (or on the other side of CE, towards B). Because $\alpha < \frac{\pi}{4}$,

$$\frac{\pi}{6} < \frac{\pi}{4} < \frac{\pi}{2} - \alpha < \frac{5\pi}{6},$$

so the point below G is on B's side and the point above G is not. Thus F is the lower point, $\angle ECF = \frac{\pi}{6}$, and the ray CF lies inside the angle ACE, hence inside $\angle ACB$. Therefore

$$\angle ACF = \angle ACE - \angle ECF = \left(\frac{\pi}{2} - \alpha\right) - \frac{\pi}{6} = \frac{\pi}{3} - \alpha = \frac{1}{3}(\pi - 3\alpha) = \frac{1}{3}\angle ACB.$$

So CF cuts off one third of $\angle ACB$ next to CA (and $\angle FCB = \frac{2}{3}\angle ACB$): the line FC trisects the angle ACB.

Checks

- With $\alpha = \frac{\pi}{6}$ the triangle has angles 30°, 60°, 90°, and $AB = (3-1)x = 2x$, the familiar hypotenuse. Then $AE = 2x\cos^2 30^\circ = \frac{3}{2}x$ and $AD = x$, so $DE = \frac{1}{2}x$. The angle at C is 90°, and $\angle ACF = 60^\circ - 30^\circ = 30^\circ$, a third of it.
- With $\alpha = 20^\circ$ and $x = 1$ we get $AB \approx 2.532$ and $\angle ACB = 120^\circ$; constructing F directly from coordinates gives $\angle ACF = 40^\circ$. All three results were also confirmed by computer for thousands of random triangles.

Common slips

- Taking F above G. That point is also distance x from C, but it lies on the far side of AC, which is why the question specifies the side.
- Forgetting to check that E lies between A and B (and on which side of D) before writing DE as a difference of lengths.

Beyond STEP

Trisecting a general angle with ruler and compasses is impossible, a fact proved with field theory: trisecting 60° would require constructing a root of an irreducible cubic, and constructible lengths only ever satisfy towers of quadratics. Here the triangle supplies extra information (the angle at B is twice that at A), and that is what lets a simple construction do the job. For more elegant triangle geometry see Coxeter and Greitzer, *Geometry Revisited*; for the impossibility proof, Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP II 2015, Question 3

Pure • Counting and Induction

STEP II 2015, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q3

The key idea

Sort the triangles by their longest rod. The triangles counted by T_n but not by T_{n-1} are exactly those that use the rod of length n, so $T_n - T_{n-1}$ is the number of pairs $a < b < n$ with $a + b \geq n$. For each choice of the middle rod b that is a simple count, and the counts form an arithmetic progression. Adding two consecutive differences gives $T_{2m} - T_{2m-2}$, which is exactly what the inductive step needs.

Solution

Counting the triangles with longest rod c

A triangle counted by T_n has rods $a < b < c \leq n$. Those with $c \leq n-1$ are precisely the triangles counted by T_{n-1}, so

$$T_n - T_{n-1} = \#\{(a,b) : 1 \leq a < b < n,\ a + b \geq n\}.$$

Fix the middle rod b. We need $n - b \leq a \leq b - 1$ (and automatically $a \geq 1$, since $b < n$), which gives $2b - n$ choices when $2b - n > 0$ and none otherwise. So b runs from the least integer above $\frac{n}{2}$ up to $n - 1$.

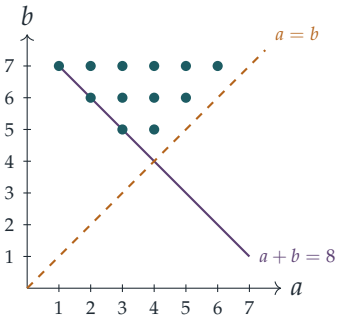
The case $n = 8$

With $c = 8$, the possible middle rods are $b = 5, 6, 7$, giving 2, 4 and 6 choices of a:

$$b = 5 :\ a \in \{3,4\}, \qquad b = 6 :\ a \in \{2,3,4,5\}, \qquad b = 7 :\ a \in \{1,\ldots,6\}.$$

($b = 4$ would need $4 \leq a \leq 3$, which is impossible.) Hence $T_8 - T_7 = 2 + 4 + 6 = 12$, as required.

The picture shows these 12 pairs (a,b): the points on or above the line $a + b = 8$ and strictly above the diagonal $a = b$.



For $T_7 - T_6$, the longest rod is 7 and the middle rod $b = 4,5,6$ gives $2b - 7 = 1,3,5$ choices, so $T_7 - T_6 = 1 + 3 + 5 = 9$. Therefore

$$T_8 - T_6 = (T_8 - T_7) + (T_7 - T_6) = 12 + 9 = 21.$$

The general differences

For $n = 2m$, the middle rod runs over $b = m+1, \ldots, 2m-1$, with $2b - 2m = 2, 4, \ldots, 2(m-1)$ choices:

$$T_{2m} - T_{2m-1} = 2 + 4 + \cdots + 2(m-1) = m(m-1).$$

For $n = 2m - 1$, the middle rod runs over $b = m, \ldots, 2m-2$, with $2b - (2m-1) = 1, 3, \ldots, 2m-3$ choices, and the sum of the first $m-1$ odd numbers is $(m-1)^2$. Adding,

$$T_{2m} - T_{2m-2} = m(m-1) + (m-1)^2 = (m-1)(2m-1).$$

(With $m = 4$ these give 12 and 21, matching the values above.) These hold whenever the smaller index is at least 3, i.e. for $m \geq 3$ in the second formula.

The induction

Let $f(m) = \frac{1}{6}m(m-1)(4m+1)$. We prove $T_{2m} = f(m)$ for all $m \geq 2$.

Base case, $m = 2$. From the rods $1,2,3,4$ the four possible choices are $(1,2,3)$, $(1,2,4)$, $(1,3,4)$, $(2,3,4)$, and all except $(1,2,4)$ satisfy $a + b \geq c$. So $T_4 = 3$, and $f(2) = \frac{1}{6} \cdot 2 \cdot 1 \cdot 9 = 3$.

Inductive step. Suppose $T_{2k-2} = f(k-1)$ for some $k \geq 3$. By the difference formula,

$$T_{2k} = f(k-1) + (k-1)(2k-1),$$

so it is enough to show $f(k) - f(k-1) = (k-1)(2k-1)$.

Ask yourself How do we tidy up $f(k) - f(k-1)$ without expanding two cubics in full?

Both terms contain the factor $k-1$, so take it out first:

$$\begin{aligned} f(k)-f(k-1) &= \tfrac{1}{6}(k-1)\big[k(4k+1)-(k-2)(4k-3)\big] \\ &= \tfrac{1}{6}(k-1)\big[(4k^2+k)-(4k^2-11k+6)\big] \\ &= \tfrac{1}{6}(k-1)(12k-6) = (k-1)(2k-1). \end{aligned}$$

Hence $T_{2k} = f(k)$. By induction, $T_{2m} = \tfrac{1}{6}m(m-1)(4m+1)$ for every $m \geq 2$.

An odd number of rods

Using $T_{2m-1} = T_{2m} - m(m-1)$,

$$T_{2m-1} = \tfrac{1}{6}m(m-1)(4m+1) - m(m-1) = \tfrac{1}{6}m(m-1)(4m-5) \qquad (m \geq 2).$$

Writing the number of rods as $2m+1$ instead (replace m by $m+1$), this is

$$T_{2m+1} = \tfrac{1}{6}m(m+1)(4m-1) \qquad (m \geq 1).$$

For example $T_3 = \tfrac{1}{6}\cdot 1\cdot 2\cdot 3 = 1$, the single triangle $(1,2,3)$.

Checks

- The formulas give $T_6 = \tfrac{1}{6}\cdot 3\cdot 2\cdot 13 = 13$, $T_7 = \tfrac{1}{6}\cdot 3\cdot 4\cdot 11 = 22$ and $T_8 = \tfrac{1}{6}\cdot 4\cdot 3\cdot 17 = 34$, so $T_8 - T_7 = 12$ and $T_8 - T_6 = 21$, as found directly.
- A computer count of all triples from $\{1,\ldots,n\}$ agrees with both closed forms for every n from 3 to 79.
- With $n = 2m$, the even formula is about $\tfrac{2}{3}m^3 = \tfrac{n^3}{12}$, and the odd one behaves the same way. That is half of the $\binom{n}{3} \approx \tfrac{n^3}{6}$ possible triples, so for large n about half of all choices of three rods make a triangle.

Common slips

- Forgetting that the rods are distinct, so $a < b$ strictly: allowing $a = b$ overcounts.
- Losing the zero-area (degenerate) triangles: the condition is $a + b \geq c$, so $(1,2,3)$ counts.
- Starting the induction at $m = 1$: T_2 is not defined, since we need $n \geq 3$.

Beyond STEP

Counting lattice points in a region, as in the picture, is the start of the theory of partitions and generating functions: T_n has a rational generating function, and the split into even and odd n reflects a factor $1 - x^2$ in its denominator. Results like this, where

a count is a polynomial in n that depends on n's residue, are called quasi-polynomials. For more on recurrences and counting arguments, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 2015, Question 4

Pure • Inverse Tangent Branches

STEP II 2015, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q4

The key idea

Knowing $\tan\theta$ only pins θ down up to a multiple of π, so each equation has infinitely many solution branches, all vertical translates of one another by π. Continuity and the value at $x = 0$ choose one branch, and continuity also forces us to *change* branch whenever the right-hand side runs off to $\pm\infty$: that is exactly the moment the angle passes through an odd multiple of $\frac{1}{2}\pi$. So in each part we first sketch the right-hand side, then follow the angle continuously.

Throughout, arctan means the principal value, with $-\frac{1}{2}\pi < \arctan u < \frac{1}{2}\pi$.

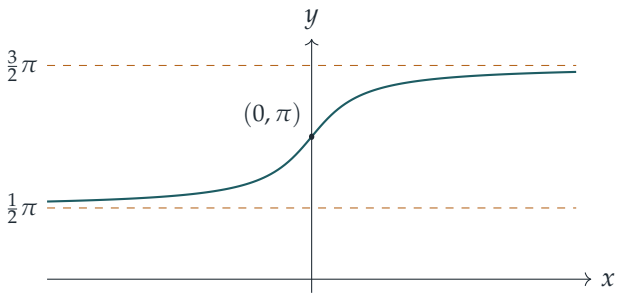
Solution

Part (i)

Every solution of $\tan\theta = x$ is $\theta = \arctan x + n\pi$ for an integer n. The function $\arctan x$ is continuous on the whole real line, so $\mathrm{f}(x) - \arctan x$ is a continuous function taking only values in $\{n\pi\}$; a continuous function cannot jump between these isolated values, so it is constant. Since $\mathrm{f}(0) = \pi$ and $\arctan 0 = 0$, the constant is π:

$$\mathrm{f}(x) = \pi + \arctan x.$$

So the curve is the usual arctan curve lifted by π: it increases strictly, passes through $(0, \pi)$ with gradient 1 (since $\mathrm{f}'(x) = \frac{1}{1+x^2}$), and has horizontal asymptotes $y = \frac{1}{2}\pi$ as $x \to -\infty$ and $y = \frac{3}{2}\pi$ as $x \to \infty$. It has rotational symmetry of order 2 about $(0, \pi)$, because $\mathrm{f}(-x) = 2\pi - \mathrm{f}(x)$.



Part (ii)

The curve $y = \dfrac{x}{1+x^2}$. Call this $q(x)$. It is odd, it is zero only at $x = 0$, and it has the sign of x. As $x \to \pm\infty$, $q(x) \approx 1/x \to 0$, so the x-axis is an asymptote, approached from above on the right and from below on the left. Differentiating,

$$q'(x) = \frac{(1+x^2) - 2x^2}{(1+x^2)^2} = \frac{1-x^2}{(1+x^2)^2},$$

so the stationary points are at $x = \pm 1$: a maximum $(1, \frac{1}{2})$ and a minimum $(-1, -\frac{1}{2})$. (Equivalently, $1 + x^2 \geq 2|x|$ by AM–GM, so $|q(x)| \leq \frac{1}{2}$.) The gradient at the origin is 1.

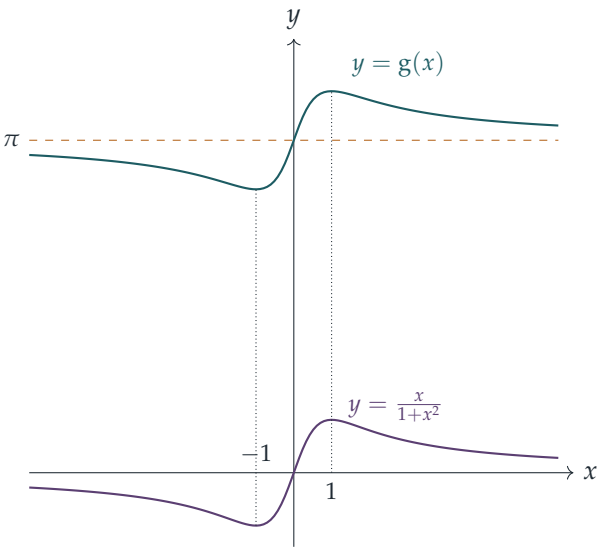
The curve $y = \mathrm{g}(x)$. Now the right-hand side $q(x)$ is continuous and bounded, so exactly the argument of part (i) gives

$$\mathrm{g}(x) = \pi + \arctan \frac{x}{1+x^2}.$$

Because arctan is increasing, g goes up and down exactly where q does. So

- $\mathrm{g}(x) \to \pi$ as $x \to \pm\infty$ (from above on the right, from below on the left);
- g has a maximum $\pi + \arctan \frac{1}{2} \approx 3.61$ at $x = 1$ and a minimum $\pi - \arctan \frac{1}{2} \approx 2.68$ at $x = -1$;
- it passes through $(0, \pi)$ with gradient $\dfrac{q'(0)}{1 + q(0)^2} = 1$.

Its shape is that of q, slightly flattened (since $\arctan u < u$ for $u > 0$) and lifted by π; it never comes near $\frac{1}{2}\pi$ or $\frac{3}{2}\pi$, so it never needs to change branch.



Part (iii)

The curve $y = \dfrac{x}{1-x^2}$**.** Call this $r(x)$. It is odd, it vanishes only at $x = 0$, and it has vertical asymptotes at $x = \pm 1$. As $x \to \pm\infty$, $r(x) \approx -1/x \to 0$. Its derivative is

$$r'(x) = \frac{(1-x^2)+2x^2}{(1-x^2)^2} = \frac{1+x^2}{(1-x^2)^2} > 0,$$

so r is strictly increasing on each of the three intervals where it is defined. Reading off the signs of x and $1-x^2$:

- on $x < -1$, r rises from 0^+ (as $x \to -\infty$) to $+\infty$ (as $x \to -1^-$);
- on $-1 < x < 1$, r rises from $-\infty$ to $+\infty$, through the origin with gradient 1;
- on $x > 1$, r rises from $-\infty$ (as $x \to 1^+$) to 0^- (as $x \to \infty$).

The curve $y = \mathrm{h}(x)$**.** On each interval separately, r is continuous, so $\mathrm{h}(x) - \arctan r(x)$ is a constant multiple of π there, but the multiple may be different on different intervals. We find it interval by interval.

On $-1 < x < 1$, the condition $\mathrm{h}(0) = \pi$ gives $\mathrm{h}(x) = \pi + \arctan r(x)$. As $x \to 1^-$, $r \to +\infty$, so $\mathrm{h}(x) \to \pi + \frac{1}{2}\pi = \frac{3}{2}\pi$; as $x \to -1^+$, $r \to -\infty$ and $\mathrm{h}(x) \to \frac{1}{2}\pi$. Continuity at $x = \pm 1$ forces $\mathrm{h}(1) = \frac{3}{2}\pi$ and $\mathrm{h}(-1) = \frac{1}{2}\pi$.

? **Ask yourself** Just to the right of $x = 1$ we have $r(x)$ large and negative, so $\arctan r(x)$ is close to $-\frac{1}{2}\pi$. Which multiple of π do we add to make h continuous at $x = 1$?

For $x > 1$ we have $\mathrm{h}(x) = n\pi + \arctan r(x)$, and as $x \to 1^+$ this tends to $n\pi - \frac{1}{2}\pi$. For this to equal $\frac{3}{2}\pi$ we need $n = 2$. Similarly, for $x < -1$ we have $\mathrm{h}(x) = m\pi +$

$\arctan r(x) \to m\pi + \frac{1}{2}\pi$ as $x \to -1^-$, and matching $\frac{1}{2}\pi$ needs $m = 0$. Hence

$$\mathrm{h}(x) = \begin{cases} \arctan\dfrac{x}{1-x^2}, & x < -1, \\ \pi + \arctan\dfrac{x}{1-x^2}, & -1 < x < 1, \\ 2\pi + \arctan\dfrac{x}{1-x^2}, & x > 1, \end{cases} \qquad \mathrm{h}(\pm 1) = \pi \pm \tfrac{1}{2}\pi.$$

So as r shoots up to $+\infty$ and reappears from $-\infty$, the angle h simply carries on increasing through $\frac{3}{2}\pi$ (and likewise through $\frac{1}{2}\pi$ at $x = -1$).

Now the features of the sketch.

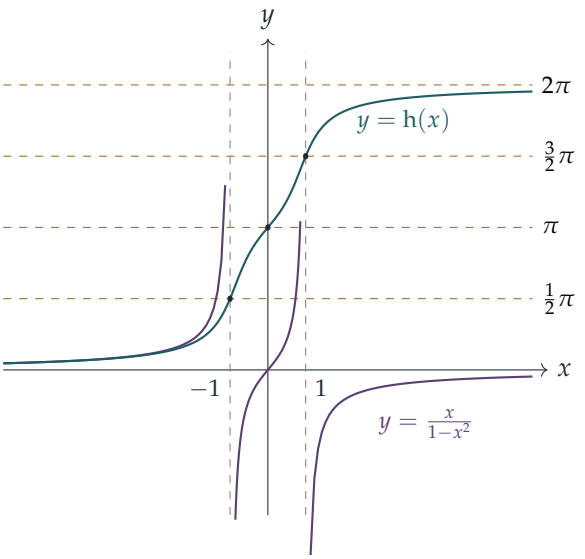
- *Asymptotes.* As $x \to -\infty$, $r \to 0^+$ so $\mathrm{h}(x) \to 0$ from above; as $x \to \infty$, $r \to 0^-$ so $\mathrm{h}(x) \to 2\pi$ from below.
- *Monotonic.* Away from ± 1, by the chain rule,

$$\mathrm{h}'(x) = \frac{r'(x)}{1 + r(x)^2} = \frac{1+x^2}{(1-x^2)^2 + x^2} = \frac{1+x^2}{x^4 - x^2 + 1} > 0.$$

  (The denominator is $(x^2 - \frac{1}{2})^2 + \frac{3}{4} > 0$.) This expression is perfectly well behaved at $x = \pm 1$, where it equals 2, so h is smooth there: the curve passes through $(1, \frac{3}{2}\pi)$ and $(-1, \frac{1}{2}\pi)$ with gradient 2, with no kink.
- *Key points.* $(0, \pi)$ with gradient 1; $(\pm 1, \pi \pm \frac{1}{2}\pi)$ with gradient 2.
- *Symmetry.* $\mathrm{h}(-x) = 2\pi - \mathrm{h}(x)$, so the curve has rotational symmetry about $(0, \pi)$.

So h increases steadily from 0 to 2π, an S-shaped curve like that of part (i) but covering twice the height. In the sketch the asymptotes $x = \pm 1$ of r line up exactly with the points where h crosses $\frac{1}{2}\pi$ and $\frac{3}{2}\pi$.

A tidy way to write all three cases at once: $\mathrm{h}(x) - \pi$ is the polar angle, in $(-\pi, \pi)$, of the point $(1 - x^2, x)$. That point never reaches the origin or the negative horizontal axis (when $1 - x^2 \le 0$ we have $x \neq 0$), which is why the angle varies continuously.



Checks

- Each formula satisfies the defining equation: for instance $\tan(2\pi+\theta)=\tan\theta$, and the values at 0 are all π.
- Spot values: $g(1)=\pi+\arctan\frac{1}{2}\approx 3.605$, and $h(2)=2\pi+\arctan(-\frac{2}{3})\approx 5.695$, which lies between $\frac{3}{2}\pi\approx 4.712$ and $2\pi\approx 6.283$ as the sketch requires.
- Continuity of h: $h(0.999)\approx 4.710$ and $h(1.001)\approx 4.714$, either side of $\frac{3}{2}\pi\approx 4.712$, consistent with gradient 2 at $x=1$. A computer check stepping through $[-10,10]$ confirmed that none of f, g, h jumps anywhere.

Common slips

- Writing $h(x)=\pi+\arctan\frac{x}{1-x^2}$ for all x: that jumps down by π at $x=1$ and up by π at $x=-1$, so it is not continuous.
- Drawing a vertical asymptote in $y=h(x)$ at $x=\pm1$ just because $\frac{x}{1-x^2}$ has one; the angle stays finite, only its tangent blows up.
- Showing a corner on $y=h(x)$ at $x=\pm1$; the gradient there is 2 from both sides.

Beyond STEP

What we have done is choose a continuous *branch* of a many-valued inverse function, and the rule "add π each time the tangent passes through infinity" is a first taste of lifting a path to a covering space. The same idea, for the complex logarithm and $\arg z$, underlies the winding number and the argument principle in complex analysis.

Needham's *Visual Complex Analysis* develops this geometrically and is a delightful read.

WORKED SOLUTION

STEP II 2015, Question 5

Pure • Arctan Series

STEP II 2015, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q5

The key idea

The tangent addition formula turns a sum of arctangents into a single fraction, so the induction for $\tan S_n$ is a matter of careful algebra. The subtle point is the step from $\tan S_n$ to S_n itself: knowing a tangent only fixes an angle up to multiples of π, so we must pin down which interval S_n lies in. In part (ii), the angles α_n turn out to be exactly the arctangents of part (i), and the infinite sum is the limit of S_n.

Solution

Part (i)

Write $\theta_m = \arctan\left(\frac{1}{2m^2}\right)$, so that $S_n = \theta_1 + \cdots + \theta_n$ and $\tan\theta_m = \frac{1}{2m^2}$.

The formula for $\tan S_n$. For $n = 1$, $\tan S_1 = \frac{1}{2} = \frac{1}{1+1}$, so the result holds.

Suppose $\tan S_n = \frac{n}{n+1}$ for some $n \geq 1$. Since $S_{n+1} = S_n + \theta_{n+1}$, the addition formula gives

$$\tan S_{n+1} = \frac{\tan S_n + \tan\theta_{n+1}}{1 - \tan S_n \tan\theta_{n+1}} = \frac{\dfrac{n}{n+1} + \dfrac{1}{2(n+1)^2}}{1 - \dfrac{n}{2(n+1)^3}}.$$

The denominator is not zero, because the product $\frac{n}{n+1} \cdot \frac{1}{2(n+1)^2}$ is less than 1. Multiplying top and bottom by $2(n+1)^3$,

$$\tan S_{n+1} = \frac{(n+1)\big(2n(n+1)+1\big)}{2(n+1)^3 - n} = \frac{(n+1)(2n^2+2n+1)}{2n^3+6n^2+5n+2}.$$

? **Ask yourself** How do we know the denominator will factorise nicely?

We are aiming for $\frac{n+1}{n+2}$, so the denominator ought to be $(n+2)(2n^2+2n+1)$. Expanding confirms it: $(n+2)(2n^2+2n+1) = 2n^3+6n^2+5n+2$. Cancelling the positive factor $2n^2+2n+1$,

$$\tan S_{n+1} = \frac{n+1}{n+2},$$

which is the result for $n+1$. By induction, $\tan S_n = \frac{n}{n+1}$ for all $n \geq 1$.

The formula for S_n. From $\tan S_n = \frac{n}{n+1}$ alone we can only say $S_n = \arctan\frac{n}{n+1} + k\pi$ for some integer k. To show $k = 0$ we prove, again by induction, that

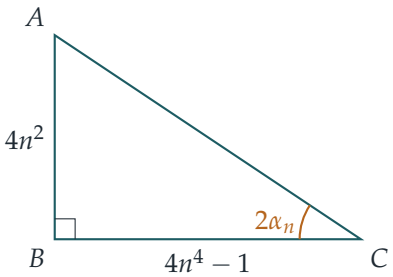
$$S_n = \arctan\frac{n}{n+1}.$$

For $n = 1$, $S_1 = \arctan\frac{1}{2}$, as required. Suppose the result holds for some n. Since $0 < \frac{n}{n+1} < 1$, we have $0 < S_n < \arctan 1 = \frac{1}{4}\pi$. Also $0 < \frac{1}{2(n+1)^2} < 1$, so $0 < \theta_{n+1} < \frac{1}{4}\pi$. Adding,

$$0 < S_{n+1} = S_n + \theta_{n+1} < \tfrac{1}{2}\pi.$$

So S_{n+1} lies in $[0, \frac{1}{2}\pi)$, which is exactly the range of arctan on non-negative numbers, and $\tan S_{n+1} = \frac{n+1}{n+2}$ by the first result. As tan is one-to-one on this interval, $S_{n+1} = \arctan\frac{n+1}{n+2}$. This completes the induction.

Part (ii)



Since the angle at B is a right angle, the side opposite angle C is AB and the adjacent side is BC (note $4n^4 - 1 > 0$ for $n \geq 1$). So

$$\tan 2\alpha_n = \frac{AB}{BC} = \frac{4n^2}{4n^4-1}.$$

Part (i) suggests trying $\tan\alpha_n = \frac{1}{2n^2}$. With $t = \frac{1}{2n^2}$, the double-angle formula gives

$$\frac{2t}{1-t^2} = \frac{\frac{1}{n^2}}{1-\frac{1}{4n^4}} = \frac{4n^2}{4n^4-1},$$

which matches. To be sure this is the right value and not merely a value, write $t = \tan\alpha_n$. The equation $\frac{2t}{1-t^2} = \frac{4n^2}{4n^4-1}$ rearranges to the quadratic

$$4n^2t^2 + 2(4n^4-1)t - 4n^2 = 0,$$

whose roots have product -1. One root is $\frac{1}{2n^2}$, so the other is $-2n^2 < 0$. But $2\alpha_n$ is an angle of a right-angled triangle other than the right angle, so $0 < 2\alpha_n < \frac{1}{2}\pi$, giving $0 < \alpha_n < \frac{1}{4}\pi$ and $t > 0$. Hence $\tan\alpha_n = \frac{1}{2n^2}$ and, as α_n lies in the range of arctan,

$$\alpha_n = \arctan\frac{1}{2n^2}.$$

Ask yourself What does an infinite sum mean here, and how does part (i) help?

The infinite sum is the limit of its partial sums, and by part (i) these are

$$\sum_{n=1}^{N}\alpha_n = S_N = \arctan\frac{N}{N+1}.$$

As $N \to \infty$, $\frac{N}{N+1} \to 1$, and since arctan is continuous, $\arctan\frac{N}{N+1} \to \arctan 1 = \frac{1}{4}\pi$. Therefore

$$\sum_{n=1}^{\infty}\alpha_n = \tfrac{1}{4}\pi,$$

as required.

Checks

- For $n = 1$, the triangle in part (ii) is the familiar 3–4–5 triangle, with $\tan 2\alpha_1 = \frac{4}{3}$. Then $\tan\alpha_1 = \frac{1}{2}$ satisfies $\frac{2\cdot\frac{1}{2}}{1-\frac{1}{4}} = \frac{4}{3}$.
- $S_2 = \arctan\frac{1}{2} + \arctan\frac{1}{8}$, and $\frac{\frac{1}{2}+\frac{1}{8}}{1-\frac{1}{16}} = \frac{10}{15} = \frac{2}{3}$, as the formula predicts.
- Numerically, adding the first 2000 terms gives $S_{2000} \approx 0.785148$, against $\frac{1}{4}\pi \approx 0.785398$. The gap is about $\frac{1}{4000}$, as it should be: the subtraction formula gives $\frac{1}{4}\pi - \arctan\frac{N}{N+1} = \arctan\frac{1}{2N+1}$ exactly.

Common slips

- Concluding $S_n = \arctan\frac{n}{n+1}$ straight from $\tan S_n = \frac{n}{n+1}$, without showing that S_n lies in $[0, \frac{1}{2}\pi)$.
- Taking $\tan\alpha_n = \frac{1}{2n^2}$ from a lucky guess without ruling out the other root of the quadratic.
- Treating the infinite sum as S_∞ without saying it is the limit of S_N.

Beyond STEP

This is a telescoping series in disguise: since $\frac{1}{2m^2} = \frac{(2m+1)-(2m-1)}{1+(2m+1)(2m-1)}$, each term is $\arctan(2m+1) - \arctan(2m-1)$, and the sum collapses to $\arctan(2n+1) - \frac{1}{4}\pi$, which

equals $\arctan \frac{n}{n+1}$. Defining an infinite sum as a limit of partial sums, and knowing when limits can be passed through continuous functions, is the starting point of a first course in analysis. A good place to read about it is Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2015, Question 6

Pure • Integration by Symmetry

STEP II 2015, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q6

The key idea

Writing $\sin x$ as $\cos\left(\frac{1}{2}\pi - x\right)$ turns $1 + \sin x$ into a double-angle expression, and then $\sec^2$ integrates at once. The substitution $y = \pi - x$ reflects $[0, \pi]$ about its midpoint, and $\sin x$ doesn't notice the reflection. So any polynomial factor in front of $f(\sin x)$ can be traded for its reflected version. In part (iii) the cubic has been chosen so that this trade gets rid of every power of x at once.

Solution

Part (i)

Since $\sin x = \cos\left(\frac{1}{2}\pi - x\right)$ and $1 + \cos 2\theta = 2\cos^2\theta$, taking $\theta = \frac{1}{4}\pi - \frac{1}{2}x$ gives

$$1 + \sin x = 1 + \cos\left(\tfrac{1}{2}\pi - x\right) = 2\cos^2\left(\tfrac{1}{4}\pi - \tfrac{1}{2}x\right).$$

Taking reciprocals and multiplying by 2 gives $\sec^2\left(\frac{1}{4}\pi - \frac{1}{2}x\right) = \dfrac{2}{1 + \sin x}$, as required. (Both sides are undefined at the same points, where $\sin x = -1$.)

Hence

$$\int \frac{1}{1 + \sin x}\,\mathrm{d}x = \frac{1}{2}\int \sec^2\left(\tfrac{1}{4}\pi - \tfrac{1}{2}x\right)\mathrm{d}x = -\tan\left(\tfrac{1}{4}\pi - \tfrac{1}{2}x\right) + C,$$

because differentiating $\tan\left(\frac{1}{4}\pi - \frac{1}{2}x\right)$ produces a factor of $-\frac{1}{2}$ by the chain rule.

Part (ii)

Let $I = \int_0^\pi x\,\mathrm{f}(\sin x)\,\mathrm{d}x$. Put $y = \pi - x$, so $\mathrm{d}x = -\mathrm{d}y$, the limits $x = 0, \pi$ become $y = \pi, 0$, and $\sin x = \sin(\pi - y) = \sin y$. Then

$$\begin{aligned} I &= \int_\pi^0 (\pi - y)\,\mathrm{f}(\sin y)\,(-\mathrm{d}y) = \int_0^\pi (\pi - y)\,\mathrm{f}(\sin y)\,\mathrm{d}y \\ &= \pi \int_0^\pi \mathrm{f}(\sin y)\,\mathrm{d}y - \int_0^\pi y\,\mathrm{f}(\sin y)\,\mathrm{d}y = \pi \int_0^\pi \mathrm{f}(\sin x)\,\mathrm{d}x - I, \end{aligned}$$

since y is only a dummy variable. So $2I = \pi \int_0^\pi \mathrm{f}(\sin x)\,\mathrm{d}x$, which is the result.

Ask yourself The unknown integral I has appeared on both sides. Is that a problem?

No. It's exactly what we want. The substitution doesn't evaluate I; it gives us a linear equation for I, and we solve that.

Now take $\mathrm{f}(t) = \dfrac{1}{1+t}$, which is continuous for $t \in [0, 1]$, the values $\sin x$ takes on $[0, \pi]$. By part (i),

$$\int_0^\pi \frac{1}{1+\sin x}\,\mathrm{d}x = \Big[-\tan\left(\tfrac{1}{4}\pi - \tfrac{1}{2}x\right) \Big]_0^\pi = -\tan\left(-\tfrac{1}{4}\pi\right) + \tan\tfrac{1}{4}\pi = 1 + 1 = 2,$$

and $\tan\left(\frac{1}{4}\pi - \frac{1}{2}x\right)$ is continuous on $[0, \pi]$, because the argument stays in $[-\frac{1}{4}\pi, \frac{1}{4}\pi]$. Therefore

$$\int_0^\pi \frac{x}{1+\sin x}\,\mathrm{d}x = \frac{\pi}{2} \times 2 = \pi.$$

Part (iii)

Use the same reflection. Let $g(x) = 2x^3 - 3\pi x^2$. Expanding,

$$\begin{aligned} g(\pi - x) &= 2(\pi^3 - 3\pi^2 x + 3\pi x^2 - x^3) - 3\pi(\pi^2 - 2\pi x + x^2) \\ &= -\pi^3 + 3\pi x^2 - 2x^3 = -\pi^3 - g(x). \end{aligned}$$

Let $J = \int_0^\pi \dfrac{g(x)}{(1+\sin x)^2}\,\mathrm{d}x$. The substitution $y = \pi - x$, exactly as in part (ii), gives

$$J = \int_0^\pi \frac{g(\pi - y)}{(1+\sin y)^2}\,\mathrm{d}y = -\pi^3 K - J, \qquad \text{where } K = \int_0^\pi \frac{\mathrm{d}x}{(1+\sin x)^2}.$$

So $J = -\frac{1}{2}\pi^3 K$.

Ask yourself Why does this particular cubic work so well?

In terms of $u = x - \frac{1}{2}\pi$, the distance from the midpoint, $g(x) = 2u^3 - \frac{3}{2}\pi^2 u - \frac{1}{2}\pi^3$. Since $\sin x = \cos u$, the function $1/(1+\sin x)^2$ is even in u, so the odd terms $2u^3$ and

$-\frac{3}{2}\pi^2 u$ integrate to zero over the symmetric interval. Only the constant $-\frac{1}{2}\pi^3$ is left, which is what we found. Part (ii) is the same thing with $x = u + \frac{1}{2}\pi$.

To find K, square the identity from part (i): $\dfrac{1}{(1+\sin x)^2} = \frac{1}{4}\sec^4\left(\frac{1}{4}\pi - \frac{1}{2}x\right)$. Substitute $u = \frac{1}{4}\pi - \frac{1}{2}x$, so $\mathrm{d}x = -2\,\mathrm{d}u$, and $x = 0, \pi$ become $u = \frac{1}{4}\pi, -\frac{1}{4}\pi$. Writing $\sec^4 u = (1+\tan^2 u)\sec^2 u$, whose integral is $\tan u + \frac{1}{3}\tan^3 u$,

$$\begin{aligned} K &= \int_{\pi/4}^{-\pi/4} \tfrac{1}{4}\sec^4 u\,(-2\,\mathrm{d}u) = \frac{1}{2}\int_{-\pi/4}^{\pi/4} (1+\tan^2 u)\sec^2 u\,\mathrm{d}u \\ &= \frac{1}{2}\Big[\tan u + \tfrac{1}{3}\tan^3 u\Big]_{-\pi/4}^{\pi/4} = \frac{1}{2}\left(\tfrac{4}{3} + \tfrac{4}{3}\right) = \frac{4}{3}. \end{aligned}$$

Therefore

$$\int_0^{\pi} \frac{2x^3 - 3\pi x^2}{(1+\sin x)^2}\,\mathrm{d}x = -\frac{\pi^3}{2}\cdot\frac{4}{3} = -\frac{2\pi^3}{3}.$$

Checks

- Differentiate $-\tan\left(\frac{1}{4}\pi - \frac{1}{2}x\right)$ and you get $\frac{1}{2}\sec^2\left(\frac{1}{4}\pi - \frac{1}{2}x\right) = 1/(1+\sin x)$.
- Signs: in part (ii) the integrand is positive, and $\pi \approx 3.14$ is believable, since $\frac{1}{2} \le 1/(1+\sin x) \le 1$ gives bounds of $\frac{1}{4}\pi^2 \approx 2.47$ and $\frac{1}{2}\pi^2 \approx 4.93$. In part (iii), $2x^3 - 3\pi x^2 = x^2(2x - 3\pi) < 0$ on $(0, \pi]$, so a negative answer is right.
- Numerical quadrature gives $\int_0^{\pi}(1+\sin x)^{-2}\,\mathrm{d}x \approx 1.33333$ and the part (iii) integral ≈ -20.6709, which agree with $\frac{4}{3}$ and $-\frac{2}{3}\pi^3$.

Common slips

- Losing the factor $-\frac{1}{2}$ from the chain rule in part (i), and getting $+2\tan(\dots)$ or $-\frac{1}{2}\tan(\dots)$.
- Forgetting that swapping the limits cancels the minus sign from $\mathrm{d}x = -\mathrm{d}y$.
- Trying integration by parts three times in part (iii). It can be done, but the reflection does it in one line.

Beyond STEP

The trick in part (ii) uses symmetry. If a weight function is unchanged by a reflection, only the part of the integrand that is even about the centre of reflection contributes. You'll meet this idea all over first-year mathematics: in Fourier series, odd functions have no cosine terms, and in probability, a distribution symmetric about μ has its mean at μ. The substitution $t = \tan\frac{1}{2}x$ is a systematic alternative for any rational function of $\sin x$ and $\cos x$. For more integrals like these, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 2015, Question 7

Pure • Intersecting Circles

STEP II 2015, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q7

The key idea

When two circles meet in two points, the line through those points (the common chord) is perpendicular to the line of centres. The circle D bisects C exactly when this chord is a diameter of C, so passes through the centre of C. Pythagoras in the right-angled triangle formed by the two centres and one meeting point then turns "bisects" into a single equation: (distance between centres)$^2 = r^2 - a^2$.

Solution

Part (i)

Let D have centre Q and radius r, and let O be the origin.

The general condition. Suppose D bisects C, meeting it at P and $P' = -P$, where $|P| = a$. Both points lie on D, so $QP = QP' = r$. Then Q is equidistant from P and P', so it lies on the perpendicular bisector of PP', which passes through O (the midpoint of PP'). Hence $OQ \perp OP$, and Pythagoras in triangle OQP gives

$$r^2 = OQ^2 + a^2. \qquad (*)$$

Conversely, if D is any circle with $OQ^2 = r^2 - a^2$ and $OQ > 0$, take P on C with $OP \perp OQ$. Then $QP^2 = OQ^2 + a^2 = r^2$, and likewise for $-P$, so D passes through two diametrically opposite points of C. Two different circles meet in at most two points, so these are the only meeting points and D bisects C.

Existence when $r > a$. Put the centre on the x-axis at distance $\sqrt{r^2 - a^2} > 0$ from O:

$$\left(x - \sqrt{r^2 - a^2}\right)^2 + y^2 = r^2.$$

Directly: subtracting $x^2 + y^2 = a^2$ from this equation gives $-2x\sqrt{r^2 - a^2} + (r^2 - a^2) = r^2 - a^2$, so $x = 0$ (as $r \neq a$), and then $y = \pm a$. The circles meet at exactly $(0, a)$ and $(0, -a)$, which are diametrically opposite on C. So this circle of radius r bisects C.

Impossibility when $r \leq a$. If some circle of radius r bisects C, then $(*)$ gives $r^2 = OQ^2 + a^2 \geq a^2$. So $r < a$ is impossible. If $r = a$, then $(*)$ forces $OQ = 0$, so D is C itself, which meets C in infinitely many points, not exactly two. Hence no circle of radius $r \leq a$ bisects C.

Part (ii)

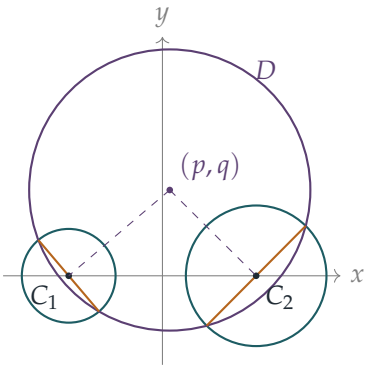
Let D have centre (p, q). Applying $(*)$ to each circle (with the origin shifted to the relevant centre),

$$(p+d)^2 + q^2 = r^2 - a_1^2,$$
$$(p-d)^2 + q^2 = r^2 - a_2^2.$$

Subtracting the second from the first, the q^2 and p^2 terms cancel, leaving $4pd = a_2^2 - a_1^2$, so

$$p = \frac{a_2^2 - a_1^2}{4d},$$

as required.



In the sketch ($d = 2$, $a_1 = 1$, $a_2 = 1.5$, $r = 3$) each common chord (amber) is a diameter of C_1 or C_2, perpendicular to the dashed line joining the centres.

For q, substitute p back into the first equation. First,

$$p + d = \frac{4d^2 + a_2^2 - a_1^2}{4d},$$

so

$$q^2 = r^2 - a_1^2 - \frac{(4d^2 + a_2^2 - a_1^2)^2}{16d^2} = \frac{16d^2r^2 - 16d^2a_1^2 - (4d^2 + a_2^2 - a_1^2)^2}{16d^2}.$$

The subtracted terms in the numerator expand to

$$\begin{aligned} 16d^2a_1^2 + (4d^2 + a_2^2 - a_1^2)^2 &= 16d^4 + 8d^2(a_2^2 - a_1^2) + 16d^2a_1^2 + (a_2^2 - a_1^2)^2 \\ &= 16d^4 + 8d^2(a_1^2 + a_2^2) + (a_2^2 - a_1^2)^2 \\ &= \left(4d^2 + (a_2 - a_1)^2\right)\left(4d^2 + (a_2 + a_1)^2\right), \end{aligned}$$

where the last line holds because $(a_2-a_1)^2+(a_2+a_1)^2 = 2(a_1^2+a_2^2)$ and $(a_2-a_1)^2(a_2+a_1)^2 = (a_2^2-a_1^2)^2$. Therefore

$$q = \pm\frac{\sqrt{16r^2d^2-\left(4d^2+(a_2-a_1)^2\right)\left(4d^2+(a_2+a_1)^2\right)}}{4d}.$$

Ask yourself Where does the inequality come from? Nothing so far looks like an inequality.

The centre (p,q) is a real point, so $q^2 \geq 0$. Since $16d^2 > 0$, this says

$$16r^2d^2 \geq \left(4d^2+(a_2-a_1)^2\right)\left(4d^2+(a_2+a_1)^2\right),$$

which is the required condition on r.

We should also check that part (i)'s requirement $r > a_1$ (and $r > a_2$) is not an extra condition. Since $d > a_1$, we have $4d^2+a_2^2-a_1^2 > 0$, so $p+d > 0$; then $r^2-a_1^2 = (p+d)^2+q^2 > 0$. Similarly $d-p > 0$ because $d > a_2$, so $r^2-a_2^2 = (p-d)^2+q^2 > 0$. So the circles D found really do bisect both C_1 and C_2.

Checks

- Symmetric case $a_1 = a_2 = a$: then $p = 0$, as symmetry about the y-axis demands, and the condition becomes $16r^2d^2 \geq 4d^2(4d^2+4a^2)$, that is $r^2 \geq d^2+a^2$. That is exactly $(*)$ with the centre at the origin, the closest possible centre to both $(\pm d, 0)$.
- In the sketch's example, $p = 0.15625$ and $q \approx 1.830$, and the intersection points of D with C_1 are approximately $(-1.353, -0.762)$ and $(-2.647, 0.762)$, whose midpoint is $(-2, 0)$, the centre of C_1. The smallest possible r there is about 2.377.
- Swapping C_1 and C_2 ($a_1 \leftrightarrow a_2$, $d \to -d$) leaves p unchanged in form and leaves the inequality unchanged, as it should.

Common slips

- Forgetting the case $r = a$ in part (i): $(*)$ allows $OQ = 0$, and you must say why $D = C$ does not count.
- Writing the condition as $OQ^2 = r^2+a^2$, which confuses bisecting with meeting at right angles (orthogonal circles). Draw the triangle: the right angle is at the centre of C, so the hypotenuse is the radius r of D.
- Giving only $q = +\sqrt{\cdots}$: reflecting in the x-axis gives a second circle.

Beyond STEP

The quantity $(p-x_0)^2+(q-y_0)^2-r^2$ for a point (p,q) and a circle with centre (x_0,y_0) and radius r is called the *power* of the point with respect to the circle, and the set of

points with equal power with respect to two circles is a straight line, the *radical axis*; subtracting two circle equations, as we did, finds it. Here the bisecting condition says the power of the centre of C with respect to D is $-a^2$. These ideas, and the related inversion in a circle, are developed beautifully in Needham, *Visual Complex Analysis*, and many STEP-style uses appear in Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP II 2015, Question 8

Pure • Vectors and Tangent Circles

STEP II 2015, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q8

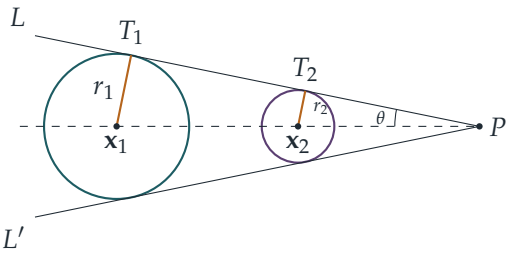
The key idea

Each centre is the same distance from both tangents (its own radius), so both centres lie on the line that bisects the angle between L and L'. Along that line, distance from P is proportional to radius, which gives the formula in (i) at once. For (ii), dividing the formula by $r_1 r_2$ makes it far more symmetric: in terms of $\mathbf{x}_i / r_i$ and $1/r_i$ the three foci satisfy one linear relation whose coefficients add up to zero, and that is exactly what collinearity means.

Solution

Part (i)

Let θ be half the angle between L and L' at P, in the angle that contains the circles.



A tangent to a circle is at perpendicular distance equal to the radius from the centre. So $\mathbf{x}_1$ is at distance r_1 from both L and L', and points equidistant from two intersecting lines lie on one of the two angle bisectors at P. Since both circles lie on the same side of L and on the same side of L', both centres are in the same one of the four angular regions cut out by L and L', so they lie on the *same* half-line from P: the bisector of that region. Let $\mathbf{u}$ be the unit vector along it, pointing from P towards the circles.

If T_1 is the point where L touches C_1, the triangle PT_1X_1 (with X_1 the centre) has a right

angle at T_1, so $r_1 = PX_1 \sin\theta$. In the same way $r_2 = PX_2 \sin\theta$. Hence

$$\mathbf{x}_1 - \mathbf{p} = \frac{r_1}{\sin\theta}\,\mathbf{u}, \qquad \mathbf{x}_2 - \mathbf{p} = \frac{r_2}{\sin\theta}\,\mathbf{u},$$

where $\mathbf{p}$ is the position vector of P. Eliminating $\mathbf{u}/\sin\theta$,

$$r_2(\mathbf{x}_1 - \mathbf{p}) = r_1(\mathbf{x}_2 - \mathbf{p}) \quad \Longrightarrow \quad (r_1 - r_2)\,\mathbf{p} = r_1\mathbf{x}_2 - r_2\mathbf{x}_1.$$

The circles have different sizes, so $r_1 \neq r_2$ and we may divide:

$$\mathbf{p} = \frac{r_1\mathbf{x}_2 - r_2\mathbf{x}_1}{r_1 - r_2},$$

as required. (Equal radii would make L and L' parallel, with no focus at all.)

Part (ii)

Ask yourself The formula for $\mathbf{p}$ is not symmetric in the obvious way. Can we rewrite it so the three foci look alike?

Divide $(r_1 - r_2)\mathbf{p} = r_1\mathbf{x}_2 - r_2\mathbf{x}_1$ by $r_1 r_2$. Writing $s_i = 1/r_i$ and $\mathbf{y}_i = \mathbf{x}_i/r_i$, it becomes

$$(s_2 - s_1)\,\mathbf{p} = \mathbf{y}_2 - \mathbf{y}_1.$$

Applying part (i) to the other two pairs of circles (allowed, as $r_3 \neq r_1$ and $r_3 \neq r_2$, and the circles do not overlap) gives, with $\mathbf{q}$ and $\mathbf{r}$ the position vectors of Q and R,

$$(s_3 - s_1)\,\mathbf{q} = \mathbf{y}_3 - \mathbf{y}_1, \qquad (s_3 - s_2)\,\mathbf{r} = \mathbf{y}_3 - \mathbf{y}_2.$$

The right-hand sides satisfy $(\mathbf{y}_2 - \mathbf{y}_1) + (\mathbf{y}_3 - \mathbf{y}_2) = \mathbf{y}_3 - \mathbf{y}_1$, so

$$(s_2 - s_1)\,\mathbf{p} + (s_3 - s_2)\,\mathbf{r} = (s_3 - s_1)\,\mathbf{q}.$$

Since $s_3 \neq s_1$, we can write

$$\mathbf{q} = \lambda\,\mathbf{p} + (1 - \lambda)\,\mathbf{r}, \qquad \text{where } \lambda = \frac{s_2 - s_1}{s_3 - s_1},$$

because the two coefficients $\frac{s_2 - s_1}{s_3 - s_1}$ and $\frac{s_3 - s_2}{s_3 - s_1}$ add up to 1. So $\mathbf{q} = \mathbf{r} + \lambda(\mathbf{p} - \mathbf{r})$, which is a point on the line through R and P. Hence P, Q and R are collinear. (If P and R happen to coincide, this gives $Q = P = R$ as well, and the statement holds trivially.)

Part (iii)

From part (ii), $\mathbf{q} = \lambda\mathbf{p} + (1 - \lambda)\mathbf{r}$. Assuming $P \neq R$, the point $\lambda\mathbf{p} + (1 - \lambda)\mathbf{r}$ is the midpoint $\frac{1}{2}(\mathbf{p} + \mathbf{r})$ exactly when $\lambda = \frac{1}{2}$ (different values of λ give different points of the line). So the condition is

$$\frac{s_2 - s_1}{s_3 - s_1} = \frac{1}{2} \iff 2s_2 = s_1 + s_3 \iff \frac{2}{r_2} = \frac{1}{r_1} + \frac{1}{r_3} \iff r_2 = \frac{2r_1 r_3}{r_1 + r_3}.$$

In words, r_2 must be the harmonic mean of r_1 and r_3. (In the degenerate case $P = R$, all three foci coincide and Q is trivially half-way, whatever the radii.)

Checks

- Part (i): putting $r_2 \to 0$ shrinks C_2 to a point, and the formula gives $\mathbf{p} = \mathbf{x}_2$: the tangents from a point pass through that point. Also $\mathbf{p}$ is $\mathbf{x}_2 + \frac{r_2}{r_1 - r_2}(\mathbf{x}_2 - \mathbf{x}_1)$, beyond the smaller circle when $r_2 < r_1$, as in the diagram.
- A numerical case for (iii): take centres $(0,0)$, $(10,1)$, $(3,8)$ with radii 1, $\frac{3}{2}$, 3, so that $\frac{2}{3/2} = \frac{4}{3} = 1 + \frac{1}{3}$. Then $P = (-20,-2)$, $R = (17,-6)$ and $Q = (-\frac{3}{2},-4)$, which is indeed the midpoint of PR.
- The formulas were also confirmed by computer: intersecting the two tangent lines directly gives the same point P for many random pairs of circles.

Common slips

- Not explaining why P lies on the line of centres, or why P is outside the segment joining them (it is the *same* half-line from P that carries both centres).
- Showing that $\mathbf{q} - \mathbf{p}$ and $\mathbf{r} - \mathbf{p}$ are parallel without checking that the relation's coefficients sum to zero; a relation $\alpha\mathbf{p} + \beta\mathbf{q} + \gamma\mathbf{r} = \mathbf{0}$ only implies collinearity when $\alpha + \beta + \gamma = 0$.

Beyond STEP

The focus P is the centre of the enlargement (homothety) with positive scale factor r_2/r_1 that maps C_1 onto C_2. Part (ii) is Monge's three-circle theorem: composing the enlargements about Q and R gives the enlargement about P, and the centre of a composite of two enlargements lies on the line through their centres. The algebra in (ii) is a first taste of barycentric (affine) coordinates, where collinearity is a linear relation with coefficients summing to zero. Strang's *Introduction to Linear Algebra* develops the linear maps behind this, and Siklos's *Advanced Problems in Mathematics* has more questions of this kind.

WORKED SOLUTION

STEP II 2015, Question 9

Mechanics • Rigid Body Rotation

STEP II 2015, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q9

The key idea

Everything about this system is controlled by one point: the centre of mass G of the two particles. A rigid body pivoted at A hangs at rest with G vertically below A, and it makes complete revolutions exactly when it has enough energy to carry G over the top. So find where G is and how far it is from A, and both parts follow quickly.

Solution

We model B and C as particles joined by light rigid rods, with a smooth axis at A, so the only forces doing work are the weights.

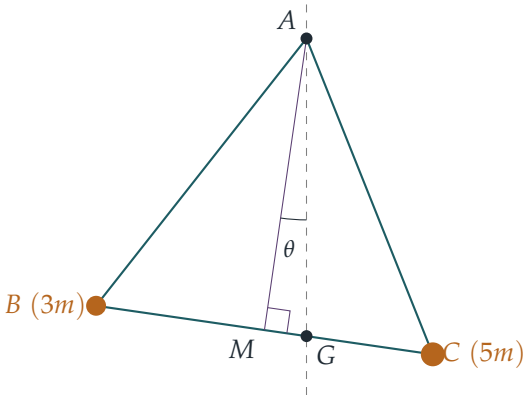
Part (i)

The rods are light, so the centre of mass of the system is the centre of mass of $3m$ at B and $5m$ at C. That point G lies on BC with

$$BG = \frac{5}{8}a, \qquad CG = \frac{3}{8}a.$$

Let M be the midpoint of BC. Then $MG = \frac{5}{8}a - \frac{1}{2}a = \frac{1}{8}a$, on the side of C. Since the triangle is equilateral, $AM \perp BC$ and $AM = \frac{\sqrt{3}}{2}a$, so by Pythagoras in triangle AMG,

$$AG^2 = \frac{3a^2}{4} + \frac{a^2}{64} = \frac{49a^2}{64}, \qquad AG = \frac{7a}{8}.$$



In equilibrium the weight $8mg$ acting at G must have zero moment about A, so G is vertically below A: the line AG is vertical. Now BC is perpendicular to AM, and the horizontal is perpendicular to the vertical AG. Rotating both lines of a pair by a right angle doesn't change the angle between them, so

$$\theta = \angle GAM.$$

In the right-angled triangle AMG (right angle at M),

$$\sin\theta = \frac{MG}{AG} = \frac{a/8}{7a/8} = \frac{1}{7},$$

as required. Note that C, the heavier particle, is the lower end of BC, as it should be.

Part (ii)

Both particles are at distance a from the axis, so they always have the same speed. If the triangle turns with angular speed ω, each particle has speed $a\omega$. Immediately after the impulse, B has speed v, so C does too, and the kinetic energy is

$$\tfrac{1}{2}(3m)v^2 + \tfrac{1}{2}(5m)v^2 = 4mv^2.$$

(Equivalently, the moment of inertia about A is $8ma^2$ and $\omega = v/a$.) The direction of the impulse doesn't matter, since only the speed enters the energy equation below.

? **Ask yourself** Do we need v to be large enough to keep something taut at the top, as for a particle on a string?

No. The rods are rigid and can push as well as pull, so the only requirement is that the angular speed never falls to zero before the system has gone right round. That is a pure energy condition.

The potential energy of the system is $8mg$ times the height of G. Since G moves on a circle of radius $AG = \frac{7}{8}a$ about A, the potential energy is greatest when G is vertically above A, at a height

$$2 \cdot \frac{7a}{8} = \frac{7a}{4}$$

above its starting position. By conservation of energy, at a position where G has risen by h,

$$4m(a\omega)^2 = 4mv^2 - 8mgh.$$

The right-hand side is smallest when $h = \frac{7}{4}a$. If it is positive there, then $\omega \neq 0$ throughout, so ω never changes sign and the system keeps turning the same way, completing revolution after revolution. If it is negative, ω must reach zero before G reaches the top and the system swings back. So the condition is

$$4mv^2 > 8mg \cdot \frac{7a}{4} \iff v^2 > \frac{7ag}{2}.$$

Hence

$$v_0 = \sqrt{\frac{7ag}{2}}.$$

(When $v = v_0$ exactly, G creeps towards the position above A and takes infinitely long to get there, which is why the condition is a strict inequality.)

Checks

- With equal masses G would be at M, giving $\theta = 0$ and $AG = \frac{\sqrt{3}}{2}a$: formulas with $MG = 0$ give exactly that.
- Minimising the potential energy numerically over all orientations gives $\sin\theta = 0.142857\ldots = \frac{1}{7}$.
- Units: ag has units $\mathrm{m^2\,s^{-2}}$, so v_0 is a speed. For $a = 1\,\mathrm{m}$ and $g = 9.8\,\mathrm{m\,s^{-2}}$, $v_0 \approx 5.86\,\mathrm{m\,s^{-1}}$.
- A step-by-step computer simulation of the motion confirms that the system goes over the top when $v = 1.01v_0$ and swings back when $v = 0.99v_0$.

Common slips

- Taking the rise of G as $2 \cdot \frac{\sqrt{3}}{2}a$ (using AM instead of AG), which gives $v_0^2 = 2\sqrt{3}\,ag$.
- Treating B and C separately and forgetting that the impulse gives C speed v as well.
- Imposing a "string stays taut" condition at the top: rods can be in thrust, so none is needed.

Beyond STEP

The energy argument here is the rigid-body version of the compound pendulum: a body pivoted at a distance d from its centre of mass behaves like a simple pendulum, with the equation of motion $I\ddot{\psi} = -Mgd\sin\psi$. In a first-year course you'll meet the phase plane for this equation, where the critical speed v_0 corresponds to the separatrix

dividing oscillations from complete rotations. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body rotation about a fixed axis and the physical pendulum in detail.

WORKED SOLUTION

STEP II 2015, Question 10

Mechanics • Constrained Motion

STEP II 2015, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

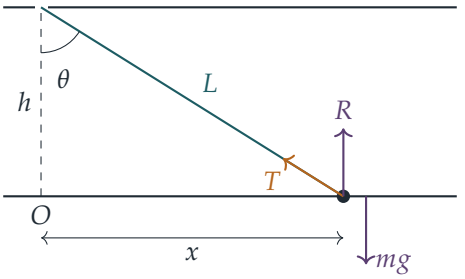
step.maths.org/questions/15-s2-q10

The key idea

This is a constraint problem: the particle's motion is completely fixed by the string, so the kinematics come first and the forces afterwards. Write the length of string below the ceiling in terms of the particle's horizontal distance from the point under the hole, and differentiate with respect to time; the rate of change of that length is known. Once the acceleration is known, Newton's second law horizontally gives the tension, and resolving vertically gives the normal reaction, which must stay non-negative while the particle is on the floor.

Solution

We model the particle as a point on a smooth horizontal floor and the string as light and inextensible, so the tension T is the same along it. Let O be the point of the floor directly below the hole, let x be the distance of the particle from O, and let L be the length of string between the hole and the particle.



The speed

By Pythagoras, and from the triangle,

$$L^2 = x^2 + h^2, \qquad x = h\tan\theta, \qquad \sin\theta = \frac{x}{L}.$$

The string is drawn in through the hole at speed V, so $\dot{L} = -V$. Differentiating $L^2 = x^2 + h^2$ with respect to time (h is constant):

$$2L\dot{L} = 2x\dot{x} \quad \Longrightarrow \quad \dot{x} = \frac{L}{x}\dot{L} = -\frac{VL}{x} = -\frac{V}{\sin\theta}.$$

So the particle moves towards O with speed

$$\boxed{v = \frac{V}{\sin\theta} = V\operatorname{cosec}\theta}$$

which is always larger than V (it is the string, not the particle, that moves at V).

Ask yourself Why isn't the particle's speed simply V, or $V\sin\theta$?

Only the component of the particle's velocity *along the string* must equal the rate at which the string is drawn in. That component is $v\sin\theta$, since the string makes angle $\frac{\pi}{2} - \theta$ with the floor. Setting $v\sin\theta = V$ gives the same answer as above, and makes clear why $v > V$.

The acceleration

Differentiate $\dot{x} = -VL/x$ again, using the quotient rule, $\dot{L} = -V$ and $\dot{x} = -VL/x$:

$$\ddot{x} = -V\,\frac{\dot{L}\,x - L\,\dot{x}}{x^2} = -V\,\frac{-Vx + VL^2/x}{x^2} = -\frac{V^2(L^2 - x^2)}{x^3} = -\frac{V^2h^2}{x^3}.$$

With $x = h\tan\theta$ this is $\ddot{x} = -\dfrac{V^2}{h\tan^3\theta}$. Since $\dot{x}$ and $\ddot{x}$ are both negative, the particle is speeding up: its acceleration has magnitude

$$\boxed{a = \frac{V^2}{h\tan^3\theta} = \frac{V^2\cos^3\theta}{h\sin^3\theta}}$$

directed towards O. (As a check, $v = V/\sin\theta$ does increase as θ decreases.)

The tension, and leaving the floor

The forces on the particle are its weight mg, the normal reaction R from the floor (vertical, as the floor is smooth) and the tension T along the string, which has horizontal component $T\sin\theta$ towards O and vertical component $T\cos\theta$ upwards.

Horizontally, by Newton's second law,

$$T\sin\theta = ma = \frac{mV^2}{h\tan^3\theta} \quad \Longrightarrow \quad \boxed{T = \frac{mV^2}{h\sin\theta\tan^3\theta} = \frac{mV^2\cos^3\theta}{h\sin^4\theta}}$$

Vertically, the particle is in equilibrium while it is on the floor, so

$$R + T\cos\theta = mg \quad \Longrightarrow \quad R = mg - \frac{mV^2\cos\theta}{h\sin\theta\tan^3\theta} = mg - \frac{mV^2}{h\tan^4\theta}.$$

Ask yourself Which way does R change as the particle is pulled in?

As the particle approaches O, θ decreases, so $\tan^4\theta$ decreases and R decreases steadily. At the start (large θ) R is close to mg; the particle stays on the floor as long as $R \geq 0$, and leaves it at the first moment that $R = 0$, which is when

$$\frac{mV^2}{h\tan^4\theta} = mg \quad \Longleftrightarrow \quad \tan^4\theta = \frac{V^2}{gh},$$

as required. For smaller θ the formula would give $R < 0$, which is impossible, so the particle has indeed left the floor.

Checks

- Dimensions: V^2/h is an acceleration, mV^2/h a force, and $V^2/(gh)$ is dimensionless, as $\tan^4\theta$ must be.
- Limits: as $\theta \to \frac{\pi}{2}$ (particle very far away) the string is almost horizontal, $v \to V$ and $a \to 0$, as expected. As $\theta \to 0$ the speed and the tension blow up, so the particle must leave the floor before reaching O.
- Numbers: with $V = 2$, $h = 1$, $\theta = \frac{\pi}{4}$ we get $v = 2\sqrt{2} \approx 2.83$ and $a = 4$; differentiating $x(t) = \sqrt{(L_0 - Vt)^2 - h^2}$ numerically gives the same values. With $g = 9.8$, the particle leaves the floor when $\theta \approx 38.6°$.

Common slips

- Taking the particle's speed to be V, or $V\sin\theta$, instead of $V/\sin\theta$.
- Forgetting that L changes when differentiating $\dot{x} = -VL/x$ a second time.
- Assuming the tension is constant because the string moves at constant speed; the particle accelerates, so it is not.

Beyond STEP

The particle here has only one degree of freedom: once the string's motion is prescribed, its position is fixed at every time. Problems with such constraints are handled systematically in Lagrangian mechanics, met in the first or second year of a university course, where the constraint is built into the choice of coordinates and constraint forces like R and T are found afterwards. Kleppner and Kolenkow, *An Introduction to Mechanics*, has many problems of this kind.

WORKED SOLUTION

STEP II 2015, Question 11

Mechanics • Momentum and Energy

STEP II 2015, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

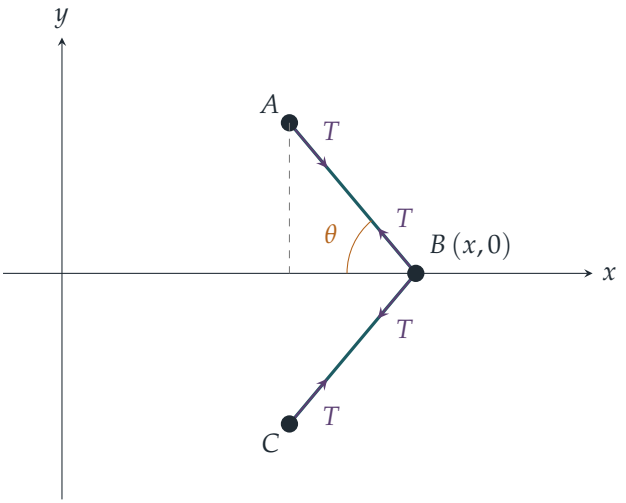
step.maths.org/questions/15-s2-q11

The key idea

Two coordinates, x and θ, describe the whole system, because the motion stays symmetric about the x-axis. The table is smooth and the string is light and inextensible, so two things are conserved: momentum in the x direction and kinetic energy. Those two conservation laws give the two equations in (ii) directly, and the second one, read carefully, says that $\dot{\theta}$ can never be zero. That single fact settles both (iii) and (iv).

Solution

We model A, B, C as particles and the string as light and inextensible. Gravity and the normal reaction are vertical, so the only horizontal forces are the tensions, which are internal to the system.



Part (i)

A is a distance a from B along a line making angle θ with the negative x direction, so

$$A = (x - a\cos\theta,\ a\sin\theta).$$

Differentiating with respect to t, using the chain rule on $\cos\theta$ and $\sin\theta$,

$$\dot{\mathbf{r}}_A = \big(\dot{x} + a\sin\theta\,\dot{\theta},\ a\cos\theta\,\dot{\theta}\big),$$

as required (and $\dot{x} = v$, the speed of B).

The set-up and the impulse are both symmetric under reflection in the x-axis, so the motion stays symmetric: $C = (x - a\cos\theta,\ -a\sin\theta)$ with velocity $\big(\dot{x} + a\sin\theta\,\dot{\theta},\ -a\cos\theta\,\dot{\theta}\big)$.

Part (ii)

First, the starting values. At the instant of the impulse both strings lie along the y-axis, perpendicular to B's new velocity, so the string does not need to jerk A or C; they start at rest, while $\dot{x} = u$. The total x-momentum is mu and the kinetic energy is $\frac{1}{2}mu^2$.

Momentum. There is no external horizontal force, so the total x-momentum stays equal to mu:

$$m\dot{x} + 2m\big(\dot{x} + a\sin\theta\,\dot{\theta}\big) = mu \quad\Longrightarrow\quad 3\dot{x} + 2a\dot{\theta}\sin\theta = u. \tag{1}$$

(The y-momenta of A and C cancel, as they must.) The right-hand side here is u, the initial speed of B. The paper prints v, but v has already been used for $\dot{x}$, so this must be a misprint for u.

Energy. The string is light and inextensible, so the tension does no net work on the system, and the only other forces do no work either. Kinetic energy is conserved until A and C collide:

$$\dot{x}^2 + 2\Big[\big(\dot{x} + a\sin\theta\,\dot{\theta}\big)^2 + a^2\cos^2\theta\,\dot{\theta}^2\Big] = u^2. \tag{2}$$

From (1), with $s = \sin\theta$ and $c = \cos\theta$,

$$\dot{x} = \tfrac{1}{3}\big(u - 2as\dot{\theta}\big), \qquad \dot{x} + as\dot{\theta} = \tfrac{1}{3}\big(u + as\dot{\theta}\big).$$

Substituting into (2), the cross terms $\mp 4uas\dot{\theta}$ cancel:

$$\tfrac{1}{9}\big(u - 2as\dot{\theta}\big)^2 + \tfrac{2}{9}\big(u + as\dot{\theta}\big)^2 + 2a^2c^2\dot{\theta}^2 = \tfrac{1}{9}\big(3u^2 + 6a^2s^2\dot{\theta}^2\big) + 2a^2c^2\dot{\theta}^2 = u^2.$$

So $a^2\dot{\theta}^2\big(\frac{2}{3}s^2 + 2c^2\big) = \frac{2}{3}u^2$, that is $a^2\dot{\theta}^2(s^2 + 3c^2) = u^2$. Since $s^2 + 3c^2 = 3 - 2s^2$,

$$\dot{\theta}^2 = \frac{u^2}{a^2(3 - 2\sin^2\theta)},$$

which is the second result (with the same reading of v as u).

Part (iii)

Because $3 - 2\sin^2\theta \leq 3$, part (ii) gives

$$\dot{\theta}^2 \geq \frac{u^2}{3a^2} > 0.$$

So $\dot{\theta}$ is never zero. Between collisions it is continuous, so it keeps the same sign and θ changes monotonically.

At the start A is at rest, so part (i) gives $\dot{x} + a\dot{\theta} = 0$, i.e. $\dot{\theta} = -u/a < 0$. So θ decreases steadily from $\frac{\pi}{2}$. A and C always have the same x-coordinate and y-coordinates $\pm a\sin\theta$, so they meet exactly when $\sin\theta = 0$. The first collision is at $\theta = 0$, behind B.

Ask yourself What does the collision do to $\dot{x}$ and $\dot{\theta}$?

At $\theta = 0$ both strings lie along the x-axis, and A and C have the same x-velocity and y-velocities $\pm a\dot{\theta}$. They approach along the y direction, so the impulse between them is along y. That is perpendicular to the strings, so no impulsive tension arises, and every x-velocity is unchanged. In an elastic collision between equal masses the velocity components along the line of centres are exchanged. So A's y-velocity changes from $a\dot{\theta}$ to $-a\dot{\theta}$: the collision simply reverses the sign of $\dot{\theta}$, and $\dot{x}$ stays the same.

Momentum and energy are the same after the collision as before, so (1) and (2), and hence both results of (ii), hold for the next stage of the motion too. Now $\dot{\theta} > 0$, and as before it cannot vanish. So θ increases steadily from 0, and the next value with $\sin\theta = 0$ is $\theta = \pi$. It gets there in a finite time, at most $\pi a\sqrt{3}/u$, because $\dot{\theta} \geq u/(a\sqrt{3})$. The second collision happens at

$$\theta = \pi,$$

with A and C meeting *in front of* B.

Part (iv)

Throughout the motion $\theta \in [0, \pi]$, so $\sin\theta \geq 0$. If $v = \dot{x} = 0$, then (1) gives $2a\dot{\theta}\sin\theta = u$, and squaring and using (ii),

$$4a^2\sin^2\theta \cdot \frac{u^2}{a^2(3 - 2\sin^2\theta)} = u^2 \iff 4\sin^2\theta = 3 - 2\sin^2\theta \iff \sin^2\theta = \tfrac{1}{2}.$$

So the possible values are $\theta = \frac{\pi}{4}$ and $\theta = \frac{3\pi}{4}$.

Ask yourself Squaring threw away a sign. Which sign do we actually need?

We squared $2a\dot{\theta}\sin\theta = u$, and since $u > 0$ we also need $\dot{\theta}\sin\theta > 0$, i.e. $\dot{\theta} > 0$. At $\sin\theta = \frac{1}{\sqrt{2}}$ we have $\dot{\theta}^2 = \frac{u^2}{2a^2}$, so $\dot{\theta} = \pm\frac{u}{a\sqrt{2}}$ and $2a\dot{\theta}\sin\theta = \pm u$. From (1):

- if $\dot{\theta} > 0$, then $\dot{x} = \frac{1}{3}(u - u) = 0$;
- if $\dot{\theta} < 0$, then $\dot{x} = \frac{1}{3}(u + u) = \frac{2}{3}u$.

So the answer is no. On the first swing (θ falling from $\frac{\pi}{2}$ to 0) θ passes $\frac{\pi}{4}$ with $v = \frac{2}{3}u$. Between the first and second collisions (θ rising from 0 to π) $v = 0$ at both $\theta = \frac{\pi}{4}$ and $\theta = \frac{3\pi}{4}$. After the second collision $\dot{\theta}$ is negative again, so on the swing back from π to 0 we have $v = \frac{2}{3}u$ at both values. The pattern then repeats: $v = 0$ at these angles only on the swings where θ is increasing.

Checks

- At $\theta = \frac{\pi}{2}$ initially, (ii) gives $\dot{\theta}^2 = u^2/a^2$, matching $\dot{\theta} = -u/a$ found directly from A being at rest.
- From (1), $\dot{x} = \frac{u}{3}\left(1 - \frac{2a\dot{\theta}\sin\theta}{u}\right)$. At $\theta = 0$ this is $\frac{u}{3}$: all three particles move together at $\frac{u}{3}$, which is the centre-of-mass velocity, as it should be when the strings are straight along the x-axis. On the rising swing, at $\theta = \frac{\pi}{2}$, $\dot{x} = -\frac{u}{3}$, so B moves backwards for a while, which explains why v passes through 0 twice.
- A computer simulation of the three particles joined by very stiff springs, with $a = u = 1$, reproduces both equations of (ii). It shows the first collision at $\theta = 0$ and the second at $\theta = \pi$, and v changing sign only near $\theta \approx 0.785$ and 2.356 on the rising swing, where $\dot{\theta} \approx 0.707$. On the first swing it gives $v \approx 0.67$ at $\theta = \frac{\pi}{4}$.

Common slips

- Taking the right-hand side of (1) to be $\dot{x}$: it is the constant u fixed by the impulse.
- Assuming the collision sends θ back up towards $\frac{\pi}{2}$ and the second collision is again at $\theta = 0$. Reversing $\dot{\theta}$ makes θ increase *through* $\frac{\pi}{2}$, and nothing can stop it before π, because $\dot{\theta}^2$ is bounded away from 0.
- Forgetting that squaring loses the sign of $\dot{\theta}$ in part (iv).

Beyond STEP

Choosing x and θ as coordinates and building the kinetic energy from them is the starting point of Lagrangian mechanics, which you meet in the first year. There, conservation of x-momentum appears automatically because x does not occur in the energy (it is an "ignorable" coordinate). The energy equation then reduces the problem to a single first-order equation for θ, just as here. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers momentum and energy methods for systems of particles very thoroughly.

WORKED SOLUTION

STEP II 2015, Question 12

Probability and Statistics • Coin Sequence Games

STEP II 2015, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q12

The key idea

Every sequence here has length three, so whether a toss ends the game depends only on that toss and the two before it. The natural thing to track is therefore the *last two tosses*: there are just four possible states (HH, HT, TH, TT), and from each one the next toss either ends the game or moves us to another state. Some states settle the winner at once; for the others we set up a pair of simple equations by conditioning on the next toss.

Solution

Throughout, tosses are independent, and each is H or T with probability $\frac{1}{2}$. No sequence can be completed in the first two tosses, so the game always reaches one of the four two-toss states, each with probability $\frac{1}{4}$. Note too that the game ends with probability 1: from any position, the next three tosses complete a given sequence with probability $\frac{1}{8}$, so the chance of the game lasting more than $3n$ further tosses is at most $(\frac{7}{8})^n \to 0$.

Part (i)

Here only HHT (for A) and THH (for B) count.

If the first two tosses are HH: the coin keeps showing H for a while, and the first T to appear completes HHT, since it is preceded by at least two Hs. Before that T no tail has appeared, so THH cannot have occurred. So A wins.

If a T appears among the first two tosses: consider the first time two consecutive heads appear after that T. The toss just before this HH cannot be H (otherwise there would be an earlier HH after the T, or the HH would be the opening pair, which we've ruled out), so it is T and THH has been completed. HHT can't have occurred earlier, because it needs an HH, and this is the first one. So B wins.

Hence A wins exactly when the first two tosses are both heads, and

$$\mathrm{P}(A \text{ wins}) = \tfrac{1}{2} \times \tfrac{1}{2} = \tfrac{1}{4},$$

as required.

Part (ii)

Now all four sequences HHT, THH, TTH, HTT are live. Condition on the first two tosses.

- **HH.** As in part (i), the tosses continue H, H, … until the first T, which completes HHT. None of the other sequences contains HH followed by H, and no tail has appeared to start THH, TTH or HTT. So A wins.
- **TT.** By the same argument with the roles of H and T swapped, the first H completes TTH, so C wins.
- **HT.** If the next toss is T we have HTT and D wins. If it is H, the last two tosses are TH (and HTH completes nothing).
- **TH.** If the next toss is H we have THH and B wins. If it is T, the last two tosses are HT (and THT completes nothing).

So once the game is in state HT or TH it bounces between these two until D or B wins. Let d_1 and d_2 be the probabilities that D wins starting from HT and from TH respectively. Conditioning on the next toss,

$$d_1 = \tfrac{1}{2} + \tfrac{1}{2}d_2, \qquad d_2 = \tfrac{1}{2}d_1.$$

Substituting, $d_1 = \frac{1}{2} + \frac{1}{4}d_1$, so $d_1 = \frac{2}{3}$ and $d_2 = \frac{1}{3}$. From these states only B or D can win, so B wins with probability $\frac{1}{3}$ from HT and $\frac{2}{3}$ from TH. Therefore

$$\mathrm{P}(A) = \tfrac{1}{4}, \qquad \mathrm{P}(C) = \tfrac{1}{4},$$
$$\mathrm{P}(D) = \tfrac{1}{4}\cdot\tfrac{2}{3} + \tfrac{1}{4}\cdot\tfrac{1}{3} = \tfrac{1}{4}, \qquad \mathrm{P}(B) = \tfrac{1}{4}\cdot\tfrac{1}{3} + \tfrac{1}{4}\cdot\tfrac{2}{3} = \tfrac{1}{4}.$$

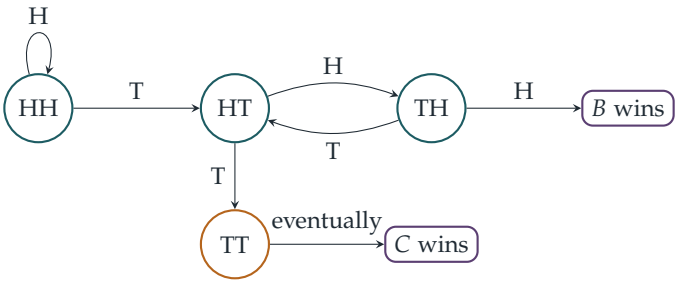
Each player wins with probability $\frac{1}{4}$.

Part (iii)

Now only THH (for B) and TTH (for C) count.

First two tosses TT. The tosses then run T, T, … until the first H, which completes TTH. THH needs a TH followed by H, and no H has appeared yet, so B can't have won first. Hence C wins with probability 1.

For the other starts it helps to see how the state (the last two tosses) moves, given that the game hasn't ended:



(From TT the next T keeps us in TT and the next H completes TTH, which is why we can treat TT as a win for C.)

With p, q, r the probabilities that C wins from HT, TH, HH:

From HT. A T gives TT, from which C wins with probability 1; an H gives HTH, which completes neither sequence and leaves us in state TH. So

$$p = \tfrac{1}{2} \cdot 1 + \tfrac{1}{2} \cdot q = \tfrac{1}{2} + \tfrac{1}{2}q,$$

as required.

From TH. An H completes THH, so B wins; a T gives THT, completing nothing, and we are in state HT. So $q = \tfrac{1}{2} \cdot 0 + \tfrac{1}{2}p = \tfrac{1}{2}p$.

Ask yourself Why does the past history not matter once we know the last two tosses?

Both target sequences have length three, so whether the next toss finishes the game depends only on that toss and the previous two. Earlier tosses are irrelevant, and future tosses are independent of the past. So the chance that C wins from a given state is the same however we got there, which is what lets q reappear on the right-hand side of the equation for p.

From HH. An H leaves us in HH; a T moves us to HT. So $r = \tfrac{1}{2}r + \tfrac{1}{2}p$, giving $r = p$.

Solving: $p = \tfrac{1}{2} + \tfrac{1}{4}p$, so

$$p = \tfrac{2}{3}, \qquad q = \tfrac{1}{3}, \qquad r = \tfrac{2}{3}.$$

Finally, conditioning on the first two tosses, each with probability $\tfrac{1}{4}$,

$$\mathrm{P}(C \text{ wins}) = \tfrac{1}{4}(1 + p + q + r) = \tfrac{1}{4}\left(1 + \tfrac{2}{3} + \tfrac{1}{3} + \tfrac{2}{3}\right) = \tfrac{1}{4} \cdot \tfrac{8}{3} = \tfrac{2}{3}.$$

Checks

- In part (ii) the four probabilities add to 1, and the answer is symmetric: swapping H and T exchanges $A \leftrightarrow C$ and $B \leftrightarrow D$, so $\mathrm{P}(A) = \mathrm{P}(C)$ and $\mathrm{P}(B) = \mathrm{P}(D)$ had to happen.
- In part (iii), $q = \tfrac{1}{3} < p$ makes sense: from TH, B is one toss away from winning.

- A computer simulation of 200,000 games of each type gave $\mathrm{P}(A) \approx 0.250$ in part (i), all four players close to 0.25 in part (ii), and $\mathrm{P}(C) \approx 0.667$ in part (iii), matching exact calculations along the Markov chain.

Common slips

- Assuming that two sequences of the same length must be equally likely to appear first. Each of HHT and THH has expected waiting time 8 tosses on its own, yet THH beats HHT three times out of four.
- Forgetting that a non-winning toss can still leave you in a useful state: from HT, the toss H doesn't just "fail", it takes you to TH.
- Writing $p = \frac{1}{2} + \frac{1}{2}q$ without explaining why HTH leaves you in the same position as starting from TH.

Beyond STEP

The method here is a first step into *Markov chains*: a process whose future depends only on its current state, analysed by solving linear equations for absorption probabilities. This particular game is a version of Penney's game, in which, for any sequence of three tosses your opponent chooses, you can always pick one that beats it with probability at least $\frac{2}{3}$, so "which sequence comes first" is not a transitive relation. Grimmett and Welsh, *Probability: An Introduction*, treats Markov chains and absorption probabilities at just this level.

WORKED SOLUTION

STEP II 2015, Question 13

Probability and Statistics • Exponential Distribution Moments

STEP II 2015, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q13

The key idea

The cost is a fixed amount ky plus a random amount that is zero unless the flood overtops the preparations. So everything comes down to the moments of $W = \max(X - y, 0)$, the excess of the flood over the prepared height. The exponential distribution is *memoryless*: given that $X > y$, the excess $X - y$ is again exponential with the same parameter. That turns both the expectation and the variance into one-line calculations.

Solution

Write $p = \mathrm{P}(X > y) = \int_y^\infty \lambda \mathrm{e}^{-\lambda x}\,\mathrm{d}x = \mathrm{e}^{-\lambda y}$, and let

$$W = \begin{cases} X - y & \text{if } X > y, \\ 0 & \text{if } X \le y, \end{cases} \qquad \text{so that} \qquad C = ky + aW.$$

Here $y \ge 0$ is a constant chosen in advance, so ky is not random.

Part (i)

Since $W = 0$ when $X \le y$,

$$\begin{aligned} \mathrm{E}(W) &= \int_y^\infty (x - y)\,\lambda \mathrm{e}^{-\lambda x}\,\mathrm{d}x = \int_0^\infty u\,\lambda \mathrm{e}^{-\lambda(u+y)}\,\mathrm{d}u \qquad (u = x - y) \\ &= \mathrm{e}^{-\lambda y} \int_0^\infty u\,\lambda \mathrm{e}^{-\lambda u}\,\mathrm{d}u = \mathrm{e}^{-\lambda y} \cdot \frac{1}{\lambda}, \end{aligned}$$

using the fact that the mean of an exponential distribution with parameter λ is $\frac{1}{\lambda}$ (or integrate by parts). Therefore

$$\mathrm{E}(C) = ky + a\,\mathrm{E}(W) = ky + \frac{a}{\lambda}\mathrm{e}^{-\lambda y},$$

as required.

Minimising. Let $g(y) = ky + \frac{a}{\lambda}e^{-\lambda y}$ for $y \geq 0$. Then

$$g'(y) = k - ae^{-\lambda y}, \qquad g''(y) = a\lambda e^{-\lambda y} > 0,$$

so g' is strictly increasing, from $g'(0) = k - a$ towards k as $y \to \infty$.

Ask yourself Why can't we just set $g'(y) = 0$ and stop?

Because y is restricted to $y \geq 0$, and the stationary point $e^{-\lambda y} = k/a$ only lies in that range when $k/a \leq 1$. So we split by the value of a/k.

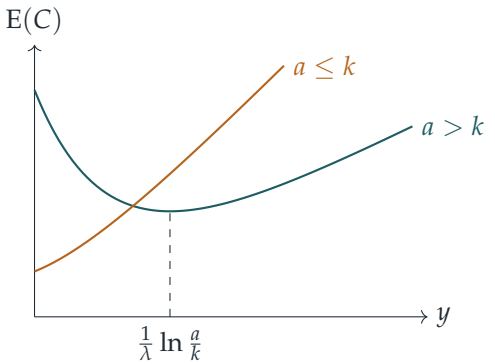
- $a/k > 1$. Then $g'(0) = k - a < 0$, and $g'(y) = 0$ at

$$y = \frac{1}{\lambda}\ln\frac{a}{k} > 0.$$

  Because g' is increasing, g decreases before this point and increases after it, so this is the global minimum on $y \geq 0$. The minimum expected cost is $\frac{k}{\lambda}\left(1 + \ln\frac{a}{k}\right)$.

- $a/k \leq 1$. Then $g'(y) > g'(0) = k - a \geq 0$ for all $y > 0$, so g is strictly increasing on $y \geq 0$ and the minimum is at $y = 0$: make no preparations at all. The expected cost is then $\frac{a}{\lambda}$.

This makes sense: each extra metre of preparation costs k for certain, and saves a only if the flood would have reached it. If $a \leq k$, the saving can never be worth the outlay.



Part (ii)

Adding the constant ky does not change the variance, so $\mathrm{Var}(C) = a^2\,\mathrm{Var}(W)$. For the second moment, with the same substitution $u = x - y$,

$$\mathrm{E}(W^2) = \int_y^\infty (x-y)^2\lambda e^{-\lambda x}\,dx = e^{-\lambda y}\int_0^\infty u^2\lambda e^{-\lambda u}\,du = e^{-\lambda y}\cdot\frac{2}{\lambda^2},$$

since an exponential variable U with parameter λ has $\mathrm{E}(U^2) = \mathrm{Var}(U) + \mathrm{E}(U)^2 = \frac{1}{\lambda^2} + \frac{1}{\lambda^2}$. Hence, with $p = e^{-\lambda y}$,

$$\mathrm{Var}(C) = a^2\left(\frac{2p}{\lambda^2} - \frac{p^2}{\lambda^2}\right) = \frac{a^2}{\lambda^2}e^{-\lambda y}\left(2 - e^{-\lambda y}\right).$$

The amount spent in advance is ky, which increases with y because $k > 0$; so we must show that $\mathrm{Var}(C)$ decreases as y increases. Differentiating,

$$\frac{\mathrm{d}}{\mathrm{d}y}\left(2\mathrm{e}^{-\lambda y} - \mathrm{e}^{-2\lambda y}\right) = -2\lambda\mathrm{e}^{-\lambda y} + 2\lambda\mathrm{e}^{-2\lambda y} = -2\lambda\mathrm{e}^{-\lambda y}\left(1 - \mathrm{e}^{-\lambda y}\right),$$

which is strictly negative for $y > 0$, since then $0 < \mathrm{e}^{-\lambda y} < 1$. So $\mathrm{Var}(C)$ is strictly decreasing in y on $y \geq 0$: the more spent on preparation, the smaller the variance of the cost, which completes the proof. (Equivalently, $p(2-p) = 1 - (1-p)^2$ increases with p on $[0,1]$, and $p = \mathrm{e}^{-\lambda y}$ decreases with y.)

Checks

- At $y = 0$, $C = aX$, so $\mathrm{E}(C) = a/\lambda$ and $\mathrm{Var}(C) = a^2/\lambda^2$, both matching the formulas.
- As $y \to \infty$ the variance tends to 0: a huge flood defence leaves essentially no uncertainty, only the fixed cost ky.
- With $k = 1$, $a = 3$, $\lambda = 0.5$, $y = 1.2$ the formulas give $\mathrm{E}(C) \approx 4.493$ and $\mathrm{Var}(C) \approx 28.67$; a computer simulation of a million years gave 4.496 and 28.68. For these values the optimal height is $2\ln 3 \approx 2.197$, confirmed by a computer search.

Common slips

- Integrating $a(x-y)\lambda\mathrm{e}^{-\lambda x}$ from 0 instead of from y: there is no flood damage when $X \leq y$.
- Forgetting the boundary $y \geq 0$ and quoting $y = \frac{1}{\lambda}\ln\frac{a}{k}$ even when $a < k$, where it is negative.
- Including ky in the variance; it is a constant and contributes nothing.

Beyond STEP

The quantity $\mathrm{E}\big(\max(X-y, 0)\big)$ is exactly the payoff of a call option with strike y, and the optimum $\mathrm{P}(X > y) = k/a$ is the classic "newsvendor" rule from operations research: choose the level at which the marginal cost of preparing equals the expected marginal saving. The memoryless property behind the neat answers is special to the exponential distribution among continuous distributions. Grimmett and Welsh, *Probability: An Introduction*, covers the exponential distribution and expectations of functions of random variables.

WORKED SOLUTION

STEP III 2015, Question 1

Pure • Reduction Formulae and Substitution

STEP III 2015, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q1

The key idea

In part (i), $I_n - I_{n+1}$ combines into a single integral with u^2 on top. Writing u^2 as $u \cdot u$ and integrating by parts brings it back to I_n. In part (ii), the expression $(x - x^{-1})^2$ doesn't change when x is replaced by $1/x$, and that substitution produces the factor x^{-2}. Averaging the two forms gives $1 + x^{-2}$, which is exactly the derivative of $x - x^{-1}$, so the substitution $u = x - x^{-1}$ then works. Part (iii) is part (ii) in disguise once the denominator is written in terms of $x - x^{-1}$.

Solution

Part (i)

Combining the two integrands over a common denominator,

$$I_n - I_{n+1} = \int_0^\infty \frac{(1+u^2)-1}{(1+u^2)^{n+1}}\,\mathrm{d}u = \int_0^\infty u \cdot \frac{u}{(1+u^2)^{n+1}}\,\mathrm{d}u.$$

The second factor has an easy antiderivative:

$$\int \frac{u}{(1+u^2)^{n+1}}\,\mathrm{d}u = -\frac{1}{2n(1+u^2)^n}.$$

So we integrate by parts, differentiating u and integrating $u/(1+u^2)^{n+1}$:

$$I_n - I_{n+1} = \left[-\frac{u}{2n(1+u^2)^n}\right]_0^\infty + \frac{1}{2n}\int_0^\infty \frac{1}{(1+u^2)^n}\,\mathrm{d}u.$$

The boundary term is 0 at $u = 0$. It also tends to 0 as $u \to \infty$, because $n \geq 1$ gives $0 \leq u/(1+u^2)^n \leq u/(1+u^2) < 1/u$. Hence

$$I_n - I_{n+1} = \frac{1}{2n} I_n,$$

as required. Rearranging,

$$I_{n+1} = \frac{2n-1}{2n} I_n \qquad (n \geq 1).$$

For the starting value, $I_1 = \left[\arctan u\right]_0^\infty = \frac{\pi}{2}$. Applying the recurrence repeatedly,

$$I_{n+1} = \frac{\pi}{2} \cdot \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)}.$$

? **Ask yourself** How do we turn a product of odd numbers into factorials?

Multiply the top and bottom by the even product $2 \cdot 4 \cdots (2n) = 2^n n!$. The numerator then becomes $1 \cdot 2 \cdot 3 \cdots (2n) = (2n)!$, and the denominator becomes $(2^n n!)^2$. So

$$I_{n+1} = \frac{\pi}{2} \cdot \frac{(2n)!}{2^{2n}(n!)^2} = \frac{(2n)!\,\pi}{2^{2n+1}(n!)^2},$$

as required. (For a formal proof, use induction: the formula holds for $n = 0$, where it gives $I_1 = \frac{\pi}{2}$. If it holds for I_n, then multiplying by $\frac{2n-1}{2n} = \frac{(2n-1)(2n)}{(2n)^2}$ turns $\frac{(2n-2)!}{2^{2n-1}((n-1)!)^2}$ into $\frac{(2n)!}{2^{2n+1}(n!)^2}$.)

Part (ii)

First equality. Substitute $x = 1/t$ in J, so $\mathrm{d}x = -t^{-2}\,\mathrm{d}t$, and x running from 0 to ∞ corresponds to t running from ∞ to 0. Also

$$\left(x - x^{-1}\right)^2 = \left(t^{-1} - t\right)^2 = \left(t - t^{-1}\right)^2,$$

because squaring removes the sign change. Therefore

$$J = \int_\infty^0 \mathrm{f}\big((t - t^{-1})^2\big)\left(-t^{-2}\right)\mathrm{d}t = \int_0^\infty t^{-2}\,\mathrm{f}\big((t - t^{-1})^2\big)\,\mathrm{d}t,$$

and renaming the dummy variable t as x gives the first form.

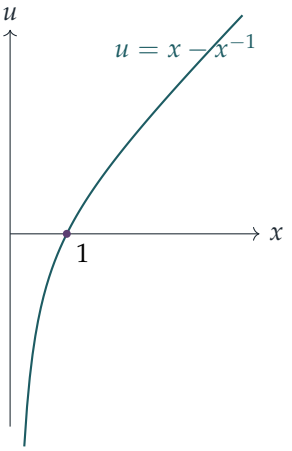
Second equality. Adding the original expression for J to this one and halving,

$$J = \frac{1}{2}\int_0^\infty \left(1 + x^{-2}\right)\mathrm{f}\big((x - x^{-1})^2\big)\,\mathrm{d}x.$$

Third equality. Now put $u = x - x^{-1}$, so that $\mathrm{d}u = (1 + x^{-2})\,\mathrm{d}x$. Since $\mathrm{d}u/\mathrm{d}x > 0$, u increases strictly with x, from $-\infty$ (as $x \to 0^+$) to $+\infty$ (as $x \to \infty$). So the substitution is a one-to-one change of variable on the whole range:

$$J = \frac{1}{2}\int_{-\infty}^\infty \mathrm{f}(u^2)\,\mathrm{d}u = \int_0^\infty \mathrm{f}(u^2)\,\mathrm{d}u.$$

The last step holds because $\mathrm{f}(u^2)$ is an even function of u, so the integrals over $(-\infty, 0)$ and $(0, \infty)$ are equal. This completes part (ii).



The curve passes through $(1,0)$ and rises steadily over $x>0$, covering every real value of u exactly once.

Part (iii)

The aim is to rewrite the integrand in the form $x^{-2}\,\mathrm{f}\big((x-x^{-1})^2\big)$. Since $(x-x^{-1})^2 = x^2-2+x^{-2}$,

$$x^4-x^2+1 = x^2\left(x^2-1+x^{-2}\right) = x^2\left(1+(x-x^{-1})^2\right).$$

Hence

$$\frac{x^{2n-2}}{(x^4-x^2+1)^n} = \frac{x^{2n-2}}{x^{2n}\big(1+(x-x^{-1})^2\big)^n} = x^{-2}\,\mathrm{f}\big((x-x^{-1})^2\big), \qquad \text{where } \mathrm{f}(y) = \frac{1}{(1+y)^n}.$$

By the first and last forms of J in part (ii),

$$\int_0^\infty \frac{x^{2n-2}}{(x^4-x^2+1)^n}\,\mathrm{d}x = \int_0^\infty \frac{1}{(1+u^2)^n}\,\mathrm{d}u = I_n.$$

Replacing n by $n-1$ in the formula from part (i) (valid for $n-1\geq 0$, i.e. every positive integer n) gives

$$\int_0^\infty \frac{x^{2n-2}}{(x^4-x^2+1)^n}\,\mathrm{d}x = \frac{(2n-2)!\,\pi}{2^{2n-1}\big((n-1)!\big)^2}.$$

Checks

- For $n=1$ the answer is $\frac{\pi}{2}$, so $\int_0^\infty \frac{\mathrm{d}x}{x^4-x^2+1} = \frac{\pi}{2} \approx 1.5708$. For $n=2$ it is $\frac{\pi}{4}$ and for $n=3$ it is $\frac{3\pi}{16} \approx 0.5890$. Numerical integration of the original integrals agrees with all of these, and with the cases up to $n=8$.
- The recurrence $I_2 = \frac{1}{2}I_1 = \frac{\pi}{4}$ matches the direct calculation with $u=\tan\theta$: $I_2 = \int_0^{\pi/2}\cos^2\theta\,\mathrm{d}\theta = \frac{\pi}{4}$.

- Part (ii) with $\mathrm{f}(y) = \mathrm{e}^{-y}$ says $\int_0^\infty \mathrm{e}^{-(x-1/x)^2}\,\mathrm{d}x = \int_0^\infty \mathrm{e}^{-u^2}\,\mathrm{d}u = \frac{\sqrt{\pi}}{2} \approx 0.88623$. A numerical evaluation of all four forms gives this value.

Common slips

- Forgetting to check that the boundary term $u/(1+u^2)^n$ vanishes at infinity. It does here, but only because $n \geq 1$.
- Substituting $u = x - x^{-1}$ but keeping the limits 0 and ∞. The new lower limit is $-\infty$, and it is the evenness of $\mathrm{f}(u^2)$ that brings the range back to $(0, \infty)$.
- Using the formula for I_{n+1} without shifting the index: the answer to part (iii) is I_n, not I_{n+1}.

Beyond STEP

The ratio $\frac{(2n)!}{2^{2n}(n!)^2} = \binom{2n}{n}/4^n$ is a central binomial probability, and the recurrence in part (i) is the same one that gives the Wallis product for π. The trick in part (ii) is known as the Cauchy–Schlömilch transformation. It evaluates many definite integrals that have no elementary antiderivative, including Gaussian-type integrals like the one in the checks. Siklos, *Advanced Problems in Mathematics*, has more integrals that yield to a well-chosen substitution, and Bartle and Sherbert, *Introduction to Real Analysis*, makes improper integrals and change of variables rigorous.

WORKED SOLUTION

STEP III 2015, Question 2

Pure • Analysis

STEP III 2015, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q2

This question has a full guided treatment in *Beyond STEP*, Section 2.1, *What "Eventually" Means*: https://beyondstep.keithmcnulty.org/ch02/eventually.html

The key idea

The relation $(s_n) \leq (t_n)$ says "eventually $s_n \leq t_n$": we may ignore any finite number of early terms. So a true statement needs a proof that names a point m beyond which the inequality is guaranteed, and a false one needs a pair of sequences that defeat every possible choice of m. Keep asking: *from which n onwards* does this work?

Solution

Part (i)

True. For a positive integer n we can divide by $n > 0$, so

$$1000n \leq n^2 \iff 1000 \leq n.$$

Hence $1000n \leq n^2$ whenever $n \geq 1000$, and $m = 1000$ works. (No smaller m does, since $1000 \cdot 999 > 999^2$.)

Part (ii)

False. The trap is that "not eventually $s_n \leq t_n$" only tells us that $s_n > t_n$ *infinitely often*, not *eventually*. Two sequences can keep overtaking each other. Take

$$s_n = 2 + (-1)^n, \qquad t_n = 2,$$

so s_n runs $1, 3, 1, 3, \ldots$ and every term of both sequences is positive.

- It is not true that $(s_n) \leq (t_n)$: whatever positive integer m is proposed, there is an even $n \geq m$, and for it $s_n = 3 > 2 = t_n$.

- It is not true that $(t_n) \leq (s_n)$ either: whatever m is proposed, there is an odd $n \geq m$, and for it $t_n = 2 > 1 = s_n$.

So the hypothesis of the statement holds but its conclusion fails.

Part (iii)

True. By hypothesis there is a positive integer m_1 with $s_n \leq t_n$ for all $n \geq m_1$, and a positive integer m_2 with $t_n \leq u_n$ for all $n \geq m_2$. Let

$$m = \max(m_1, m_2).$$

If $n \geq m$ then $n \geq m_1$ and $n \geq m_2$, so both inequalities hold at n, giving

$$s_n \leq t_n \leq u_n.$$

Thus $s_n \leq u_n$ whenever $n \geq m$, which is $(s_n) \leq (u_n)$.

Ask yourself Why not simply take $m = m_1$, or $m = m_1 + m_2$?

Taking $m = m_1$ fails because for $m_1 \leq n < m_2$ we know nothing about t_n versus u_n. We need n to be past *both* thresholds at once, and $\max(m_1, m_2)$ is the smallest choice that guarantees this. ($m_1 + m_2$ also works, but it is wasteful.)

Part (iv)

True, with $m = 4$. A short table shows where to start:

n	1	2	3	4	5	6
n^2	1	4	9	16	25	36
2^n	2	4	8	16	32	64

The inequality holds at $n = 1, 2$, fails at $n = 3$ and holds again from $n = 4$. So $m = 4$ is the smallest possible choice, and we prove by induction that $n^2 \leq 2^n$ for all $n \geq 4$.

Base case. At $n = 4$, $n^2 = 16 = 2^4$.

Inductive step. Suppose $n \geq 4$ and $n^2 \leq 2^n$. Since $n \geq 3$ we have $n^2 - 2n - 1 = (n-1)^2 - 2 \geq 2 > 0$, so $2n + 1 < n^2$. Therefore

$$\begin{aligned}
(n+1)^2 &= n^2 + 2n + 1 \\
&< n^2 + n^2 && \text{since } 2n + 1 < n^2 \\
&= 2n^2 \\
&\leq 2 \cdot 2^n && \text{by the inductive hypothesis} \\
&= 2^{n+1}.
\end{aligned}$$

So the result for n implies it for $n + 1$, and by induction $n^2 \leq 2^n$ for every $n \geq 4$. Hence $(n^2) \leq (2^n)$ with $m = 4$.

Ask yourself Where exactly did the induction need n to be large?

Only in the step $2n+1 < n^2$, which needs $n \geq 3$. That is why the induction cannot run from $n = 1$: the step from $n = 2$ to $n = 3$ uses $5 < 4$, which is false, and indeed $3^2 > 2^3$.

Checks

- Part (i): at $n = 1000$ both sides equal 10^6; at $n = 999$, $999\,000 > 998\,001$.
- Part (iv): checking $n^2 \leq 2^n$ by computer for every n from 4 to 2000 confirms it, and the table above confirms the failure at $n = 3$. For a feel of how fast 2^n wins: $20^2 = 400$ but $2^{20} = 1\,048\,576$.
- Part (ii): any m you try is beaten by the next even n (for one order) and the next odd n (for the other).

Common slips

- Treating "not eventually $\leq$" as "eventually $>$". The negation of "there is an m such that for all $n \geq m\ldots$" is "for every m there is some $n \geq m$ with $s_n > t_n$".
- Starting the induction in (iv) at $n = 1$ or $n = 2$: the inductive step genuinely fails there.
- In (iii), using the same letter m for both hypotheses, as though the two thresholds had to coincide.

Beyond STEP

The relation here is the "eventually" quantifier that runs through all of real analysis: a sequence converges if, for every tolerance, it is *eventually* within that tolerance of its limit. Part (ii) shows that, unlike the order on real numbers, this ordering of sequences is not total: two sequences need not be comparable at all. Computer scientists use the same idea for big-O notation, where (iv) is the familiar fact that exponentials eventually beat polynomials. Abbott's *Understanding Analysis* is a friendly place to see these quantifier arguments made systematic.

WORKED SOLUTION

STEP III 2015, Question 3

Pure • Polar Curves and Area

STEP III 2015, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q3

The key idea

The curve $r = a\sec\theta$ is just the straight line $x = a$, because $r\cos\theta = x$. So $r - a\sec\theta$ measures, along the ray from the origin at angle θ, how far a point is from where that ray meets the line, and the equation says that this distance is b. That is exactly the definition of a conchoid, with the pole as the fixed point. For the sketches, unpick the modulus into $r = a\sec\theta + b$ and $r = a\sec\theta - b$ and ask, for each, which values of θ give $r \geq 0$.

Solution

Throughout, $\sec\theta$ is undefined at $\theta = \pm\frac{\pi}{2}$, so those angles are excluded. Removing the modulus, $(*)$ says

$$r = a\sec\theta + b \qquad \text{or} \qquad r = a\sec\theta - b,$$

and in either case we also need $r \geq 0$.

Part (i)

Why $\sec\theta > 0$. Suppose $\sec\theta < 0$. Then $|\cos\theta| \leq 1$ gives $\sec\theta \leq -1$, so $a\sec\theta \leq -a$. Both possible values of r then satisfy

$$r = a\sec\theta \pm b \leq -a + b < 0,$$

using $a > b$. This contradicts $r \geq 0$, so every point satisfying $(*)$ has $\sec\theta > 0$, that is, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$.

The conchoid. Take A to be the pole O, L to be the line $x = a$ (in polar form $r\cos\theta = a$, or $r = a\sec\theta$), and $d = b$. Note that A is not on L, since $a > 0$.

Let P satisfy $(*)$, with polar angle θ, so $\cos\theta > 0$. The line OP is not parallel to L, and it meets L at the point Q on the same ray with $OQ = a\sec\theta$, i.e. $Q = (a, a\tan\theta)$ in Cartesians. Since O, Q and P all lie on one ray from O, at distances $a\sec\theta$ and r from

O,

$$PQ = |r - a\sec\theta| = b.$$

So P is at distance b from L, measured along a line through $A = O$: it lies on the conchoid of Nicomedes with this L, A and $d = b$.

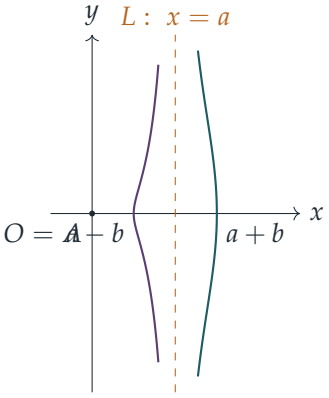
(Conversely, every point of that conchoid satisfies $(*)$: on any line through O meeting L at Q, the two points at distance b from Q are at distances $a\sec\theta \pm b$ from O along the ray OQ, both positive because $a\sec\theta \geq a > b$. So the locus is the whole conchoid.)

Sketch. In Cartesian coordinates, $x = r\cos\theta = a \pm b\cos\theta$ and $y = r\sin\theta$. Multiplying $r - a\sec\theta = \pm b$ by $\cos\theta$ gives $x - a = \pm b\cos\theta = \pm bx/r$, so the curve is

$$(x-a)^2(x^2+y^2) = b^2x^2.$$

There are two branches, both symmetric about the x-axis (replace θ by $-\theta$) and both asymptotic to L as $\theta \to \pm\frac{\pi}{2}$, where $r \to \infty$:

- the outer branch $r = a\sec\theta + b$, with $x = a + b\cos\theta \in (a,\, a+b]$, crossing the axis at $(a+b, 0)$;
- the inner branch $r = a\sec\theta - b$, with $x = a - b\cos\theta \in [a-b,\, a)$, crossing the axis at $(a-b, 0)$, which is to the right of O.



(The picture uses $a = 2$, $b = 1$: the outer branch in teal, the inner in plum.)

Part (ii)

Now $a < b$. Go through the two cases again, checking where $r \geq 0$.

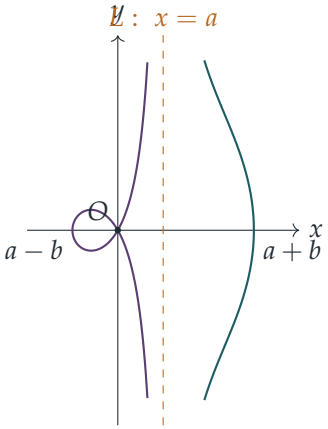
- $r = a\sec\theta + b$. If $\sec\theta > 0$ this is always positive: the same outer branch as before, to the right of L. If $\sec\theta < 0$, then $r \geq 0$ exactly when $-a\sec\theta \leq b$, i.e. $|\sec\theta| \leq b/a$, i.e. $\cos\theta \leq -a/b$.
- $r = a\sec\theta - b$. If $\sec\theta < 0$ this is negative, so there are no points. If $\sec\theta > 0$, then $r \geq 0$ exactly when $\sec\theta \geq b/a$, i.e. $0 < \cos\theta \leq a/b$.

Let $\alpha = \arccos(a/b)$, which lies in $(0, \frac{\pi}{2})$ because $0 < a/b < 1$.

Ask yourself What does the piece with $\sec\theta < 0$ look like?

It is $r = b + a\sec\theta$ for $\cos\theta \le -a/b$, that is, $\pi - \alpha \le |\theta| \le \pi$. At the ends, $\cos\theta = -a/b$ gives $r = b - b = 0$, so it starts and finishes at the origin. At $\theta = \pi$ we get $r = b - a$, the point $(a-b, 0)$ on the negative x-axis. In between $r > 0$, and the curve is symmetric about the x-axis, so this piece is a closed loop to the left of the origin, lying in $a - b \le x \le 0$ (since $x = a + b\cos\theta$).

The inner branch $r = a\sec\theta - b$, for $\alpha \le |\theta| < \frac{\pi}{2}$, also starts at the origin (where $\cos\theta = a/b$) and runs out to the asymptote L as $|\theta| \to \frac{\pi}{2}$, staying in $0 \le x < a$. So the inner branch and the loop together form one curve which crosses itself at O, with tangent directions $\theta = \pm\alpha$ there.



(The picture uses $a = 1$, $b = 2$, so $\alpha = \frac{\pi}{3}$. The outer branch is in teal; the inner branch together with the loop is in plum.)

Area of the loop for $a = 1$, $b = 2$. Here $\cos\theta \le -\frac{1}{2}$ means $\frac{2\pi}{3} \le |\theta| \le \pi$, and on the loop $r = 2 + \sec\theta$. As θ runs through this range the ray sweeps the loop exactly once, so the area is $\frac{1}{2}\int r^2\,\mathrm{d}\theta$. Using the symmetry about the x-axis, we take twice the upper half, $\frac{2\pi}{3} \le \theta \le \pi$:

$$\begin{aligned}\text{Area} &= 2\cdot\frac{1}{2}\int_{2\pi/3}^{\pi}(2+\sec\theta)^2\,\mathrm{d}\theta = \int_{2\pi/3}^{\pi}\left(4 + 4\sec\theta + \sec^2\theta\right)\mathrm{d}\theta \\ &= \Big[\,4\theta + 4\ln|\sec\theta + \tan\theta| + \tan\theta\,\Big]_{2\pi/3}^{\pi}.\end{aligned}$$

The given antiderivative of $\sec\theta$ is valid here because the interval does not contain $\frac{\pi}{2}$. At $\theta = \pi$: $\sec\theta = -1$, $\tan\theta = 0$, so the bracket is $4\pi + 4\ln 1 + 0 = 4\pi$. At $\theta = \frac{2\pi}{3}$: $\sec\theta = -2$, $\tan\theta = -\sqrt{3}$, so the bracket is $\frac{8\pi}{3} + 4\ln(2+\sqrt{3}) - \sqrt{3}$. Subtracting,

$$\text{Area of loop} = \frac{4\pi}{3} + \sqrt{3} - 4\ln\left(2+\sqrt{3}\right).$$

Checks

- Numerically the area is $4.1888 + 1.7321 - 5.2678 \approx 0.6530$. It is positive, as it must be, and small compared with the box $-1 \le x \le 0$, $|y| \le 0.45$ that the loop fits in (whose area is 0.9). A direct numerical integration of $\frac{1}{2}(2+\sec\theta)^2$ over $[\frac{2\pi}{3}, \frac{4\pi}{3}]$ gives 0.65301 as well.
- The loop passes through O and $(-1,0)$: at $\theta = \frac{2\pi}{3}$, $r = 2 - 2 = 0$; at $\theta = \pi$, $r = 2 - 1 = 1$.
- Pick a point on the curve in (i), say $a = 2$, $b = 1$, $\theta = \frac{\pi}{4}$: then $r = 2\sqrt{2} + 1$ and the ray meets L at $(2,2)$, distance $2\sqrt{2}$ from O, so the point is $1 = b$ beyond it.

Common slips

- Forgetting the condition $r \ge 0$. It is what forces $\sec\theta > 0$ in (i), and it fixes the range of θ for the loop in (ii).
- Integrating the loop over $\frac{\pi}{2} < \theta < \pi$, through the asymptote direction, instead of starting where $r = 0$ at $\theta = \frac{2\pi}{3}$.
- Sign errors with $\ln|\sec\theta + \tan\theta|$ in the second quadrant, where both $\sec\theta$ and $\tan\theta$ are negative.

Beyond STEP

The conchoid was invented by Nicomedes around 200 BC for trisecting angles and finding cube roots, problems that are impossible with straight edge and compass alone; the impossibility proofs come from field theory, met in a first course in algebra (see Allenby, *Rings, Fields and Groups*). The self-crossing at the origin in (ii) is a *node*, an example of a singular point on an algebraic curve, here the quartic $(x-a)^2(x^2+y^2) = b^2x^2$. For many more STEP-style curve-sketching problems, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 2015, Question 4

Pure • Roots of Cubic Equations

STEP III 2015, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q4

The key idea

The three conditions in part (iii) look like a trigonometric puzzle, but each is the imaginary part of something: $r_k^n \sin(n\theta_k)$ is the imaginary part of z_k^n when $z_k = r_k e^{i\theta_k}$. So the conditions say that the power sums S_1, S_2, S_3 of three complex numbers are real. Part (ii) turns real power sums into a cubic with real coefficients, and part (i) then hands us a real root.

Solution

Part (i)

Let $f(x) = x^3 + ax^2 + bx + c$ for real x; this is a continuous real function. Put $R = 1 + |a| + |b| + |c|$. For $x \geq R$ we have $x \geq 1$, so $|x| \leq x^2$ and $1 \leq x^2$, giving $|ax^2 + bx + c| \leq (|a| + |b| + |c|)x^2$. Hence

$$f(x) \geq x^3 - (|a| + |b| + |c|)x^2 = x^2\big(x - |a| - |b| - |c|\big) > 0.$$

In the same way, for $x \leq -R$ we get $f(x) \leq -|x|^2\big(|x| - |a| - |b| - |c|\big) < 0$. So $f(-R) < 0 < f(R)$, and by the intermediate value theorem f has a zero between $-R$ and R. That zero is a real root of $(*)$.

(Equivalently: non-real roots of a real polynomial come in conjugate pairs, and three roots cannot all be paired off.)

Part (ii)

Since z_1, z_2, z_3 are the roots,

$$z^3 + az^2 + bz + c = (z - z_1)(z - z_2)(z - z_3),$$

and comparing coefficients gives

$$a = -(z_1 + z_2 + z_3), \qquad b = z_1z_2 + z_2z_3 + z_3z_1, \qquad c = -z_1z_2z_3.$$

So $a = -S_1$. Squaring the sum, $S_1^2 = S_2 + 2(z_1z_2 + z_2z_3 + z_3z_1) = S_2 + 2b$, so

$$a = -S_1, \qquad b = \tfrac{1}{2}\left(S_1^2 - S_2\right).$$

For c, rather than expanding S_1^3, use the fact that each root satisfies the equation: $z_k^3 + az_k^2 + bz_k + c = 0$ for $k = 1, 2, 3$. Adding these three equations,

$$S_3 + aS_2 + bS_1 + 3c = 0.$$

Substituting for a and b and multiplying by 2:

$$\begin{aligned} 6c &= -2S_3 + 2S_1S_2 - S_1\left(S_1^2 - S_2\right) \\ &= -S_1^3 + 3S_1S_2 - 2S_3, \end{aligned}$$

as required.

Part (iii)

Define $z_k = r_k e^{i\theta_k}$ for $k = 1, 2, 3$. By de Moivre's theorem $z_k^n = r_k^n\left(\cos n\theta_k + i \sin n\theta_k\right)$, so the three given conditions say exactly

$$\operatorname{Im} S_1 = \operatorname{Im} S_2 = \operatorname{Im} S_3 = 0,$$

where S_1, S_2, S_3 are the power sums of z_1, z_2, z_3 as in part (ii). So S_1, S_2, S_3 are all real.

Ask yourself Part (ii) started from a cubic. Where is the cubic here?

We build it: let a, b, c be defined by $(z - z_1)(z - z_2)(z - z_3) = z^3 + az^2 + bz + c$. Then z_1, z_2, z_3 are the roots of this cubic, so part (ii) applies and gives

$$a = -S_1, \qquad b = \tfrac{1}{2}\left(S_1^2 - S_2\right), \qquad c = \tfrac{1}{6}\left(-S_1^3 + 3S_1S_2 - 2S_3\right),$$

all of which are real. By part (i) the cubic has a real root, and its only roots are z_1, z_2, z_3, so some z_k is real. Now $z_k = r_k e^{i\theta_k}$ with $r_k > 0$ is real only if $\sin\theta_k = 0$, that is θ_k is a multiple of π; the only such value in $-\pi < \theta_k < \pi$ is $\theta_k = 0$. So $\theta_k = 0$ for at least one k.

Now suppose $\theta_1 = 0$, so $z_1 = r_1$ is real and non-zero. Then

$$z_2 + z_3 = S_1 - z_1 \quad \text{and} \quad z_2z_3 = \frac{-c}{z_1}$$

are both real. Hence z_2 and z_3 are the roots of the quadratic $z^2 - (z_2 + z_3)z + z_2z_3 = 0$, which has real coefficients. There are two cases.

- If the roots are non-real, they are complex conjugates: $z_3 = \overline{z_2} = r_2 e^{-i\theta_2}$. Since $-\theta_2$ also lies in $(-\pi, \pi)$ and the argument in that interval is unique, $\theta_3 = -\theta_2$ (and $r_3 = r_2$).
- If the roots are real, then exactly as above (each has positive modulus and argument in $(-\pi, \pi)$) we get $\theta_2 = \theta_3 = 0$, so again $\theta_2 = -\theta_3$.

In either case $\theta_2 = -\theta_3$, as required.

Checks

- Part (ii) with roots $1, 2, 3$: $S_1 = 6$, $S_2 = 14$, $S_3 = 36$, and $-216 + 252 - 72 = -36 = 6c$ with $c = -6$, as it should be for $(z-1)(z-2)(z-3)$.
- Part (iii) the other way round: with $\theta_1 = 0$, $\theta_3 = -\theta_2$ and $r_3 = r_2$, the second and third terms of each sum cancel and the first is zero, so all three conditions hold. A computer search solving the three conditions numerically from thousands of random starting points found only solutions with one angle zero and the other two opposite.

Common slips

- In part (i), saying "a cubic always crosses the axis" without justification; the sign change at large $|x|$ needs a line of argument.
- In part (iii), forgetting that $\theta_k = \pm\pi$ would also make z_k real; it is the open interval $(-\pi, \pi)$ that rules this out.
- Concluding $\theta_2 = -\theta_3$ only from the conjugate case and overlooking the case where z_2, z_3 are both real.

Beyond STEP

The coefficients of a polynomial are the elementary symmetric functions of its roots, and the identities in part (ii) are the first of Newton's identities linking them to the power sums. A first course in algebra shows that every symmetric polynomial in the roots is a polynomial in the elementary symmetric functions, which is the starting point of Galois theory. Allenby's *Rings, Fields and Groups* is a good place to read about this.

WORKED SOLUTION

STEP III 2015, Question 5

Pure • Irrationality by Descent

STEP III 2015, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q5

This question has a full guided treatment in *Beyond STEP*, Section 1.3, *The Smallest Counterexample*: https://beyondstep.keithmcnulty.org/ch01/well-ordering.html

The key idea

This is proof by *infinite descent*: if $\sqrt{2}$ were rational there would be a smallest positive integer that turns $\sqrt{2}$ into an integer when multiplied by it, and we then build a smaller one. Multiplying by $\sqrt{2}-1$ does the job, because it is a number between 0 and 1 that keeps us inside the set. For the cube root of 2, the same trick needs a set that also keeps track of $2^{2/3}$, which is why the question first asks you to tie the two numbers together.

Solution

Part (i)

Step 3. By the assumption in step 1, $\sqrt{2} = \frac{p}{q}$ for some integers p and q with $q \neq 0$. Since $\sqrt{2} > 0$ we may take p and q both positive. Then

$$q\sqrt{2} = p$$

is an integer, and q is a positive integer, so $q \in S$. Hence S is not empty.

Step 5. Since $k \in S$, the number $m = k\sqrt{2}$ is an integer. We need to check two things about $n = (\sqrt{2}-1)k$.

- n is a positive integer: $n = k\sqrt{2} - k = m - k$ is a difference of integers, and $n > 0$ because $k > 0$ and $\sqrt{2} > 1$.
- $n\sqrt{2}$ is an integer:

  $$n\sqrt{2} = (\sqrt{2}-1)k\sqrt{2} = 2k - k\sqrt{2} = 2k - m,$$

  again a difference of integers.

So $(\sqrt{2}-1)k \in S$.

Step 6. Since $1 < \sqrt{2} < 2$ (as $1 < 2 < 4$), we have $0 < \sqrt{2}-1 < 1$, and so, as $k > 0$,

$$(\sqrt{2}-1)k < k.$$

Step 5 therefore gives a positive integer in S that is smaller than k, contradicting the choice of k in step 4 as the smallest element of S. (Step 4 is legitimate, since every non-empty set of positive integers has a smallest element, and step 3 showed S is non-empty.) The only assumption made was step 1, so it must be false: $\sqrt{2}$ is irrational.

Part (ii)

Write $a = 2^{1/3}$, so $a^2 = 2^{2/3}$ and $a^3 = 2$.

The equivalence. If a is rational then so is a^2, a product of rationals. Conversely, if a^2 is rational, then

$$a = \frac{a^3}{a^2} = \frac{2}{a^2}$$

is rational, since $a^2 \neq 0$. So $2^{1/3}$ is rational if and only if $2^{2/3}$ is.

Irrationality. Suppose, for contradiction, that a is rational. By the equivalence, a^2 is rational as well. Define T to be the set of positive integers n such that *both* na and na^2 are integers.

> **Ask yourself** Why not just copy part (i) with $S = \{n : na \text{ is an integer}\}$?

In part (i) the check that $(\sqrt{2}-1)k \in S$ needed $k(\sqrt{2})^2 = 2k$ to be an integer, which is automatic. Here, multiplying $(a-1)k$ by a gives $ka^2 - ka$, and nothing in the condition "ka is an integer" says that ka^2 is an integer. So we build that into the set, and this is where we need the first half of the part.

T is non-empty. Write $a = \frac{p}{q}$ with p, q positive integers. Then $q^2 a = pq$ and $q^2 a^2 = p^2$ are integers, so $q^2 \in T$.

Descent. Let k be the smallest element of T, so $u = ka$ and $v = ka^2$ are integers. Consider $n = (a-1)k$.

- $n = u - k$ is an integer, and $n > 0$ since $a > 1$ (as $1^3 < 2$).
- $na = ka^2 - ka = v - u$ is an integer.
- $na^2 = ka^3 - ka^2 = 2k - v$ is an integer, using $a^3 = 2$.

So $n \in T$. But $a < 2$ (as $2^3 = 8 > 2$), so $0 < a - 1 < 1$ and $n = (a-1)k < k$, contradicting the minimality of k. Hence $2^{1/3}$ is irrational, and by the equivalence $2^{2/3}$ is irrational too.

Checks

- Try the descent on a near miss. The fraction $\frac{7}{5}$ is close to $\sqrt{2}$; the map $(k, m) \mapsto (m-k, 2k-m)$ from step 5 sends $(5,7)$ to $(2,3)$, and $3^2 - 2\cdot 2^2 = 1 = -(7^2 - 2\cdot 5^2)$. The quantity $m^2 - 2k^2$ only changes sign, so if it were 0 (that is, $m = k\sqrt{2}$) it would stay 0 while k kept shrinking, which is impossible.
- Numerically, $\sqrt{2} - 1 \approx 0.414$ and $2^{1/3} - 1 \approx 0.260$, both strictly between 0 and 1, as the descent needs.
- A computer search over all $n \leq 2 \times 10^5$ finds no n with $2n^2$ a perfect square, or with $2n^3$ or $4n^3$ a perfect cube.

Common slips

- Forgetting to show that $(\sqrt{2}-1)k$ is *positive*: S contains only positive integers, so this is needed.
- In part (ii), using a set defined by na alone and then asserting without reason that ka^2 is an integer.
- Not saying why the smallest element k exists: it needs S (or T) to be non-empty, which is the point of step 3.

Beyond STEP

The argument is really about the well-ordering of the positive integers, the principle behind induction, and it generalises: if $\sqrt[r]{N}$ is not an integer then it is irrational, as one sees by taking the set of n for which $n\alpha, n\alpha^2, \ldots, n\alpha^{r-1}$ are all integers and multiplying its least element by $\alpha - \lfloor \alpha \rfloor$, which lies strictly between 0 and 1. In a first course in algebra this becomes the statement that the integers form an *integrally closed* ring: a rational root of a monic polynomial with integer coefficients must be an integer. Davenport's *The Higher Arithmetic* is a lovely place to read more about the integers and their properties.

WORKED SOLUTION

STEP III 2015, Question 6

Pure • Symmetric Functions of Roots

STEP III 2015, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q6

The key idea

Everything about the pair w, z is controlled by its sum and its product, because w and z are the two roots of $t^2 - (w+z)t + wz = 0$. Part (i) is then a statement about when a quadratic with real coefficients has real roots, and the condition $u^2 \leq 2v$ is the discriminant in disguise. In part (ii) the first two equations fix the product wz, and the third then becomes a cubic in u alone, with a root you can spot.

Solution

Part (i)

The identity that drives everything is

$$2v - u^2 = 2(w^2 + z^2) - (w+z)^2 = w^2 - 2wz + z^2 = (w-z)^2.$$

We also note that $wz = \frac{1}{2}\big((w+z)^2 - (w^2+z^2)\big) = \frac{1}{2}(u^2 - v)$.

If w and z are real: then $u = w+z$ and $v = w^2 + z^2$ are certainly real, and $2v - u^2 = (w-z)^2 \geq 0$ because $w - z$ is real. So $u^2 \leq 2v$.

Conversely, suppose u and v are real and $u^2 \leq 2v$. Then $p = wz = \frac{1}{2}(u^2 - v)$ is real too. Since

$$(t-w)(t-z) = t^2 - ut + p,$$

the numbers w and z are precisely the two roots of the quadratic $t^2 - ut + p = 0$, whose coefficients are real. Its discriminant is

$$u^2 - 4p = u^2 - 2(u^2 - v) = 2v - u^2 \geq 0,$$

so both roots are real. Hence w and z are real, which completes the proof of the equivalence.

? **Ask yourself** Why isn't "u and v real" enough on its own?

Because a conjugate pair also has real sum and real sum of squares. For example $w = 1 + 2\mathrm{i}$, $z = 1 - 2\mathrm{i}$ give $u = 2$ and $v = -6$, both real, but $u^2 = 4 > -12 = 2v$. The inequality is exactly what rules out a non-real conjugate pair.

Part (ii)

Reducing to a cubic. The first equation says $u = w + z$. Writing $v = w^2 + z^2$, the second says $v = u^2 - \frac{2}{3}$, so

$$wz = \tfrac{1}{2}(u^2 - v) = \tfrac{1}{3}.$$

For the cubes we use the standard identity

$$w^3 + z^3 = (w+z)^3 - 3wz(w+z) = u^3 - u.$$

The third equation therefore becomes $u^3 - u - \lambda u = -\lambda$, that is

$$u^3 - (1+\lambda)u + \lambda = 0.$$

Substituting $u = 1$ gives $1 - 1 - \lambda + \lambda = 0$, so $u - 1$ is a factor, and

$$u^3 - (1+\lambda)u + \lambda = (u-1)(u^2 + u - \lambda).$$

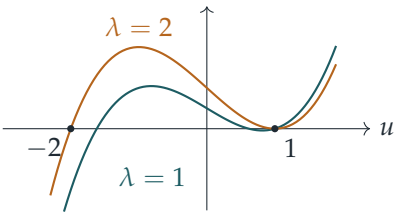
Counting the roots. The quadratic factor has discriminant $1 + 4\lambda > 0$, since $\lambda > 0$. So it has two distinct real roots,

$$u = \frac{-1 \pm \sqrt{1+4\lambda}}{2}.$$

One of these equals 1 exactly when $1^2 + 1 - \lambda = 0$, that is when $\lambda = 2$. So:

- if $\lambda \neq 2$, the possible values of u are 1 and $\frac{1}{2}\left(-1 \pm \sqrt{1+4\lambda}\right)$: three distinct values, all real;
- if $\lambda = 2$, the cubic is $(u-1)^2(u+2)$ and there are only two values, $u = 1$ and $u = -2$.

Each of these values really does occur: given u, take w and z to be the roots of $t^2 - ut + \frac{1}{3} = 0$. Then $w + z = u$ and $wz = \frac{1}{3}$, and reversing the steps above shows that all three equations hold. So the exceptional value is $\lambda = 2$.



The sketch (vertically scaled) shows $u^3 - (1+\lambda)u + \lambda$ for $\lambda = 1$, with three separate crossings, and for $\lambda = 2$, where the curve just touches the axis at $u = 1$.

Are w and z necessarily real? No. Since u is real and $v = u^2 - \frac{2}{3}$ is real, part (i) tells us that w and z are real if and only if $u^2 \le 2v = 2u^2 - \frac{4}{3}$, that is, if and only if $u^2 \ge \frac{4}{3}$. The root $u = 1$ is present for every λ, and $1 < \frac{4}{3}$, so it always gives non-real w and z.

Explicitly, $u = 1$ gives $t^2 - t + \frac{1}{3} = 0$, so

$$w = \frac{1}{2} + \frac{\mathrm{i}}{2\sqrt{3}}, \qquad z = \frac{1}{2} - \frac{\mathrm{i}}{2\sqrt{3}}.$$

As a check: $w + z = 1 = u$; $wz = \frac{1}{4} + \frac{1}{12} = \frac{1}{3}$, so $w^2 + z^2 = 1 - \frac{2}{3} = u^2 - \frac{2}{3}$; and $w^3 + z^3 = u^3 - 3wz\,u = 1 - 1 = 0$, so $w^3 + z^3 - \lambda u = -\lambda$ for every $\lambda > 0$. This is a counterexample: the three values of u are all real, but w and z need not be.

Checks

- With $\lambda = 1$ the values of u are 1 and $\frac{1}{2}(-1 \pm \sqrt{5}) \approx 0.618$, -1.618, and each satisfies $u^3 - 2u + 1 = 0$.
- With $\lambda = 2$ the cubic $u^3 - 3u + 2$ vanishes at $u = -2$ ($-8 + 6 + 2 = 0$), and both it and its derivative $3u^2 - 3$ vanish at $u = 1$, confirming the double root.
- A computer check over many values of λ built w and z from each u and confirmed all three equations to rounding error; it also found three distinct values of u except at $\lambda = 2$.

Common slips

- In part (i), proving only one direction. The converse needs the product wz to be real, which is where the reality of v is used.
- Forgetting to check that the quadratic's roots are different from 1, and so missing the exceptional value $\lambda = 2$.
- Assuming that because u is real, w and z must be. Real symmetric functions of w and z are equally consistent with a conjugate pair.

Beyond STEP

The idea that a pair (or any set) of numbers is determined by its symmetric functions is the starting point of the theory of symmetric polynomials: every symmetric polynomial in w and z can be written in terms of $w + z$ and wz, and Newton's identities relate power sums like $w^3 + z^3$ to them. The same reasoning, with discriminants, decides when all roots of a real polynomial are real. Needham's *Visual Complex Analysis* is a beautiful companion for complex numbers, and Allenby's *Rings, Fields and Groups* takes symmetric polynomials further.

WORKED SOLUTION

STEP III 2015, Question 7

Pure • Differential Operators

STEP III 2015, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q7

The key idea

Powers of x are eigenfunctions of $\mathrm{D} = x\frac{\mathrm{d}}{\mathrm{d}x}$: applying D to x^a just multiplies it by a. So D^n acts on a polynomial term by term, multiplying the coefficient of x^r by r^n, which is exactly where the r^n in the final sum comes from. For the divisibility, notice that each application of D can lower the power of a factor $(1-x)$ by at most one.

Solution

The opening result

For any constant a (and $x > 0$ if a is not an integer),

$$\mathrm{D}x^a = x \cdot ax^{a-1} = ax^a.$$

Applying D again and using this result with the constant a pulled out,

$$\mathrm{D}^2x^a = \mathrm{D}\left(ax^a\right) = a\,\mathrm{D}x^a = a^2x^a,$$

as required. Since D is linear ($\mathrm{D}(\alpha f + \beta g) = \alpha\,\mathrm{D}f + \beta\,\mathrm{D}g$ for constants α, β), repeating the step gives, by induction,

$$\mathrm{D}^n x^a = a^n x^a \qquad (n \geq 1).$$

Part (i)

Write $\mathrm{P}(x) = \sum_{k=0}^{r} c_k x^k$ with $c_r \neq 0$. By linearity and the result above,

$$\mathrm{D}^n\mathrm{P}(x) = \sum_{k=0}^{r} c_k\, k^n x^k.$$

This is a polynomial with no terms beyond x^r, and its x^r coefficient is $r^n c_r$. Because $r \geq 1$, we have $r^n \neq 0$, and $c_r \neq 0$, so $r^n c_r \neq 0$. Hence $\mathrm{D}^n\mathrm{P}(x)$ is a polynomial of degree exactly r.

(The condition $r \geq 1$ matters: a constant polynomial is sent to 0, because $0^n = 0$.)

Part (ii)

Fix m. We prove by induction on n that for $1 \le n \le m$,

$$\mathrm{D}^n(1-x)^m = (1-x)^{m-n}\,\mathrm{Q}_n(x) \quad \text{for some polynomial } \mathrm{Q}_n.$$

Base case. $\mathrm{D}(1-x)^m = x\cdot\big(-m(1-x)^{m-1}\big) = (1-x)^{m-1}\cdot(-mx)$, so $\mathrm{Q}_1(x) = -mx$.

Inductive step. Suppose the statement holds for some n with $n < m$, so that $m - n \ge 1$. By the product rule,

$$\begin{aligned}\mathrm{D}^{n+1}(1-x)^m &= x\frac{\mathrm{d}}{\mathrm{d}x}\Big[(1-x)^{m-n}\mathrm{Q}_n(x)\Big]\\ &= x\Big[-(m-n)(1-x)^{m-n-1}\mathrm{Q}_n(x) + (1-x)^{m-n}\mathrm{Q}_n'(x)\Big]\\ &= (1-x)^{m-n-1}\cdot x\Big[(1-x)\mathrm{Q}_n'(x) - (m-n)\mathrm{Q}_n(x)\Big].\end{aligned}$$

The bracket is a polynomial, so this has the required form with $\mathrm{Q}_{n+1}(x) = x\big[(1-x)\mathrm{Q}_n'(x) - (m-n)\mathrm{Q}_n(x)\big]$.

Ask yourself Where did we use $n < m$ in the inductive step?

Taking out the factor $(1-x)^{m-n-1}$ only makes sense as a polynomial factor if $m - n - 1 \ge 0$, that is, $n < m$. At $n = m$ there is no factor of $(1-x)$ left to lose, so the induction (correctly) stops there.

So for every positive integer $n < m$, $\mathrm{D}^n(1-x)^m$ is divisible by $(1-x)^{m-n}$.

Part (iii)

Expand by the binomial theorem:

$$(1-x)^m = \sum_{r=0}^{m}(-1)^r\binom{m}{r}x^r.$$

Applying D^n term by term, exactly as in part (i),

$$\mathrm{D}^n(1-x)^m = \sum_{r=0}^{m}(-1)^r\binom{m}{r}r^n x^r.$$

By part (ii), with $n < m$, the left-hand side equals $(1-x)^{m-n}\mathrm{Q}_n(x)$ with $m - n \ge 1$, so it vanishes at $x = 1$. Setting $x = 1$ on the right-hand side therefore gives

$$\sum_{r=0}^{m}(-1)^r\binom{m}{r}r^n = 0,$$

as required. (The $r = 0$ term is $0^n = 0$, consistent with D killing the constant term.)

Checks

Take $m = 3$, $n = 2$: the sum is $0 - 3 \cdot 1 + 3 \cdot 4 - 1 \cdot 9 = 0$. Take $m = 4$, $n = 3$: $0 - 4 + 6 \cdot 8 - 4 \cdot 27 + 64 = 0$. The identity was also confirmed by computer for all $1 \leq n < m \leq 19$, together with the fact that $(1-x)^{m-n}$ is the *exact* power dividing $\mathrm{D}^n(1-x)^m$ for $m \leq 11$. As a contrast, when $n = m$ the sum is no longer zero: for $m = 2$ it is $0 - 2 + 4 = 2 = 2!$, and in general it is $(-1)^m m!$.

Common slips

- Writing $\mathrm{D}^2 x^a$ as $x^2 \frac{\mathrm{d}^2}{\mathrm{d}x^2} x^a = a(a-1)x^a$: D^2 is D applied twice, not x^2 times the second derivative.
- In part (i), forgetting to say why the leading coefficient $r^n c_r$ is non-zero; this is where $r \geq 1$ is used.
- In part (iii), substituting $x = 1$ into $(1-x)^m$ *before* applying D^n; the point is that the factor $(1-x)$ survives $n < m$ applications.

Beyond STEP

The operator $x\frac{\mathrm{d}}{\mathrm{d}x}$ is sometimes called the Euler operator, and x^a is an eigenfunction with eigenvalue a; it is the natural tool for Euler–Cauchy equations such as $x^2y'' + bxy' + cy = 0$, which become constant-coefficient equations under $x = e^t$. The identity in part (iii) says that the mth finite difference of a polynomial of degree less than m is zero, and it is closely linked to Stirling numbers and inclusion–exclusion counts of surjections. Boyce and DiPrima, *Elementary Differential Equations*, covers Euler equations; Siklos, *Advanced Problems in Mathematics*, has more problems in this style.

WORKED SOLUTION

STEP III 2015, Question 8

Pure • Polar Differential Equations

STEP III 2015, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q8

This question has a full guided treatment in *Beyond STEP*, Section 8.4, *Spiralling onto a Circle*: https://beyondstep.keithmcnulty.org/ch08/limit-cycle.html

The key idea

Both differential equations are built from the combinations $x\,\mathrm{d}y - y\,\mathrm{d}x$ and $x\,\mathrm{d}x + y\,\mathrm{d}y$, and these are exactly what polar coordinates simplify: they become $r^2\,\mathrm{d}\theta$ and $r\,\mathrm{d}r$. So we treat θ as the parameter along the curve, write $\mathrm{d}y/\mathrm{d}x$ as a ratio of θ-derivatives, and let the trigonometry collapse. In part (ii) the extra terms turn the equation into a Bernoulli equation for r, which the substitution $u = r^{-2}$ makes linear.

Solution

Write r' for $\mathrm{d}r/\mathrm{d}\theta$, $c = \cos\theta$ and $s = \sin\theta$. Along a curve $r = r(\theta)$,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = r'c - rs, \qquad \frac{\mathrm{d}y}{\mathrm{d}\theta} = r's + rc, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}\theta}{\mathrm{d}x/\mathrm{d}\theta}.$$

Multiplying either equation through by $\mathrm{d}x/\mathrm{d}\theta$, an equation of the form $P\dfrac{\mathrm{d}y}{\mathrm{d}x} = Q$ becomes

$$P\frac{\mathrm{d}y}{\mathrm{d}\theta} - Q\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0.$$

Two identities do all the work:

$$x\frac{\mathrm{d}y}{\mathrm{d}\theta} - y\frac{\mathrm{d}x}{\mathrm{d}\theta} = rc(r's + rc) - rs(r'c - rs) = r^2(c^2 + s^2) = r^2,$$
$$x\frac{\mathrm{d}x}{\mathrm{d}\theta} + y\frac{\mathrm{d}y}{\mathrm{d}\theta} = rc(r'c - rs) + rs(r's + rc) = rr'.$$

Part (i)

Here $P = y + x$ and $Q = y - x$, so

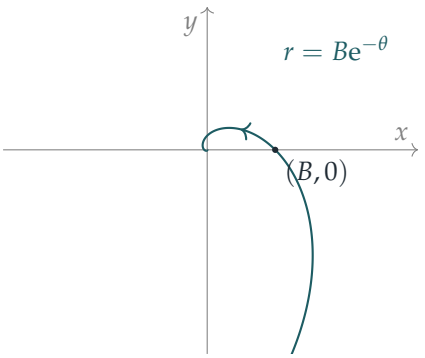
$$\begin{aligned}(y+x)\frac{\mathrm{d}y}{\mathrm{d}\theta} - (y-x)\frac{\mathrm{d}x}{\mathrm{d}\theta} &= \left(x\frac{\mathrm{d}y}{\mathrm{d}\theta} - y\frac{\mathrm{d}x}{\mathrm{d}\theta}\right) + \left(x\frac{\mathrm{d}x}{\mathrm{d}\theta} + y\frac{\mathrm{d}y}{\mathrm{d}\theta}\right)\\ &= r^2 + rr'.\end{aligned}$$

So the equation is $r(r' + r) = 0$, and since $r > 0$ we may divide by r to get

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} + r = 0,$$

as required.

This is separable (or just recognise exponential decay): $r = B\mathrm{e}^{-\theta}$ with $B > 0$. This is an *equiangular* (logarithmic) spiral. As θ increases the curve winds anticlockwise into the origin, never reaching it; as θ decreases it winds outwards without bound. Since $r/r' = -1$, the curve cuts every ray from the origin at the same angle, 45° to the inward radius. Different B give rotations of the same spiral, because $B\mathrm{e}^{-\theta} = \mathrm{e}^{-(\theta - \ln B)}$.



The arrow shows the direction of increasing θ. The spiral shrinks by a factor $\mathrm{e}^{2\pi} \approx 535$ on each turn, so only about one turn is visible at any scale; the rest is packed tightly around the origin.

Part (ii)

Now $P = y + x - x(x^2 + y^2)$ and $Q = y - x - y(x^2 + y^2)$, with $x^2 + y^2 = r^2$. The new terms contribute

$$-r^2\left(x\frac{\mathrm{d}y}{\mathrm{d}\theta} - y\frac{\mathrm{d}x}{\mathrm{d}\theta}\right) = -r^2 \cdot r^2 = -r^4,$$

so, using part (i) for the old terms, the equation becomes

$$r^2 + rr' - r^4 = 0 \quad \Longrightarrow \quad \frac{\mathrm{d}r}{\mathrm{d}\theta} = r^3 - r,$$

again dividing by $r > 0$.

Ask yourself How do we deal with the r^3 term, which stops the equation being linear?

This is a Bernoulli equation. The target answer is in terms of r^2, which suggests working with $u = r^{-2}$. Then

$$\frac{\mathrm{d}u}{\mathrm{d}\theta} = -2r^{-3}\frac{\mathrm{d}r}{\mathrm{d}\theta} = -2r^{-3}(r^3 - r) = -2 + 2u.$$

So $u' - 2u = -2$, a linear equation with constant coefficients: $u = 1$ is a particular solution and $A\mathrm{e}^{2\theta}$ solves the homogeneous equation, giving

$$\frac{1}{r^2} = 1 + A\mathrm{e}^{2\theta} \qquad \Longrightarrow \qquad r^2 = \frac{1}{1 + A\mathrm{e}^{2\theta}},$$

which is the required form. (Separating variables and using partial fractions on $\dfrac{1}{r(r-1)(r+1)}$ gives the same result with more effort.) The curve exists only where $1 + A\mathrm{e}^{2\theta} > 0$.

The shapes of the solutions

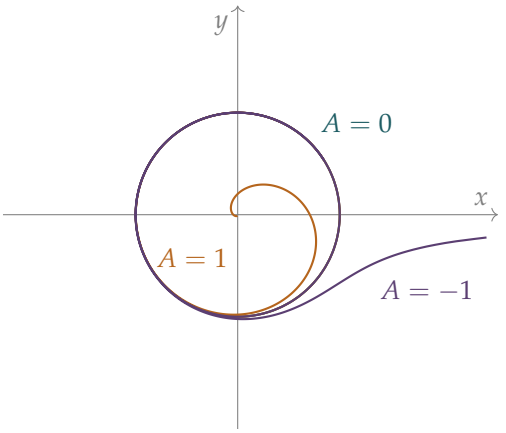
First note that if $A = \pm\mathrm{e}^{-2\alpha}$ then $A\mathrm{e}^{2\theta} = \pm\mathrm{e}^{2(\theta-\alpha)}$, so the curve for A is the curve for $A = \pm 1$ rotated through α. So there are really only three shapes, according to the sign of A.

- $A = 0$: $r = 1$, the unit circle.
- $A > 0$: $1 + A\mathrm{e}^{2\theta} > 1$, so $r < 1$ and the curve is defined for every θ. Since $r' = r(r^2 - 1) < 0$, r decreases steadily. As $\theta \to -\infty$, $r \to 1$, so the curve winds out towards the unit circle from inside; as $\theta \to +\infty$, $r \approx A^{-1/2}\mathrm{e}^{-\theta} \to 0$, so it spirals into the origin, looking more and more like the spiral of part (i).
- $A < 0$: we need $\mathrm{e}^{2\theta} < -1/A$, that is $\theta < \theta_0 = \frac{1}{2}\ln(-1/A)$. Now $r > 1$ and $r' > 0$. As $\theta \to -\infty$, $r \to 1$ from above, so the curve winds in towards the unit circle from outside; as $\theta \to \theta_0^-$, $r \to \infty$. Near θ_0, $1 + A\mathrm{e}^{2\theta} \approx 2(\theta_0 - \theta)$ (since $A\mathrm{e}^{2\theta_0} = -1$), so $r \approx \big(2(\theta_0 - \theta)\big)^{-1/2}$, and the distance from the curve to the line $\theta = \theta_0$ is

  $$r\sin(\theta_0 - \theta) \approx r(\theta_0 - \theta) \approx \frac{1}{2r} \to 0.$$

  So the curve goes off to infinity with the line through the origin at angle θ_0 as an asymptote, approaching it from the clockwise side. For $A = -1$, $\theta_0 = 0$ and the curve ends up close to $y = -\frac{1}{2x}$ for large x.

The unit circle is a limit cycle: every other solution spirals towards it as $\theta \to -\infty$, and away from it as θ increases.



The plum curve ($A = -1$) runs off to infinity along the positive x-axis, just below it.

Checks

- Substituting $r = Be^{-\theta}$ back into the original equation of (i) at a few points (convert to x, y and $\mathrm{d}y/\mathrm{d}x$) gives zero; likewise for the formula of (ii) with several values of A of both signs.
- $A = 0$ gives $r = 1$, and indeed $r' = r^3 - r$ is satisfied by the constant $r = 1$; in x, y, the circle $x^2 + y^2 = 1$ makes the extra terms cancel the old ones exactly.
- Starting from $r(0) = \frac{1}{2}$ (so $A = 3$), the formula gives $r(2) = 1/\sqrt{1 + 3e^4} \approx 0.0779$, which agrees with a step-by-step numerical solution of $r' = r^3 - r$ to many decimal places.
- For $A = -1$ at $\theta = -10^{-4}$, $r \approx 70.71$ and the distance to the x-axis is $0.00707 \approx 1/(2r)$, confirming the asymptote.

Common slips

- Forgetting that $\mathrm{d}y/\mathrm{d}x$ must be found as $(\mathrm{d}y/\mathrm{d}\theta)/(\mathrm{d}x/\mathrm{d}\theta)$ with r varying, i.e. dropping the r' terms.
- Missing the restriction $1 + Ae^{2\theta} > 0$ when $A < 0$, and so not noticing that these curves go off to infinity at a finite angle.
- Sketching only one curve for each sign of A without saying that the others are rotations of it.

Beyond STEP

The equation $r' = r^3 - r$ is a polar picture of a planar dynamical system, and the unit circle is a *limit cycle*: an isolated closed orbit that nearby trajectories spiral towards or

away from. Systems of this kind, where the radial equation decouples, are the standard first examples in courses on nonlinear differential equations and phase-plane analysis, leading to the Poincaré–Bendixson theorem. Boyce and DiPrima, *Elementary Differential Equations*, has a good chapter on nonlinear systems and limit cycles.

WORKED SOLUTION

STEP III 2015, Question 9

Mechanics • Nonlinear Elastic Oscillations

STEP III 2015, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q9

The key idea

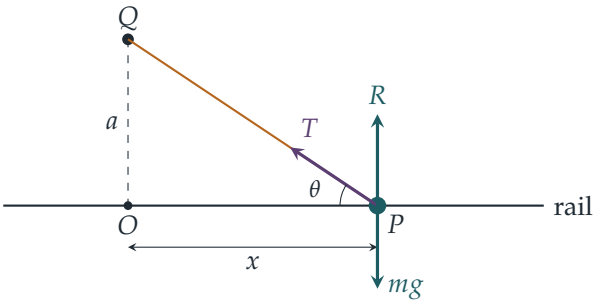
The rail is smooth and the normal reaction does no work, so energy is conserved, and that gives the first-order equation straight away. The string is never slack, because $PQ \geq a$ at every point of the rail. Turning points are where $\dot{x} = 0$. The period is four times the time to go from the centre out to x_0. For small v the particle stays close to its start, so $\sqrt{x^2 + a^2} - a \approx x^2/(2a)$ and the substitution $x = x_0 u$ tidies up the integral.

Solution

The energy equation

Model P as a particle and the string as light and elastic, obeying Hooke's law. Take the rail as the x-axis with P starting at the origin, the foot of the perpendicular from Q. When P is at x, its distance from Q is $PQ = \sqrt{x^2 + a^2} \geq a$, so the string is taut, with extension

$$e = \sqrt{x^2 + a^2} - a \quad (\geq 0).$$



The forces on P are its weight, the normal reaction R of the rail (both perpendicular to the motion, so they do no work) and the tension T. The tension is a conservative force with elastic potential energy $\lambda e^2/(2a)$. At the start $e = 0$ and the speed is v, so

conservation of energy gives

$$\tfrac{1}{2}m\dot{x}^2 + \frac{\lambda}{2a}\Big(\sqrt{x^2+a^2} - a\Big)^2 = \tfrac{1}{2}mv^2.$$

Multiplying by $2/m$ and writing $k^2 = \lambda/(ma)$,

$$\dot{x}^2 = v^2 - k^2\Big(\sqrt{x^2+a^2} - a\Big)^2,$$

as required.

The greatest displacement and the acceleration there

The extension e increases with $|x|$, so $\dot{x}^2$ decreases as P moves out, and P moves out until $\dot{x} = 0$. At that point $k\big(\sqrt{x_0^2+a^2} - a\big) = v$ (taking the positive root, since $e \geq 0$). So

$$\sqrt{x_0^2+a^2} = a + \frac{v}{k} \quad\Longrightarrow\quad x_0^2 = \Big(a + \frac{v}{k}\Big)^2 - a^2 = \frac{2av}{k} + \frac{v^2}{k^2},$$

giving

$$x_0 = \frac{\sqrt{v\,(2ak+v)}}{k}.$$

For the acceleration, resolve along the rail. If θ is the angle between PQ and the rail, then $\cos\theta = x/\sqrt{x^2+a^2}$, and the component of the tension along the rail is $T\cos\theta$ towards O. With $T = \lambda e/a$,

$$m\ddot{x} = -\frac{\lambda}{a}\Big(\sqrt{x^2+a^2} - a\Big)\frac{x}{\sqrt{x^2+a^2}}, \qquad \text{so} \qquad \ddot{x} = -k^2\Big(\sqrt{x^2+a^2} - a\Big)\frac{x}{\sqrt{x^2+a^2}}.$$

(Differentiating the energy equation with respect to t and cancelling $2\dot{x}$ gives the same thing.) At $x = x_0$ we have $\sqrt{x_0^2+a^2} - a = v/k$ and $\sqrt{x_0^2+a^2} = (ak+v)/k$, so

$$\ddot{x} = -k^2 \cdot \frac{v}{k} \cdot \frac{k\,x_0}{ak+v} = -\frac{k^2 v\,x_0}{ak+v} = -\frac{kv\sqrt{v\,(2ak+v)}}{ak+v}.$$

So the acceleration at x_0 has magnitude $\dfrac{kv\sqrt{v(2ak+v)}}{ak+v}$ and is directed back along the rail towards the initial position.

The period

The equation for $\dot{x}^2$ depends only on x^2, so the motion is symmetric about O. The particle oscillates between $-x_0$ and x_0, and each of the four quarters $O \to x_0 \to O \to -x_0 \to O$ takes the same time. On the first quarter $\dot{x} > 0$, so $\mathrm{d}t = \mathrm{d}x/\dot{x}$ and the period is

$$\tau = 4\int_0^{x_0} \frac{\mathrm{d}x}{\sqrt{v^2 - k^2\big(\sqrt{x^2+a^2} - a\big)^2}}.$$

The integrand blows up at x_0, but only like $(x_0 - x)^{-1/2}$, so the integral converges.

Ask yourself What does $v \ll ka$ tell us about the size of x?

Since $x_0^2 = 2av/k + v^2/k^2$, if $v/k \ll a$ then $x_0^2 \approx 2av/k$, so $x_0^2/a^2 \approx 2v/(ka) \ll 1$. Every x in the range of integration therefore satisfies $|x| \ll a$, and the binomial expansion gives

$$\sqrt{x^2+a^2} - a = a\left(1 + \frac{x^2}{a^2}\right)^{1/2} - a \approx \frac{x^2}{2a}.$$

The period becomes

$$\tau \approx 4\int_0^{x_0} \frac{\mathrm{d}x}{\sqrt{v^2 - k^2x^4/(4a^2)}}, \qquad x_0 \approx \sqrt{\frac{2av}{k}},$$

and this approximate x_0 is exactly where the new denominator vanishes, since $k^2x_0^4/(4a^2) = v^2$. Now substitute $x = x_0u$, so that $\mathrm{d}x = x_0\,\mathrm{d}u$ and $k^2x^4/(4a^2) = v^2u^4$:

$$\tau \approx 4\int_0^1 \frac{x_0\,\mathrm{d}u}{v\sqrt{1-u^4}} = \frac{4}{v}\sqrt{\frac{2av}{k}}\int_0^1 \frac{\mathrm{d}u}{\sqrt{1-u^4}} = \sqrt{\frac{32a}{kv}}\int_0^1 \frac{\mathrm{d}u}{\sqrt{1-u^4}},$$

since $\frac{16}{v^2}\cdot\frac{2av}{k} = \frac{32a}{kv}$. This is the required approximation.

Checks

- Units: $k^2 = \lambda/(ma)$ has units of s^{-2}, so v/k is a length (as it must be, since it is the extension at x_0) and $\sqrt{a/(kv)}$ is a time.
- With $m = 1$, $\lambda = 2$, $a = 1$, $v = 1$ we get $k = \sqrt{2}$, $x_0 = \sqrt{1+2\sqrt{2}} \approx 1.3836$ and acceleration ≈ -1.1462 at x_0. Stepping the equation of motion forward numerically gives the same turning point, and a period of about 7.612, which matches the integral.
- For $a = 1$, $k = 2$, $v = 0.001$, the exact period integral is 165.862, against 165.833 from the approximation, where $\int_0^1 (1-u^4)^{-1/2}\,\mathrm{d}u \approx 1.3110$. The ratio tends to 1 as $v \to 0$.

Common slips

- Using the extension x or $\sqrt{x^2+a^2}$ instead of $\sqrt{x^2+a^2} - a$; the string starts at its natural length.
- Forgetting to resolve the tension: only the component $T\cos\theta$ acts along the rail.
- Taking the period as $2\int_0^{x_0}$ or $\int_0^{x_0}$; the oscillation is between $-x_0$ and x_0, so four quarter-periods are needed.

Beyond STEP

For small v the restoring force is roughly proportional to x^3, not x, so this is a nonlinear oscillator whose period depends on its amplitude, growing like $v^{-1/2}$ as the energy shrinks, unlike simple harmonic motion. The integral $\int_0^1 (1-u^4)^{-1/2}\,\mathrm{d}u$ is an elliptic integral (the lemniscate constant divided by two), the kind that appears for the exact period of a pendulum too. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers energy methods and oscillations in this spirit.

WORKED SOLUTION

STEP III 2015, Question 10

Mechanics • Rotating Rigid Rod

STEP III 2015, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

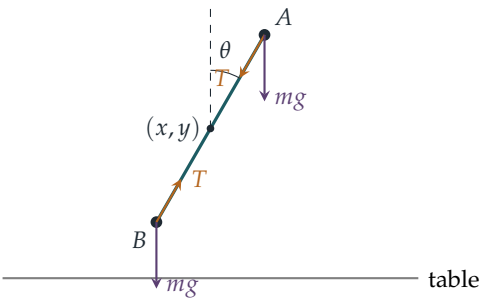
step.maths.org/questions/15-s3-q10

The key idea

Describe the system by the motion of the midpoint and the rotation of the rod. Write each particle's position in terms of x, y and θ, differentiate twice, and apply Newton's second law to each particle. Adding the two equations removes the tension and gives the motion of the centre of mass. Subtracting them and resolving perpendicular to the rod removes it again and leaves the rotation. Once all three accelerations are known, the final part is projectile motion plus steady rotation.

Solution

We model the two masses as particles joined by a light rigid rod. The only forces on each particle are its weight and the force from the rod, which acts along the rod. Let T be the tension, so $T > 0$ means the rod pulls each particle towards the midpoint. (A rod can also push, and then $T < 0$; nothing below depends on the sign.)



Equations of motion for particle A

Let A be the particle on the end of the rod that starts above the midpoint, so the vector from the midpoint to A is $a(\sin\theta, \cos\theta)$. Then A is at

$$\mathbf{r}_A = \begin{pmatrix} x + a\sin\theta \\ y + a\cos\theta \end{pmatrix}, \qquad \text{so} \qquad \dot{\mathbf{r}}_A = \begin{pmatrix} \dot{x} + a\dot{\theta}\cos\theta \\ \dot{y} - a\dot{\theta}\sin\theta \end{pmatrix}$$

by the chain rule. This is the required velocity. Differentiating again, using the product rule on $\dot{\theta}\cos\theta$ and $\dot{\theta}\sin\theta$,

$$\ddot{\mathbf{r}}_A = \begin{pmatrix} \ddot{x} + a\ddot{\theta}\cos\theta - a\dot{\theta}^2\sin\theta \\ \ddot{y} - a\ddot{\theta}\sin\theta - a\dot{\theta}^2\cos\theta \end{pmatrix}.$$

The rod pulls A towards the midpoint, in the direction $-(\sin\theta, \cos\theta)$, with force of size T. Gravity contributes $-mg(0,1)$. So Newton's second law for A gives exactly the stated equation:

$$m\begin{pmatrix} \ddot{x} + a\ddot{\theta}\cos\theta - a\dot{\theta}^2\sin\theta \\ \ddot{y} - a\ddot{\theta}\sin\theta - a\dot{\theta}^2\cos\theta \end{pmatrix} = -T\begin{pmatrix} \sin\theta \\ \cos\theta \end{pmatrix} - mg\begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{A}$$

The other particle

For B, at the other end, replace a by $-a$ throughout. The rod now pulls B in the direction $+(\sin\theta, \cos\theta)$:

$$m\begin{pmatrix} \ddot{x} - a\ddot{\theta}\cos\theta + a\dot{\theta}^2\sin\theta \\ \ddot{y} + a\ddot{\theta}\sin\theta + a\dot{\theta}^2\cos\theta \end{pmatrix} = T\begin{pmatrix} \sin\theta \\ \cos\theta \end{pmatrix} - mg\begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{B}$$

Here B has velocity $(\dot{x} - a\dot{\theta}\cos\theta,\ \dot{y} + a\dot{\theta}\sin\theta)$.

Deducing the accelerations

Adding (A) and (B) cancels every term containing a or T:

$$2m\begin{pmatrix} \ddot{x} \\ \ddot{y} \end{pmatrix} = -2mg\begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad \text{so} \qquad \ddot{x} = 0, \quad \ddot{y} = -g.$$

Subtracting (B) from (A) and dividing by 2:

$$\begin{aligned} ma\ddot{\theta}\cos\theta - ma\dot{\theta}^2\sin\theta &= -T\sin\theta, \\ -ma\ddot{\theta}\sin\theta - ma\dot{\theta}^2\cos\theta &= -T\cos\theta. \end{aligned}$$

Ask yourself How do we get rid of T, which we don't know?

The tension acts along the rod, so we resolve perpendicular to it, in the direction $(\cos\theta, -\sin\theta)$. That means taking $\cos\theta$ times the first equation minus $\sin\theta$ times the

second. The right-hand sides give $-T\sin\theta\cos\theta + T\cos\theta\sin\theta = 0$, and the $\dot\theta^2$ terms cancel in the same way. What is left is

$$ma\ddot\theta(\cos^2\theta + \sin^2\theta) = 0,$$

and since $m, a \neq 0$ this gives $\ddot\theta = 0$. (Resolving along the rod instead gives $T = ma\dot\theta^2$, the tension needed for circular motion about the midpoint.)

The final part

At $t = 0$ the rod is vertical, so $\theta = 0$. Particle A is then the higher one, with velocity $(\dot x + a\dot\theta,\ \dot y)$, and B is the lower one, with velocity $(\dot x - a\dot\theta,\ \dot y)$. Matching these with the given initial velocities:

$$\dot x + a\dot\theta = u, \qquad \dot x - a\dot\theta = 0, \qquad \dot y = v,$$

so initially $\dot x = \frac{1}{2}u$, $\dot\theta = \dfrac{u}{2a}$ and $\dot y = v$.

By the previous part, $\dot\theta$ stays constant, so $\theta = \dfrac{ut}{2a}$. We take $u > 0$, so that θ increases. Also $\ddot y = -g$ with $y(0) = h$, so

$$y = h + vt - \tfrac{1}{2}gt^2.$$

The rod has turned through $\frac{1}{2}\pi$ when $\dfrac{ut}{2a} = \dfrac{\pi}{2}$, that is, at $t = \dfrac{\pi a}{u}$. The rod is then horizontal, so both particles are at the height y of the midpoint. They both hit the table at that moment exactly when $y = 0$ there:

$$h + v\cdot\frac{\pi a}{u} - \frac{1}{2}g\cdot\frac{\pi^2a^2}{u^2} = 0.$$

Multiplying by $2u^2$ and rearranging gives

$$2hu^2 = \pi^2a^2g - 2\pi uva,$$

as required.

Checks

- With $v = 0$, the time to fall is $\sqrt{2h/g}$, and this must equal $\pi a/u$. Squaring gives $2hu^2 = \pi^2a^2g$, which matches.
- Units: every term in the result has dimensions $\mathrm{L}^3\mathrm{T}^{-2}$.
- We simulated two particles joined by a rigid rod on a computer, for several random choices of a, u, v with $g = 9.8$ and h given by the formula. For example, $a \approx 1.754$ gives $h \approx 17.93$. In every case both particles reached the table together at $t = \pi a/u$, to within 10^{-10}, with the rod horizontal.

Common slips

- Getting the direction of the tension wrong for one of the particles. The rod pulls each particle towards the midpoint, so the two forces are equal and opposite.
- Using a different value of T in each equation. The rod is light, so the net force on it must be zero, and the tension is the same at both ends.
- Forgetting the $-a\dot{\theta}^2$ terms when differentiating $\dot{\theta}\cos\theta$. Without them the subtraction step does not come out cleanly.

Beyond STEP

This question is a small instance of a general theorem about systems of particles. The centre of mass moves as if all the mass and all the external force were concentrated there. Internal forces cancel in pairs, so with no external torque about the centre of mass the angular momentum about it is conserved. In first-year mechanics you would also meet the Lagrangian approach, which takes x, y, θ as generalised coordinates and never needs T at all. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers the centre of mass and rigid-body rotation thoroughly.

WORKED SOLUTION

STEP III 2015, Question 11

Mechanics • Circular Motion

STEP III 2015, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q11

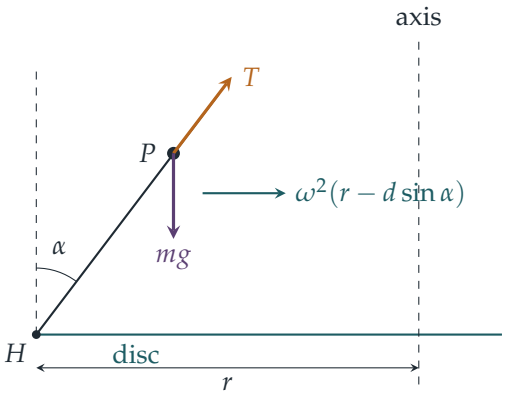
The key idea

A light rod has no mass, so the forces and moments acting on it must balance exactly, even though it is moving. In part (i) only two forces act on the rod, and the one from the hinge has no moment about the hinge, so the force from P must point straight at the hinge. After that, P is a particle moving in a horizontal circle, and resolving does the rest. In part (ii) the forces from the two particles need not lie along the rod, but taking moments about the hinge still gives a single equation for β.

Solution

Model. The particles are point masses, the rod is light and rigid, and the hinge is smooth, so it exerts no couple about its own axis (the horizontal line through the hinge perpendicular to the vertical plane containing the rod). Take axes in that plane, with origin at the hinge H, x horizontal pointing towards the axis of rotation, and y vertically upwards. A point of the rod at distance s from H is at $(s\sin\theta,\ s\cos\theta)$, where θ is the rod's angle with the upward vertical, and it is at horizontal distance $r - s\sin\theta$ from the axis.

Part (i)



The forces on P are drawn above: its weight, and the force from the rod, which we are about to show acts along the rod. The teal arrow is the acceleration of P.

The force from P is along the rod. Only two forces act on the rod: the force from the hinge, acting at H, and the force **G** from P, acting at P. The rod is light, so the total moment on it about the hinge axis is zero (mass times angular acceleration is zero), whatever its motion. The hinge force has no moment about H, so neither does **G**.

Next, P moves in a horizontal circle centred on the axis, so its acceleration is horizontal and points towards the axis, which lies in the vertical plane of the rod. Its weight is in that plane too, so the force on P from the rod, and hence **G** (equal and opposite, by Newton's third law), also lies in that plane. A force in the plane with zero moment about H has its line of action through H. That line also passes through P, which is on the rod, so **G** is parallel to the rod, as required.

The angle. So the rod pushes or pulls P along the rod with a force T, which we take to point from H towards P, i.e. components $(T\sin\alpha,\ T\cos\alpha)$. P is at distance $r - d\sin\alpha$ from the axis, which is positive since $d\sin\alpha < r$, so P is on the same side of the axis as H and its acceleration $\omega^2(r - d\sin\alpha)$ points in the $+x$ direction. Newton's second law for P:

$$\begin{aligned} \text{vertically:} \quad & T\cos\alpha - mg = 0, \\ \text{horizontally:} \quad & T\sin\alpha = m\omega^2(r - d\sin\alpha). \end{aligned}$$

Dividing the second by the first (we have $\cos\alpha \neq 0$, since otherwise $T\cos\alpha = 0 \neq mg$):

$$\tan\alpha = \frac{\omega^2(r - d\sin\alpha)}{g} = \frac{r - d\sin\alpha}{a}.$$

Multiplying by $a\cot\alpha$ gives $a = r\cot\alpha - d\cos\alpha$, that is,

$$r\cot\alpha = a + d\cos\alpha,$$

as required.

The hinge force. The first equation gives $T = mg/\cos\alpha > 0$, so the rod pushes P away from the hinge: the rod is in compression. By Newton's third law P pushes the rod along the rod towards H with force T. The rod is light, so the forces on it sum to zero, and the hinge force on the rod is equal and opposite: it acts **along the rod, away from the hinge towards** P (upwards, and inwards towards the axis), with magnitude

$$\frac{mg}{\cos\alpha} = mg\sec\alpha.$$

Part (ii)

Now the particle of mass m_i $(i = 1, 2)$ is at $(d_i\sin\beta,\ d_i\cos\beta)$, at distance $r - d_i\sin\beta > 0$ from the axis.

> **Ask yourself** Each particle now pulls on the rod in some direction we don't know. How can we avoid finding these forces individually?

We don't need them individually: Newton's second law for each particle gives the force the rod exerts on it, and the moment equation for the light rod about H then removes the unknown hinge force. Let $\mathbf{F}_i$ be the force on particle i from the rod. Its acceleration is $\omega^2(r - d_i\sin\beta)$ in the $+x$ direction, so

$$\mathbf{F}_i + (0, -m_ig) = \big(m_i\omega^2(r - d_i\sin\beta),\ 0\big) \quad\Longrightarrow\quad \mathbf{F}_i = \big(m_i\omega^2(r - d_i\sin\beta),\ m_ig\big).$$

By Newton's third law, particle i exerts $-\mathbf{F}_i$ on the rod, at the point $(x_i, y_i) = (d_i\sin\beta,\ d_i\cos\beta)$. The anticlockwise moment about H of a force (X, Y) acting at (x, y) is $xY - yX$, so the moment of $-\mathbf{F}_i$ is

$$-d_i\sin\beta\cdot m_ig + d_i\cos\beta\cdot m_i\omega^2(r - d_i\sin\beta).$$

The hinge force has no moment about H, and the rod is light, so these two moments sum to zero. Dividing by $\omega^2\sin\beta$ (which is non-zero: if $\beta = 0$ the moment equation would read $\sum m_id_i\omega^2 r = 0$, which is false) and using $g/\omega^2 = a$:

$$\sum_{i=1}^{2} m_id_i\big(-a + r\cot\beta - d_i\cos\beta\big) = 0,$$

that is,

$$(m_1d_1 + m_2d_2)\,r\cot\beta = (m_1d_1 + m_2d_2)\,a + (m_1d_1^2 + m_2d_2^2)\cos\beta.$$

Dividing by $m_1d_1 + m_2d_2 > 0$,

$$r\cot\beta = a + b\cos\beta, \qquad b = \frac{m_1d_1^2 + m_2d_2^2}{m_1d_1 + m_2d_2}.$$

Checks

- With one particle ($m_2 = 0$, say) we get $b = d_1$, recovering part (i). If $d_1 = d_2 = d$, then $b = d$ too: two particles at the same point act like one.

- b is a weighted average of d_1 and d_2 (with weights $m_i d_i$), so it lies between them, as it should.
- As $\omega \to 0$, $a \to \infty$, so $\cot\alpha \to \infty$ and the rod tends to the vertical, as we would expect when the disc barely turns. As $\omega \to \infty$ with $d > r$, $a \to 0$ and the equation tends to $r = d\sin\alpha$: the particle approaches the axis, where it needs almost no centripetal force.
- With $r = 1\,\text{m}$, $\omega = 5\,\text{rad}\,\text{s}^{-1}$, $d = 0.6\,\text{m}$ and $g = 9.81\,\text{m}\,\text{s}^{-2}$, we have $a = 0.3924\,\text{m}$ and the equation gives $\alpha \approx 0.9251$ (about 53.0°); then $r\cot\alpha \approx 0.7538 \approx a + d\cos\alpha$. A computer simulation of the rod's motion started at rest at this angle stays there.

Common slips

- In part (ii), assuming each particle's force on the rod is along the rod. That was special to a single particle; with two, only the total moment about H is zero.
- Using radius r (or $d\sin\alpha$) for the particle's circle instead of $r - d\sin\alpha$.
- Giving the hinge force as acting towards the hinge: the rod is in compression, so the hinge pushes the rod towards P.

Beyond STEP

The moment equation in part (ii) is the condition that the effective potential $\sum m_i\left(gd_i\cos\theta - \frac{1}{2}\omega^2(r - d_i\sin\theta)^2\right)$ is stationary: in the rotating frame, the centrifugal force behaves like an extra potential. This is the starting point for analysing rotating systems such as the centrifugal governor, and for asking whether a steady state is stable. Here the effective potential has a maximum at $\theta = \alpha$, so the steady state of part (i) is in fact unstable: nudge the rod slightly and it swings away. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rigid-body moments and rotating frames.

WORKED SOLUTION

STEP III 2015, Question 12

Probability and Statistics • Probability Generating Functions

STEP III 2015, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q12

The key idea

A remainder on division by m only cares about the exponent modulo m, so in the generating function of a remainder we may replace x^m by 1 wherever it appears. Adding independent throws multiplies generating functions, and then we reduce the powers of x again. With this rule, the whole calculation becomes simple polynomial algebra in which $x \cdot y$ and y^2 collapse to multiples of y.

Solution

Throughout, let $X_1, X_2, \ldots$ be the outcomes of the throws, which are independent and each uniform on $\{1, \ldots, 6\}$.

The reduction rule. Let m be a positive integer. If U and V are independent and take values in $\{0, 1, \ldots, m-1\}$, the pgf of $U+V$ is $\mathrm{G}_U(x)\mathrm{G}_V(x)$, a polynomial in which the term x^k carries the probability that $U+V=k$. The remainder of $U+V$ on division by m equals j exactly when $U+V$ is j or $j+m$. So we get the pgf of that remainder by replacing every x^k with x^{k-m} for $k \geq m$; in other words, by setting $x^m = 1$. Since the remainder of S_n is the remainder of (remainder of S_{n-1}) + (remainder of X_n), and these two are independent, we get the pgf of each remainder from the one before by multiplying and then setting $x^m = 1$. We write $\equiv$ for "equal once $x^m = 1$ is used".

Part (i)

Throws of $1, \ldots, 5$ leave remainders $1, \ldots, 5$, and a 6 leaves remainder 0. Each happens with probability $\frac{1}{6}$, so

$$\mathrm{G}(x) = \tfrac{1}{6}\left(1 + x + x^2 + x^3 + x^4 + x^5\right) = \tfrac{1}{6}z, \qquad \text{where } z = 1 + x + \cdots + x^5.$$

With $x^6 = 1$, multiplying z by x just moves each term round one place: $xz = x + x^2 + \cdots + x^5 + x^6 \equiv z$. Hence $x^k z \equiv z$ for every k, and

$$z^2 = \sum_{k=0}^{5} x^k z \equiv 6z.$$

So the pgf of R_2 is

$$\tfrac{1}{36} z^2 \equiv \tfrac{1}{36} \cdot 6z = \tfrac{1}{6} z = \mathrm{G}(x),$$

as required.

The same step then works at every stage: if R_{n-1} has pgf $\mathrm{G}(x)$, then R_n has pgf $\mathrm{G}(x)^2 \equiv \mathrm{G}(x)$. By induction, every R_n has pgf $\mathrm{G}(x) = \frac{1}{6} z$. The probability that $6 \mid S_n$ is $\mathrm{P}(R_n = 0)$, the constant term of $\mathrm{G}(x)$:

$$\mathrm{P}(S_n \text{ divisible by } 6) = \tfrac{1}{6}.$$

Part (ii)

Now $m = 5$. Throws of $1, 2, 3, 4$ give remainders $1, 2, 3, 4$; a 5 gives 0; a 6 gives 1. So remainder 1 has probability $\frac{2}{6}$ and the others $\frac{1}{6}$ each:

$$\mathrm{G}_1(x) = \tfrac{1}{6}\left(1 + 2x + x^2 + x^3 + x^4\right) = \tfrac{1}{6}(x + y), \qquad y = 1 + x + x^2 + x^3 + x^4.$$

Exactly as in part (i), with $x^5 = 1$ we have $xy \equiv y$, so $x^k y \equiv y$ for all k and

$$y^2 = \sum_{k=0}^{4} x^k y \equiv 5y.$$

Therefore

$$\begin{aligned} \mathrm{G}_2(x) &\equiv \tfrac{1}{36}(x + y)^2 = \tfrac{1}{36}\left(x^2 + 2xy + y^2\right) \\ &\equiv \tfrac{1}{36}\left(x^2 + 2y + 5y\right) = \tfrac{1}{36}\left(x^2 + 7y\right), \end{aligned}$$

which is the required result. (As a sanity check, putting $x = 1$ gives $\frac{1}{36}(1 + 35) = 1$.)

The general pgf. By the reduction rule, $\mathrm{G}_n(x) \equiv \mathrm{G}_1(x)^n = \frac{1}{6^n}(x + y)^n$. Expand by the binomial theorem:

$$(x + y)^n = x^n + \sum_{k=1}^{n} \binom{n}{k} x^{n-k} y^k.$$

For $k \geq 1$, using $y^k \equiv 5^{k-1} y$ (induction on $y^2 \equiv 5y$) and then $x^{n-k} y \equiv y$,

$$x^{n-k} y^k \equiv 5^{k-1} y.$$

So

$$(x + y)^n \equiv x^n + y \sum_{k=1}^{n} \binom{n}{k} 5^{k-1} = x^n + \frac{y}{5}\left[(1 + 5)^n - 1\right] = x^n + \frac{6^n - 1}{5}\, y.$$

Ask yourself Where did $(6^n-1)/5$ come from?

The sum $\sum_{k=0}^{n}\binom{n}{k}5^k$ is $(1+5)^n = 6^n$ by the binomial theorem; we are missing the $k=0$ term (which is 1) and have an extra factor $\frac{1}{5}$. Hence

$$G_n(x) = \frac{1}{6^n}\left(x^r + \frac{6^n-1}{5}y\right),$$

where r is the remainder when n is divided by 5 (since $x^n \equiv x^r$). So every remainder has probability $\dfrac{6^n-1}{5\cdot 6^n}$, except remainder r, which gets an extra $\frac{1}{6^n}$.

Divisibility by 5. We need $P(T_n = 0)$, the constant term of $G_n(x)$. The constant term of y is 1.

- If $5 \nmid n$, then $r \neq 0$ and x^r contributes nothing to the constant term, so

$$P(S_n \text{ divisible by } 5) = \frac{6^n-1}{5\cdot 6^n} = \frac{1}{5}\left(1-\frac{1}{6^n}\right),$$

  as required.
- If $5 \mid n$, then $r = 0$ and $x^r = 1$ adds $\frac{1}{6^n}$:

$$P(S_n \text{ divisible by } 5) = \frac{6^n-1}{5\cdot 6^n} + \frac{1}{6^n} = \frac{6^n+4}{5\cdot 6^n} = \frac{1}{5}\left(1+\frac{4}{6^n}\right).$$

Checks

- $n=1$: the formula gives $\frac{1}{5}\cdot\frac{5}{6} = \frac{1}{6}$, and indeed only a throw of 5 is divisible by 5.
- $n=2$: the formula gives $\frac{1}{5}\cdot\frac{35}{36} = \frac{7}{36}$. Directly, sums of 5 (four ways) and 10 (three ways) give $\frac{7}{36}$, matching the constant term of $\frac{1}{36}(x^2+7y)$.
- $n=5$: the answer is $\frac{6^5+4}{5\cdot 6^5} = \frac{389}{1944} \approx 0.2001$, just above $\frac{1}{5}$. Full enumeration of all 6^n outcomes by computer for $n \leq 8$ confirms the whole pgf $G_n(x)$ and both cases.
- As $n \to \infty$ both answers tend to $\frac{1}{5}$: the remainder becomes uniform.

Common slips

- Forgetting that a 6 leaves remainder 1 on division by 5, so that G_1 has coefficient $\frac{2}{6}$ for x.
- Reducing x^n to x^0 for every n: the term x^n reduces to x^r with r the remainder of n, which is exactly why the case $5 \mid n$ is different.
- Writing $G_n(x) = G_1(x)^n$ with "=" and then reading off coefficients without reducing modulo $x^5 = 1$.

Beyond STEP

Setting $x^m = 1$ amounts to working in the ring of polynomials modulo $x^m - 1$, the group algebra of the cyclic group of order m. An equivalent tool is the roots-of-unity filter: $\mathrm{P}(m \mid S_n) = \frac{1}{m}\sum_{\omega^m=1} \mathrm{G}_S(\omega)$, which is the discrete Fourier transform at work. That viewpoint also explains why sums of random steps on a cycle become uniform, the start of the theory of Markov chains and random walks on groups. Grimmett and Welsh, *Probability: An Introduction*, covers generating functions and random walks at the right level.

WORKED SOLUTION

STEP III 2015, Question 13

Probability and Statistics • Functions of Uniform Variables

STEP III 2015, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q13

The key idea

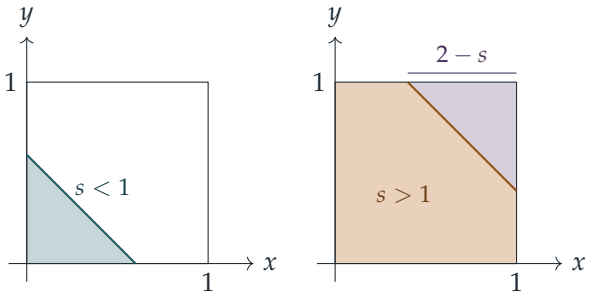
Because X and Y are independent and uniform on $[0,1]$, the pair (X,Y) is uniformly distributed over the unit square, so every probability about X and Y is simply an *area*. Events such as $X+Y \le s$ or $Y \le rX$ cut the square along a straight line, and the area on one side is a triangle or a square minus a triangle. After that, the distributions of $1/(X+Y)$ and $X/(X+Y)$ follow by rewriting their events in terms of the ones we have already measured.

Solution

Throughout, (X,Y) is uniform on the unit square $[0,1]^2$, which has area 1, so for any region A of the square, $\mathrm{P}\big((X,Y)\in A\big)$ equals the area of A.

Part (i)

Let $S = X+Y$, which takes values in $[0,2]$, and let $F(s) = \mathrm{P}(S \le s)$. The event $S \le s$ is the part of the square on or below the line $x+y=s$.



If $0 \le s \le 1$, the line meets the axes at $(s,0)$ and $(0,s)$, and the region below it is a right-angled isosceles triangle with legs s. If $1 \le s \le 2$, the region *above* the line

is a triangle in the corner $(1,1)$; its legs run from the line to the corner, of length $1-(s-1)=2-s$. Hence

$$F(s)=\mathrm{P}(X+Y\le s)=\begin{cases}0 & s<0,\\ \tfrac{1}{2}s^2 & 0\le s\le 1,\\ 1-\tfrac{1}{2}(2-s)^2 & 1\le s\le 2,\\ 1 & s>2.\end{cases}$$

The two middle formulas agree at $s=1$ (both give $\tfrac{1}{2}$), as they must.

Part (ii)

Let $T=1/S$. Since $0<S\le 2$ (the value $S=0$ has probability zero), T takes values in $[\tfrac{1}{2},\infty)$. For $t>0$, because $s\mapsto 1/s$ is decreasing on $s>0$,

$$\mathrm{P}(T\le t)=\mathrm{P}\big(S\ge \tfrac{1}{t}\big)=1-F\big(\tfrac{1}{t}\big),$$

using the fact that S is continuous, so $\mathrm{P}(S=1/t)=0$.

> **Ask yourself** Which formula for F do we use? It depends on where $1/t$ lies, not where t lies.

If $t\ge 1$ then $0<1/t\le 1$, so we use the first branch: $\mathrm{P}(T\le t)=1-\tfrac{1}{2}t^{-2}$. If $\tfrac{1}{2}\le t\le 1$ then $1\le 1/t\le 2$, so we use the second branch: $\mathrm{P}(T\le t)=\tfrac{1}{2}\big(2-t^{-1}\big)^2$. And $\mathrm{P}(T\le t)=0$ for $t<\tfrac{1}{2}$.

Differentiating gives the density. For $t\ge 1$,

$$f(t)=\frac{\mathrm{d}}{\mathrm{d}t}\Big(1-\tfrac{1}{2}t^{-2}\Big)=t^{-3},$$

and for $\tfrac{1}{2}\le t\le 1$, by the chain rule,

$$f(t)=\big(2-t^{-1}\big)\cdot t^{-2}=2t^{-2}-t^{-3},$$

with $f(t)=0$ otherwise. This is the required density.

For the expectation,

$$\begin{aligned}\mathrm{E}\Big(\frac{1}{X+Y}\Big)=\int_{1/2}^{\infty} t f(t)\,\mathrm{d}t &= \int_{1/2}^{1}\big(2t^{-1}-t^{-2}\big)\,\mathrm{d}t+\int_{1}^{\infty} t^{-2}\,\mathrm{d}t\\ &=\Big[2\ln t+t^{-1}\Big]_{1/2}^{1}+\Big[-t^{-1}\Big]_{1}^{\infty}\\ &=(0+1)-(-2\ln 2+2)+1\\ &=2\ln 2.\end{aligned}$$

The improper integral converges because t^{-2} is integrable on $[1,\infty)$.

Part (iii)

Let $R = Y/X$, which takes values in $[0,\infty)$ (the event $X = 0$ has probability zero). For $r \geq 0$, the event $R \leq r$ is $Y \leq rX$: the part of the square on or below the line $y = rx$ through the origin.

If $0 \leq r \leq 1$, this line leaves the square through the right-hand side at $(1, r)$, and the region below it is a triangle with base 1 and height r, area $\frac{1}{2}r$. If $r \geq 1$, the line leaves through the top at $(1/r, 1)$, and the region *above* it is a triangle with base $1/r$ along the top edge and height 1, area $\frac{1}{2r}$. So

$$H(r) = \mathrm{P}\Big(\frac{Y}{X} \leq r\Big) = \begin{cases} 0 & r < 0, \\ \frac{1}{2}r & 0 \leq r \leq 1, \\ 1 - \dfrac{1}{2r} & r \geq 1. \end{cases}$$

Now let $Z = \dfrac{X}{X+Y} = \dfrac{1}{1+R}$, which takes values in $[0,1]$. For $0 < z \leq 1$, since $r \mapsto 1/(1+r)$ is decreasing on $r \geq 0$,

$$\mathrm{P}(Z \leq z) = \mathrm{P}\Big(1 + R \geq \frac{1}{z}\Big) = \mathrm{P}\Big(R \geq \frac{1-z}{z}\Big) = 1 - H\Big(\frac{1-z}{z}\Big).$$

Again the branch depends on where $(1-z)/z$ lies: it is at most 1 exactly when $z \geq \frac{1}{2}$.

- For $\frac{1}{2} \leq z \leq 1$: $\mathrm{P}(Z \leq z) = 1 - \dfrac{1-z}{2z} = \dfrac{3}{2} - \dfrac{1}{2z}$.
- For $0 < z \leq \frac{1}{2}$: $\mathrm{P}(Z \leq z) = 1 - \Big(1 - \dfrac{z}{2(1-z)}\Big) = \dfrac{z}{2(1-z)}$.

Both give $\frac{1}{2}$ at $z = \frac{1}{2}$. Differentiating (for the second, write $\frac{z}{2(1-z)} = \frac{1}{2}\big(\frac{1}{1-z} - 1\big)$), the density of Z is

$$g(z) = \begin{cases} \dfrac{1}{2(1-z)^2} & 0 \leq z \leq \frac{1}{2}, \\ \dfrac{1}{2z^2} & \frac{1}{2} \leq z \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Notice that $g(1-z) = g(z)$: the density is symmetric about $\frac{1}{2}$.

The expectation. We can write it down without integrating. The pairs (X, Y) and (Y, X) have the same distribution, since X and Y are independent with the same distribution. So $\dfrac{X}{X+Y}$ and $\dfrac{Y}{X+Y}$ have the same expectation. Their sum is exactly 1, so

$$\mathrm{E}\Big(\frac{X}{X+Y}\Big) = \frac{1}{2}.$$

Verification by integration.

$$\mathrm{E}(Z) = \int_0^{1/2} \frac{z}{2(1-z)^2}\,\mathrm{d}z + \int_{1/2}^{1} \frac{1}{2z}\,\mathrm{d}z.$$

The second integral is $\frac{1}{2}\big[\ln z\big]_{1/2}^{1} = \frac{1}{2}\ln 2$. In the first, substitute $u = 1 - z$, so $\mathrm{d}z = -\mathrm{d}u$, and $z = 0, \frac{1}{2}$ correspond to $u = 1, \frac{1}{2}$:

$$\begin{aligned}\int_0^{1/2} \frac{z}{2(1-z)^2}\,\mathrm{d}z &= \frac{1}{2}\int_{1/2}^{1} \frac{1-u}{u^2}\,\mathrm{d}u = \frac{1}{2}\Big[-\frac{1}{u} - \ln u\Big]_{1/2}^{1} \\ &= \frac{1}{2}\big(-1 - (-2 + \ln 2)\big) = \frac{1}{2}(1 - \ln 2).\end{aligned}$$

Adding, $\mathrm{E}(Z) = \frac{1}{2}(1 - \ln 2) + \frac{1}{2}\ln 2 = \frac{1}{2}$, confirming the value above.

Checks

- Each distribution function rises from 0 to 1 and is continuous where the branches meet: $F(1) = \frac{1}{2}$, $\mathrm{P}(T \le 1) = \frac{1}{2}$, $H(1) = \frac{1}{2}$, $\mathrm{P}(Z \le \frac{1}{2}) = \frac{1}{2}$. Each of these is $\frac{1}{2}$ by symmetry of the square about a diagonal.
- The density f integrates to 1: $\int_{1/2}^{1}(2t^{-2} - t^{-3})\,\mathrm{d}t = 2 - \frac{3}{2} = \frac{1}{2}$ and $\int_1^{\infty} t^{-3}\,\mathrm{d}t = \frac{1}{2}$.
- $2\ln 2 \approx 1.386$ lies between 1 and 2, as it should, and exceeds $1/\mathrm{E}(X+Y) = 1$, as Jensen's inequality for the convex function $1/s$ predicts. A direct numerical evaluation of $\iint (x+y)^{-1}\,\mathrm{d}x\,\mathrm{d}y$ over the square gives 1.3863. The distribution functions and $\mathrm{E}(Z) = \frac{1}{2}$ were also confirmed by computer simulation.

Common slips

- Choosing the branch of F or H according to t or z rather than according to $1/t$ or $(1-z)/z$, which swaps the two formulas.
- Forgetting that $1/s$ and $1/(1+r)$ are decreasing, so that the inequality must be reversed: $T \le t$ means $S \ge 1/t$.
- Getting the leg of the corner triangle in part (i) wrong: it is $2 - s$, not $s - 1$.

Beyond STEP

Turning a statement about (X, Y) into an area and then differentiating the distribution function is the general *distribution function method* for finding the law of a function of random variables. At university it is joined by the change-of-variables (Jacobian) formula for joint densities, which handles $X + Y$ and $X/(X+Y)$ together; in particular, the density of $X + Y$ here is the *convolution* of two uniform densities. Grimmett and Welsh, *Probability: An Introduction*, covers both methods, and Blitzstein and Hwang, *Introduction to Probability*, has many examples in the same spirit.

Beyond STEP

STEP is the hardest mathematics examination most school students ever sit. It is also one of the best: its questions are short, unfamiliar and deep, and the best of them are small pieces of university mathematics in disguise.

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